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A fast Quasi-Bayes recursive Procedure for Hierarchical Dirichlet processes

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« Pourquoi veut-on que je sois aujourd'hui de la même opinion qu'il y a six semaines ? En ce cas, mon opinion serait mon tyran. »

Le Rouge et le Noir, II, IX (1830)

Stendhal

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Introduction

Speed and statistical accuracy are two faces of the same coin: in Bayesian statistics, someone has always to trade one for the other. The modern response to this tension has taken many forms, from variational Bayes (Blei et al., 2017) to importance sampling (MacEachern et al., 1999) for streaming data to cite a few; yet all these methods share a common starting point: a fully specified likelihood-prior model, of which they approximate the posterior or predictive distribution. A different and less understood class of algorithms proceeds in the opposite direction: fast recursive procedures, notably Newton's algorithm (Newton et al., 1998), are defined directly as updating rules rather than obtained from an explicit model; then the question is no longer on their approximation properties of a Bayesian procedure but on the same existence of a Bayesian justification.

Newton's algorithm is a fast online procedure for density estimation in nonparametric mixture models. Its frequentist consistency has been established (Ghosh and Tokdar (2006), Martin and Ghosh (2008) , Tokdar et al. (2009)), but consistency alone is an uninformative criterion for Bayesian validation: any consistent estimator agrees asymptotically with the Bayes solution, thus in this sense they all are indistinguishable. The question that remained open was a different one: which, if any, hierarchical model does Newton's algorithm imply and which is its relationship with a proper Bayesian model?

The proper framework to address this question is the Bayesian predictive approach, which characterizes a probability law for a stochastic process entirely through one-step-ahead predictive kernels. Unlike the usual Bayesian learning scheme starting from a model and a prior and updating it to a posterior distribution, here one starts from the algorithm itself, reads its updating rule as a probabilistic predictive learning rule , and may recover the implied statistical model. An algorithm is then called *quasi-Bayesian* (or, precisely, quasi-exchangeable Bayesian) if the statistical model it implies is, at least asymptotically, a genuine exchangeable Bayesian model: that is, if the sequence of observations generated under the algorithm's law is asymptotically exchangeable, with a (possibly implicitly defined) directing random measure $\tilde{P}$ playing the role of the asymptotic model, and the algorithm's estimates converge almost surely to $\tilde{P}$.

Within this framework, Fortini and Petrone (2020) showed that, when read as a probabilistic learning rule, Newton's recursions imply that the sequence of observations is *conditionally identically distributed* (c.i.d.) and hence asymptotically exchangeable, with

directing random measure $P_{\tilde{G}} = \int F(\cdot \mid \theta)\, d\tilde{G}(\theta)$, a mixture model where $\tilde{G}$ is the almost sure weak limit of the Newton estimates G_n thus has an implicit prior law.

These results both resolve an open question and open a new direction: the same predictive approach can be used to study recursive procedures for models beyond exchangeability, where exact posterior sampling is even more costly.
Partially exchangeable models, in which observations are exchangeable within groups but not across them, are an important such class. Bayesian nonparametric priors for partially exchangeable data, most notably the Hierarchical Dirichlet process (HDP) of Teh et al. (2006), are popular precisely because they formalize borrowing of strength across groups through shared latent components.

This work addresses the following questions.

1. Is it possible to construct a recursive algorithm for HDP mixtures that retains the computational simplicity of Newton's recursion while enabling borrowing of strength across groups?

2. Does such an algorithm admit a quasi-Bayesian interpretation in the sense that the statistical model it implies is asymptotically partially exchangeable?

3. How does the algorithm's one-step predictive *learning rule* compare to exact HDP inference, and what are the sources and magnitude of the approximation?

The main contribution of this work is the *Newton-HDP algorithm*, a novel recursive procedure that extends Newton's estimation principle to the HDP setting. The algorithm maintains group-level estimates $G_{j,t}$ and a global estimate $G_{0,t}$, coupled through a feedback loop that mirrors the hierarchical structure of the HDP and enables borrowing of strength at a computational cost of $\mathcal{O}(NJK)$, compared to $\mathcal{O}(N^2 \cdot C_{\text{MCMC}})$ for exact inference. We characterize the information losses of the algorithm relative to exact HDP inference, validate it empirically against both independent Newton recursions and a collapsed Gibbs sampler, and establish its theoretical foundations: under some restrictions and assumptions the group and global estimates converge almost surely, and the implied array $[X_{n,j}]_{n\geq 1, j=1,\ldots,J}$ is asymptotically partially exchangeable, and the procedure is a quasi (*partially-exchangeable*) Bayesian approximation of a HDP mixture model in the same sense that Newton's algorithm approximates a Dirichlet process mixture model.

Thesis structure. The thesis is structured in two parts. Part I introduces the foundational principles and develops the recursive predictive procedures in the framework of exchangeability. An original contribution here is in Chapter 3. Part II extends to partial exchangeability, and is the main original part of the thesis. More specifically,

Chapter 1 introduces the foundational framework: exchangeability, de Finetti's representation theorem, and the Bayesian predictive approach, which is compared to the inferential approach through the example of the Dirichlet process and its predictive characterization via the Chinese Restaurant Process. Lastly the Dirichlet process mixture models are introduced as the most popular semiparametric model for continuous data.

Chapter 2 presents Newton's algorithm as a quasi-Bayes procedure for Dirichlet process mixtures, following Fortini and Petrone (2020): we identify the statistical model implied by the recursions, establish the c.i.d. property and asymptotic exchangeability of the implied sequence, and discuss uncertainty quantification through the asymptotic posterior distribution of $\tilde{G}$.

Chapter 3 addresses some open theoretical questions through two empirical contributions: a clustering procedure based on the modal structure of the density estimate g_n, and a warm-start scheme using a preliminary Gibbs sampling step to reduce permutation sensitivity.

Chapter 4 extends the framework to partial exchangeability, introduces the HDP and its Chinese Restaurant Franchise predictive construction, and derives the exact one-step-ahead predictive distinguishing the group and global information contribution as a benchmark.

Chapter 5 presents the Newton-HDP algorithm, characterizes its information losses relative to exact HDP inference, and compares its one-step predictive with the exact expression of Chapter 4.

Chapter 6 evaluates the algorithm empirically across balanced, unbalanced, and heterogeneous settings.

Chapter 7 establishes the theoretical foundations: almost sure convergence of the estimates (Theorem 7.2), asymptotic partial exchangeability of the implied array (Theorem 7.3), and a discussion of the p-CID obstruction and directions for future work.

A final discussion of the results and of open directions concludes the thesis.

1 Foundational concepts. Inference and prediction in Bayesian statistics

In this chapter we start from foundational principles of Bayesian Statistics, to discuss the interpretation of inference and prediction in the Bayesian approach. The aim is to introduce the *predictive approach* and explain how it differs from the usual *inferential approach*, and how they are related in the context of exchangeability. This chapter provides the general framework within which we will develop our discussion.

1.1 Exchangeability, information and learning.

Bayesian Statistics has risen in popularity both from a theoretical and an applied point of view, for its fascinating probabilistic coherence, its flexible modeling properties, its optimality properties in many problems; and for giving the principles of rational reasoning under uncertainty, based on the subject's information. From a philosophical point of view, the developments around Bayesian methods stream from the ancient controversy about induction, or (very) roughly speaking generalization.

In fact, the debate on the rational justification of inductive processes, which is fundamentally what statistical inference is about, has a long history. David Hume in 1739 writes:

> *"Is there a rational justification of inductive methods, i.e. of methods that predict or infer that instances of which we have had no experience resemble those of which we have had experience? We tend to believe that things behave in a regular manner, i.e. that patterns in the behaviour of objects will persist into the future."*

This has been called the *Principle of Uniformity of Nature* and is, in metaphysical terms, what allows inferential techniques to have some sort of meaning; because, if there are no symmetries in what one observes and what does not, clearly inference is not meaningful.

Even though Hume gives a negative answer, the inductive process (the somehow principled generalization made on a phenomenon, having at hand only partial information on it), is crucial both in scientific inquiry and everyday life. The problem was reformulated by de Finetti (de Finetti (1937), de Finetti (1959)). In de Finetti (1975, p. 201), he noted

"In the philosophical arena, the problem of induction, its meaning, use and justification, has given rise to endless controversy, which, in the absence of an appropriate probabilistic framework, has inevitably been fruitless, leaving the major issues unresolved."

The appropriate probabilistic framework to express the Uniformity Principle is *exchangeability.*

Definition 1.1 (Infinite exchangeability)**.** A sequence of random variables $(X_i)_{i\geq 1}$, each taking values in $\mathbb{X}$, is said to be *infinitely exchangeable* if for every finite permutation σ of $\mathbb{N}$, i.e. a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that $\sigma(n) = n$ for all but finitely many n, it holds that

$$(X_1, X_2, \ldots) \stackrel{d}{=} (X_{\sigma(1)}, X_{\sigma(2)}, \ldots).$$

Thus exchangeability expresses a probabilistic symmetry; if one judges information on what is not observed to be similar to the one conveyed by what is observed than the latter can be used to say something about the former and generalize beyond finite samples. Incomplete information, and acting under it, is the core basic problem underlying Bayesian statistics and it is formalized through probability. Thus the focus is not on the physical process generating the data, rather on the observer assessment on the information structure, judged on the data, that is used to learn (on a latent parameter or the next data arriving).

In the context of random sampling i.e. homogeneous replicates of an experiment, judging observations as independent would imply no learning. Instead one would specify a joint law on $(X_1, X_2, \ldots, X_n)$ that allows for learning. In this case the most natural choice is that of an exchangeable law (one under which Theorem 1.1 holds); not because one thinks that the physics generating the data is as such but to specify his learning process.

1.2 Exchangeability and inference

Usually the assignment of a joint law is carried out by introducing latent parameters through which the data in some way are dependent; thus one has a sequence of random variables (r.v.'s) $(X_n)_{n\geq 1}$, specifies a conditional model $(X_1, \ldots, X_n) \mid \tilde{\theta} \sim p(x_1, \ldots, x_n \mid \tilde{\theta})$, $n \geq 1$ and a prior distribution π on $\tilde{\theta}$, then jointly:

$$p(x_1, \ldots, x_n) = \int p(x_1, \ldots, x_n \mid \theta)\, d\pi(\theta), \tag{1}$$

the focus then is to obtain the posterior $\pi(\cdot \mid x_{1:n})$ and compute the predictive distribution

$$p(x_{n+1}|x_{1:n}) = \int_\Theta p(x_{n+1}|x_{1:n}, \theta)\, d\pi(\theta|x_{1:n}),$$

as an average of all the possible models weighted by the posterior. The logical flow from prior to posterior and then predictive is what characterizes the so called inferential scheme in Bayesian statistics and emphasis is on the random $\tilde{\theta}$. However for years the issue on the mathematical justification of the introduction of a prior law on a latent quantity was controversial. The fundamental result that linked a probabilistic judgment on observables, i.e. exchangeability, and a hierarchical *representation as above* was the so called *de Finetti representation theorem* for which assigning an exchangeable law to the observables is *equivalent* to specifying a hierarchical structure for which the data are *conditionally iid* given a latent random parameter theoretically justifying a Bayesian inferential scheme.

To be rigorous on the objects involved in the theorem it is useful to give a definition of *random probability measure* (see Billingsley (1995)).

Definition 1.2 (Random probability measure)**.** Let $(\Omega, \mathcal{F})$ be a measurable space, a *random probability measure* $\tilde{P}$ on the space $(\mathcal{S}, \mathcal{G})$ is a measurable map

$$\tilde{P} : (\Omega, \mathcal{F}) \to (\mathcal{P}, \mathcal{H})$$

where $\mathcal{P}$ is the space of all probability measures on $\mathcal{S}$ and $\mathcal{H}$ is a suitable sigma algebra on it. [1]

We now state the theorem in its general form.

Theorem 1.3 (de Finetti's representation theorem)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathbb{X}$ be a Polish space and let $\mathcal{P}$ denote the space of all probability measures on $\mathbb{X}$, endowed with its Borel σ–algebra.*

[1] In particular $\mathcal{H} = \sigma\big(\{\varphi_A^{-1}(B) : A \in \mathcal{G},\, B \in \mathcal{B}([0,1])\}\big)$ where $\varphi_A(P) = P(A)$ denotes the evaluation map (for details see Billingsley (1995)) .

A sequence of random variables $(X_n)_{n\geq 1}$ taking values in $\mathbb{X}$ is infinitely exchangeable if and only if there exists a unique random probability measure $\tilde{P}$, taking values in $\mathcal{P}$, such that for all $n \geq 1$ and measurable sets $A_1, \ldots, A_n$,

$$\mathbb{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \tilde{P}(X_i \in A_i)\, Q(d\tilde{P})$$

Moreover, the law of $\tilde{P}$ is a probability measure Q on $(\mathcal{P}, \mathcal{H})$, and Q is called the de Finetti measure *(or* directing measure*) of the sequence. In addition with $\mathbb{P}$-probability one, the sequence of the empirical distributions $\hat{P}_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}$ converges weakly as $n \to \infty$ to a random distribution $\tilde{P}$,*

$$\hat{P}_n \to \tilde{P}.$$

In more familiar terms:

$$\begin{aligned} X \mid \tilde{P} &\overset{iid}{\sim} \tilde{P} \\ \tilde{P} &\sim Q, \end{aligned}$$

is the usual form in which an *exchangeable* Bayesian model is specified. De Finetti result puts emphasis on the law $\mathbb{P}$ which shapes the model $\tilde{P}$ through the prior Q. In practice one then chooses $\mathbb{P}$ and thus defines Q or, usually vice versa. In applications one may have enough information to restrict the support of Q so that

$$Q(\{P_\theta : \theta \in \Theta\}^c) = 0$$

meaning putting all prior mass over a family of models parametrized by θ. For our work it is useful to keep the general result in mind. For this reason we will indulge bit more on some aspects characterizing more practically the prior Q.

Q, puts mass over all possible statistical models which endows *nonparametric* specifications with a higher degree of flexibility with respect to adaptation to the data. This property of nonparametric priors is referred as *full weak support*. Indeed by endowing $\mathcal{P}$ with the metric induced by weak convergence [2] we can define the *weak support* for a

[2] A sequence of probability measures $\{P_n\}_{n\geq 1}$ on $\mathbb{X}$ is said to converge *weakly* to P, and we write $P_n \Rightarrow P$, if

$$\int_{\mathbb{X}} f\, dP_n \longrightarrow \int_{\mathbb{X}} f\, dP \quad \text{for all } f \in C_b(\mathbb{X}),$$

probability measure Q on $\mathcal{P}$ as :

$$\text{supp}^*(Q) = \Big\{P \in \mathcal{P} : \forall \varepsilon > 0,\ \forall f \in C(\mathbb{X}),\ Q\Big(W(P, \varepsilon, f)\Big) > 0\Big\}$$

Here $W(P, \varepsilon, f)$ denotes a weak neighborhood of P, defined by

$$W(P, \varepsilon, f) = \left\{P' \in \mathcal{P} : \left|\int_{\mathbb{X}} f\, dP - \int_{\mathbb{X}} f\, dP'\right| < \varepsilon\right\}$$

Thus a prior Q with full (weak) support places positive mass over each set (neighborhood) of probability measures on the space $\mathbb{X}$ i.e.

$$\text{supp}^*(Q) = \mathcal{P}$$

Here lies the flexibility of nonparametric models, or the rigidness of parametric ones, for which if $Q(\{A : A \in \mathcal{P}\}) = 0$ then $Q(A \mid X_1, \ldots, X_n) = 0$ (see Ghosal and Vaart (2017) for a more detailed explanation of the above). Thus when restricting Q too much, as happens in the parametric case, renders impossible to correctly recover the structure of the data if the model is mispecified.

1.3 Predictive approach and modeling

So far we have worked in the standard Bayesian learning scheme with prior and posteriors. Looking back at Equation 1, when assigning a probability law to specify a learning process, through the rules of probability, one can express the joint law of the sample using one step ahead predictive kernels so that for every n:

$$p(x_1, \ldots, x_n) = p(x_1)p(x_2|x_1) \cdots p(x_n|x_{1:n-1}). \tag{2}$$

This is at the core of the *predictive modeling* where one considers only observable quantities and on what information from the sample is relevant for prediction. The clear advantage lies in the bypass of latent parameters which, in complex settings tend to lose interpretability and for which assigning a prior is a tricky task; whereas specifying predictive rules on the data is less cumbersome.

This approach is justified also in a more foundational argument. De Finetti noticed that the usual Bayesian prior-likelihood interpretation was insufficient since, as de Finetti

noted, $\tilde{\theta}$ is of a "metaphysical" nature and probability statements should only be on observables (Bernardo and Smith (2009)). We already said that the central idea in Bayesian modeling is that of incomplete information, indeed as noted in Fong et al. (2023) the source of statistical uncertainty is the lack of $(X_{n+1}, \ldots, X_\infty)$ the presence of which would pin down even a latent quantity of interest as a function of the whole sample $\theta = g(X_1, \ldots, X_\infty)$. In this sense the hierarchal formulation with uncertainty expressed through the prior is opaque. Intuitively one of the advantages of this approach is that it does not require to specify a prior; this hints that the probabilistic requirements on the data, to keep enjoying the relevant advantages of Bayesian statistics, may be looser. We will follow up on this.

The theoretical architecture of predictive modeling start by specifying:

- an initial distribution $P_0(\cdot) = \mathbb{P}(X_1 \in \cdot)$,
- for $n \geq 1$ the one step ahead distributions $P_n(\cdot) = \mathbb{P}(X_{n+1} \in \cdot \mid X_{1:n})$

A well know fact in probability establishes that by specifying a sequence of transition kernels (as (P_n) is) one uniquely characterizes a probability law $\mathbb{P}$ over a sequence (X_n).

Theorem 1.4 (Ionesco-Tulcea)**.** *For all n, let $(S_n, \mathcal{G}_n)$ be a measurable space. Let Q_1 be a probability measure on $\mathcal{G}_1$ and, for all $n = 2, 3, \ldots$, let $\tilde{Q}_n$ be a transition kernel on $S_1 \times \cdots \times S_{n-1} \times \mathcal{G}_n$. Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random sequence (X_n) on it such that*

$$X_1 \sim Q_1$$

and, for all $n = 2, 3, \ldots$,

$$X_n \mid X_1 = x_1, \ldots, X_{n-1} = x_{n-1} \sim \tilde{Q}_n.$$

Moreover, the law of (X_n) is uniquely determined by $(Q_1, \tilde{Q}_2, \tilde{Q}_3, \ldots)$.

This result, then, allows the researcher to specify (P_n) and have the law $\mathbb{P}$ pinned down. So far we have given general results; however, given that the sequence (P_n) specifies $\mathbb{P}$, one may ask under what conditions on (P_n) the law $\mathbb{P}$ is exchangeable. The conditions are established in the following theorem of Fortini et al. (2000).

Theorem 1.5. *Let $(X_n) \sim \mathbb{P}$ be an infinite sequence of r.v.'s, with predictive rule $(P_n)_{n\geq 0}$. Then $(X_n)_{n\geq 1}$ is exchangeable if and only if, for every $n \geq 0$, the following conditions hold, with $P_0(\cdot|x_{1:0})$ meant as $P_0(\cdot)$:*

(i) For every A, $P_n(A|x_{1:n})$ is a symmetric function of $x_1, \ldots, x_n$.

(ii) The set function $(A, B) \to \int_A P_{n+1}(B|x_{1:n+1})\, dP_n(x_{n+1}|x_{1:n})$ is symmetric in A and B.

Condition (i) requires that the predictive distribution depends only on the *empirical distribution* $\hat{F}_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$, not on the ordering of observations, a natural requirement for exchangeability. Condition (ii) is more subtle: it demands that the predictive distribution for the *next two* observations (X_{n+1}, X_{n+2}) be jointly symmetric. This is weaker than requiring full joint exchangeability of all future observations but sufficient to characterize the entire exchangeable structure.

Moreover Aldous (1985) shows that the sequence (P_n) under an exchangeable law converge to the same limit as the empirical distribution $\hat{P}_n$ reconciling predictions with facts further strengthening the point made by Fong et al. (2023) on missing observations.

Proposition 1.6. *Let $(X_n)_{n\geq 1} \sim \mathbb{P}$ be an exchangeable sequence with predictive rule (P_n). Then $\mathbb{P}$-a.s., as $n \to \infty$, the sequence of predictive distributions converges to the random measure:*

$$P_n \to \tilde{P}, \quad \mathbb{P}\text{-a.s.},$$

where $\tilde{P}$ is as in Theorem 1.3.

These results allow us to establish an interesting connection between prediction and inference. Through a sequence of predictives that characterizes an exchangeable law over a sequence as in (2) , through Theorem 1.3 one can represent the sequence as Theorem 1.3. Thus by specifying distribution *just on observables* one can *implicitly* define a prior $\tilde{P}$ that can be of autonomous interest.

1.4 An example: the Dirichlet process

The symmetry between the *inferential approach* and the *predictive* one in the exchangeable case is best shown with the example of the predictive construction of the *Dirichlet process*

first as a distribution over $\mathcal{P}$ and then as the law of the random $\tilde{P}$ under the exchangeable law specified by the *Chinese Restaurant process* predictives.

A significant advancement in the literature and employment of nonparametric methods was made when a tractable and computationally convenient explicit definition of the form of the law Q over $\mathcal{P}$. This was done by Ferguson (1974) through the definition of the famous *Dirichlet process*.

Definition 1.7 (Dirichlet process)**.** Let $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$ be a Polish measurable space and let α be a scalar in $\mathbb{R}$ and P_0 a non-random probability distribution on $\mathcal{B}(\mathbb{X})$. A random probability measure $\tilde{P}$ on $\mathbb{X}$ is said to follow a *Dirichlet process* with base measure P_0, denoted

$$\tilde{P} \sim \mathrm{DP}(\alpha P_0),$$

if for every finite measurable partition $\{B_1, \ldots, B_n\}$ of $\mathbb{X}$, it holds that

$$\Big(\tilde{P}(B_1), \ldots, \tilde{P}(B_n)\Big) \sim \mathrm{Dirichlet}\Big(\alpha P_0(B_1), \ldots, \alpha P_0(B_n)\Big),$$

Probably the most appealing property of the *DP* as a nonparametric prior is its conjugacy in a model of the type:

$$\begin{aligned} X \mid \tilde{P} &\overset{iid}{\sim} \tilde{P} \\ \tilde{P} &\sim DP \end{aligned}$$

Seminal literature (see Ferguson (1974) and Sethuraman (1994)) proves that:

$$Q(\tilde{P} \mid X_1, \ldots, X_n) = DP\left(\frac{\alpha}{\alpha + n} P_0 + \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{X_i}\right)$$

The updated base measure of the posterior is a convex combination of the prior base measure P_0 and the empirical distribution. In this sense α plays the role of a shrinkage parameter, which pulls the posterior distribution towards the initial "guess". Clearly in the limit, as it is for every Bayesian model, the data "sweep" the prior as $\frac{\alpha}{\alpha+n} \to 0$. This property of the *DP* has rendered it as the most used and computationally friendly nonparametric prior.

A predictive characterization as extension of *Pòlya sequences* was given by Blackwell and MacQueen (1973), who proved that an exchangeable sequence $(X_n)_{n\geq 1}$ arises from a *DP* prior if and only if it satisfies the following predictive rule:

$$X_1 \sim P_0, \quad X_{n+1} \mid X_1, \ldots, X_n \sim \frac{\alpha}{\alpha+n} P_0 + \frac{1}{\alpha+n} \sum_{i=1}^{n} \delta_{X_i}. \tag{3}$$

The predictive distribution is thus a mixture: with probability $\frac{\alpha}{\alpha+n}$, the next observation is drawn from the base measure P_0, while with probability $\frac{n}{\alpha+n}$, it replicates one of the previous observations uniformly at random. This "rich get richer" dynamics generates *clustering*: observations naturally group into ties, even when P_0 is diffuse.

Importantly, this predictive rule operates entirely at the *observable* level, no explicit reference to $\tilde{P}$ appears. Yet by the representation theorem, such a sequence is exchangeable with directing random measure $\tilde{P} \sim \mathrm{DP}(\alpha P_0)$.

The clustering induced by (3) can be made explicit through the *Chinese Restaurant Process* (CRP). Imagine customers entering a restaurant sequentially and choosing tables according to the rule:

- The first customer sits at table 1.
- Customer $n+1$ joins table k (where n_k customers already sit) with probability $\frac{n_k}{\alpha+n}$, or sits at a new table with probability $\frac{\alpha}{\alpha+n}$.

This metaphor captures the predictive dynamics: "tables" correspond to distinct observed values, and the seating rule matches (3). Crucially, the table assignments induce a *random partition* ρ_n of $\{1, \ldots, n\}$, where customers i and j are in the same block if $X_i = X_j$.

The probability of a particular partition $\rho_n = (A_1, \ldots, A_{k_n})$ with block sizes $(n_1, \ldots, n_{k_n})$ can be derived directly from the predictives. Following the seating process:

$$\mathbb{P}(\rho_n) = \frac{\alpha}{\alpha} \cdot \frac{\alpha}{\alpha+1} \cdots \frac{\alpha}{\alpha+k_n-1} \cdot \prod_{j=1}^{k_n} \frac{(n_j-1)!}{(\alpha+n_{j,\mathrm{start}}) \cdots (\alpha+n_{j,\mathrm{end}}-1)},$$

where the first term accounts for creating k_n tables, and the product captures within-table growth. Simplifying yields the *exchangeable partition probability function* (EPPF) for the

Dirichlet process:

$$\mathbb{P}(\rho_n) = \frac{\alpha^{k_n}}{\alpha^{[n]}} \prod_{j=1}^{k_n} (n_j - 1)!, \tag{4}$$

where $\alpha^{[n]} = \alpha(\alpha+1)\cdots(\alpha+n-1)$ is the rising factorial. This compact formula depends only on the partition structure $(n_1, \ldots, n_{k_n})$, not on which specific values appear. The concept of EPPF was introduce in the literature by Pitman (1996).

The EPPF (4) thus emerges from the predictive rule (3), without ever invoking the inferential construction via Dirichlet distributions. This exemplifies the power of the predictive approach: complex hierarchical structures can be characterized through simple sequential learning rules. More examples involving predictive constructions and random partitions can be found in section 3.5 of Fortini and Petrone (2025).

1.5 Mixture priors for continous data

The predictive rule (3) naturally handles discrete observations, as the δ_{X_i} assign positive mass to previously seen values. For continuous data, this would yield degenerate predictions concentrated on the sample points. The classical remedy is to introduce a *smoothing kernel*, leading to the *Dirichlet process mixture model* (DPMM) Lo (1984):

$$\begin{aligned} X_i \mid \theta_i &\overset{ind}{\sim} f(\cdot \mid \theta_i) \\ \theta_i \mid \tilde{G} &\overset{iid}{\sim} \tilde{G} \\ \tilde{G} &\sim \mathrm{DP}(\alpha P_0). \end{aligned} \tag{5}$$

Marginally, this gives $X_i \mid \tilde{G} \overset{iid}{\sim} f_{\tilde{G}}(x) = \int f(x \mid \theta)\, d\tilde{G}(\theta)$. Exchangeability is assumed for the latent (θ_n): observations sharing the same θ_i form a cluster, while the kernel $f(\cdot \mid \theta)$ spreads mass continuously.

This latent clustering can be made explicit through the partition representation. Since $\tilde{P}$ is discrete, the sequence $(\theta_i)_{i\geq 1}$ induces a random partition ρ_n where $i \sim j$ if $\theta_i = \theta_j$.

Equivalently, the generative process becomes:

$$
\begin{aligned}
&X_i \mid \theta^*_{c(i)} \sim f(\cdot \mid \theta^*_{c(i)}),\\
&\theta^*_j \overset{iid}{\sim} P_0 \text{ for } j = 1, \dots, k_n,\\
&\rho_n \sim \text{EPPF}(\alpha),
\end{aligned}
$$

where $c(i) \in \{1, \dots, k_n\}$ identifies the cluster containing observation i. This formulation clarifies that DPMMs are *clustering models*: aleatoric randomness lives in the emission kernel f, while learning occurs through posterior inference on the partition structure ρ_n.

A natural attempt to adapt Pólya sequences to continuous data might be to directly smooth the empirical distribution in (3), replacing point masses δ_{X_i} with probability kernels $K(\cdot|X_i)$:

$$P_n(\cdot) = \frac{\alpha}{\alpha + n} P_0(\cdot) + \frac{1}{\alpha + n} \sum_{i=1}^{n} K(\cdot|X_i).$$

However, the exchangeability constraint severely restricts admissible choices of K (see Berti et al. (2023)). This hints at the fact that exchangeability might be too strict of a requirement and one could accept a more general form of dependence, that still preserves some results of exchangeable sequences (such as Theorem 1.6) as a result of an *approximate* procedure.

2 Newton's recursions as a quasi-Bayes procedure for mixture models

In this chapter we develop the discussion on the computational difficulties to estimate *DPMM*. We have seen that preserving exchangeability using Kernel-based Dirichlet sequences is not a satisfactory direction. Thus, we introduce a recursive procedure, referred to as *Newton* 's algorithm (Newton, 2002), as a fast and online-friendly recursive approximation for the mixing density. Then, following Fortini and Petrone (2020) and Fortini and Petrone (2025), we will re-interpret Newton's recursive rule as a predictive learning rule, that give rise to a sequence (X_n) that is no longer exchangeable. We will introduce the notion of *conditionally identically distributed* sequences and leveraging it clarify in which sense Newton's rule is a *quasi-Bayes* approximation for a DPMM.

2.1 DPMM's computational complexities

The posterior of (5) takes the form

$$\tilde{G} \mid x_{1:n} \sim \int \mathrm{DP}\left(\alpha G_0 + \sum_{i=1}^{n} \delta_{\theta_i}\right) \mathbb{P}(d\theta_{1:n} \mid x_{1:n}),$$

where the posterior distribution over latent cluster assignments $\mathbb{P}(\theta_{1:n} \mid x_{1:n})$ is intractable. The Bayesian posterior mean $\mathbb{E}(\tilde{P} \mid x_{1:n})$ can be expressed recursively as shown in Fortini and Petrone (2020) as :

$$\begin{aligned}
\tilde{G}_n^{(\mathrm{Bayes})}(\cdot) &= \mathbb{E}\Big(\tilde{G}(\cdot) \mid x_{1:n}\Big) = \mathbb{P}(\theta_{n+1} \in \cdot \mid x_{1:n}) \\
&= \frac{\alpha}{\alpha+n} P_0(\cdot) + \frac{1}{\alpha+n} \sum_{i=1}^{n} \mathbb{P}(\theta_i \in \cdot \mid x_{1:n}) \\
&= \frac{\alpha+n-1}{\alpha+n} \frac{\alpha P_0(\cdot) + \sum_{i=1}^{n-1} \mathbb{P}(\theta_i \in \cdot \mid x_{1:n})}{\alpha+n-1} + \frac{1}{\alpha+n} q_{\tilde{G}_{n-1}^{(\mathrm{Bayes})}}(\theta_n \in \cdot \mid x_n) \qquad (6)
\end{aligned}$$

where the posterior cluster probabilities $\mathbb{P}(\theta_i \in \cdot \mid x_{1:n})$ depend on *all* observations and

$$q_{\tilde{G}_{n-1}^{(\mathrm{Bayes})}}(\theta_n \in A \mid x_n) = \frac{\int_A f(x_n \mid \theta)\, d\tilde{G}_{n-1}^{(\mathrm{Bayes})}(\theta)}{\int_\Theta f(x_n \mid \theta)\, d\tilde{G}_{n-1}^{(\mathrm{Bayes})}(\theta)}.$$

Evaluating (6) requires recomputing cluster assignments for *all* previous observations at each step, the partition must be globally reassessed as each new datum arrives. Stan-

dard MCMC algorithms for DPMMs, such as Algorithm 8 in Neal (2000), operate by sampling cluster assignments for all n observations across $h = k_n + m$ components (including auxiliary "empty" clusters), yielding computational cost $\mathcal{O}(nh)$ per iteration. This becomes prohibitive for large n and entirely unsuitable for streaming data, where observations arrive sequentially and rapid updating is essential.

2.2 Newton's recursions

In Newton et al. (1998), Newton and Zhang (1999) and Newton (2002) the Bayesian posterior estimate of the mixing measure $\mathbb{E}(\tilde{G} \mid x_{1:n})$ is approximated at each n starting from an initial guess G_0 and for any A by:

$$G_n(A) = (1 - \alpha_n)G_{n-1}(A) + \alpha_n \frac{\int_A f(x_n|\theta)\, dG_{n-1}(\theta)}{\int_\Theta f(x_n|\theta)\, dG_{n-1}(\theta)} \tag{7}$$

where (α_n) is a sequence in $(0, 1)$ so that $\alpha_n \to 0$ with $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$. In analogy with DPMMs, α_n is usually chosen as $\alpha_n = 1/(\alpha + n)$. Newton's algorithm does not update G_{n-1} when a new observation arrives; rather, x_n enters only in the inference about θ_n through the term

$$q_{G_{n-1}}(A \mid x_n) = \frac{\int_A f(x_n|\theta)\, dG_{n-1}(\theta)}{\int_\Theta f(x_n|\theta)\, dG_{n-1}(\theta)}, \tag{8}$$

treating the previous estimate G_{n-1} as an informed prior for θ_n. This constitutes a fundamental departure from exact Bayesian updating, where the arrival of x_n would trigger a re-weighting of all past latent parameters $\theta_1, \ldots, \theta_{n-1}$ through the posterior $\mathbb{E}(\tilde{G} \mid x_{1:n})$.

Even at $n = 1$, Newton's rule introduces an additional simplification with respect to (6). The exact Bayesian posterior is (2.1), a mixture of Dirichlet processes where each component corresponds to a possible value of the latent parameter θ_1 weighted by its posterior probability. Newton's algorithm collapses this mixture into a single Dirichlet process $\mathrm{DP}(\alpha G_0 + \tilde{P}(\theta_1 \mid x_1))$, effectively replacing the integration over θ_1 with its posterior distribution directly added to the base measure.

Moreover if G_0 has density g_0 with respect to a measure λ, then G_n has density g_n with respect to the same λ, and rule Equation 7 implies that g_n satisfies the recursive

equation

$$g_n(\theta) = (1-\alpha_n)g_{n-1}(\theta) + \alpha_n \frac{g_{n-1}(\theta)f(x_n|\theta)}{\int f(x_n|\theta')g_{n-1}(\theta')\,d\lambda(\theta')} \tag{9}$$

Clearly the bookkeeping of this procedure is much less involved then what is needed in an exact sampler. Indeed if one evaluates G_n on a grid of K points and computes the normalization constant with a trapezoid rule the cost is $\mathcal{O}(Kn)$. In addition all the information with a dataset of size n is stored in G_n, this makes updating at the arrival of x_n much faster.

2.3 Newton recursion as a probabilistic learning rule

A powerful consequence of the framework of predictive modeling is that it only requires to specify one-step ahead prediction rules. Newton algorithm takes the form: $X_1 \sim P_0$ and for every $n \geq 1$ $X_{n+1}|X_1, \ldots, X_n \sim P_n$, with

$$\begin{cases} P_n = p_n(T_n), \\ T_n = h_n(T_{n-1}, X_n), \end{cases} \tag{10}$$

where p_n and h_n are given functions and T_n is a predictive sufficient summary of $X_1, \ldots, X_n$. In our case G_n encodes the information up to datum x_n and $h_n = \frac{\int_A f(x_n|\theta)\,dG_{n-1}(\theta)}{\int_\Theta f(x_n|\theta')\,dG_{n-1}(\theta')}$ and lastly $p_n = \int f(x \mid \theta) dG_n(\theta)$. By the Ionesco-Tulcea theorem, the sequence (P_n) uniquely determines $\mathbb{P}$ on the array $((X_n, \theta_n))_{n\geq 1}$, thereby defining a coherent probabilistic model for the joint evolution of observations and latent parameters. Clearly (7) does not meet the requirements of Theorem 1.5 since clearly the order in $\omega = (x_1, \ldots, x_n)$ change the realization of $G_n(\omega)$. One then may ask which kind of dependence is encoded through P_n into $\mathbb{P}$ and, under $\mathbb{P}$, which properties are preserved with respect to exchangeability and most importantly if, at least in the limit, it resembles a Bayesian exchangeable model. All these questions find answer in the approach proposed by Fortini and Petrone (2020) which specify a statistical model starting from the Newton's recursions understood in the form specified by (10).

The point estimate Equation 7 can be reinterpreted as the predictive distribution of θ_{n+1}

given $x_{1:n}$, and one may see an underlying latent variable model:

$$\begin{aligned} X_n \mid \theta_n &\overset{\text{indep}}{\sim} f(x \mid \theta_n), \\ \theta_{n+1} \mid x_{1:n} &\sim G_n(\cdot), \quad n \geq 1. \end{aligned} \tag{11}$$

with $\theta_1 \sim G_0$ and G_n as in (7).

Clearly (11) is different from (5) thus when using Newton's rule one is implicitly assuming a different statistical model pinned down by $\mathbb{P}$. From (11) it follows that $X_1 \sim P_{G_0}(\cdot) = \int f(\cdot \mid \theta) dG_0(\theta)$ and for any $n \geq 1$:

$$X_{n+1} \mid x_{1:n} \sim P_n = \int F(\cdot \mid \theta) dG_n(\theta) = (1 - \alpha_n) P_{n-1}(\cdot) + \alpha_n P_{G_{n-1}}(\cdot \mid x_n), \tag{12}$$

where $P_{G_{n-1}}(\cdot \mid x_n) = \int F(\cdot \mid \theta) dq_{G_{n-1}}(\theta \mid x_n)$, where $F(\cdot \mid \theta)$ is the distribution function associated with the density kernel $f(x \mid \theta)$ and we denote with the same symbol the corresponding measure. This provides a predictive characterization of the sequence (X_n).

2.4 Beyond exchangeability: CID sequences

The question on the stochastic dependence implied by these predictive rules is still open and to answer it we introduce the concept of *conditionally identically distributed* (c.i.d.) sequences, first studied in the exchangeable setting by Kallenberg (1988) and formalized in full generality by Berti et al. (2004). For a start we report Kallenberg (1988)'s Proposition 2.1.

Proposition 2.1. *A stationary sequence* (X_n) *that satisfies*

$$(X_1, \ldots, X_n, X_{n+1}) \overset{d}{=} (X_1, \ldots, X_n, X_{n+k}) \quad \textit{for all integers } k \geq 1 \textit{ and } n \geq 1 \tag{13}$$

is exchangeable.

Notice that (13) is equivalent to $(X_1, \ldots, X_n, X_{n+1}) \overset{d}{=} (X_1, \ldots, X_n, X_{n+2})$ for all $n \geq 1$. Extending this notion, Berti et al. (2004) give the following definition.

Definition 2.2 (Conditionally identically distributed sequence)**.** Let (X_n) be a sequence adapted to a filtration $\mathcal{G} = (\mathcal{G}_n)_{n \geq 0}$. The sequence (X_n) is said to be *conditionally*

identically distributed with respect to the filtration $\mathcal{G}$, or $\mathcal{G}$-c.i.d., if

$$\mathbb{E}[f(X_{n+k}) \mid \mathcal{G}_n] = \mathbb{E}[f(X_{n+1}) \mid \mathcal{G}_n] \quad \mathbb{P}\text{-a.s.}$$

for all $k \geq 1$, $n \geq 0$ and all bounded measurable functions $f : \mathbb{X} \to \mathbb{R}$.

Roughly speaking, the future observations X_{n+k} are identically distributed, given the past $\mathcal{G}_n$. In particular, the X_n are marginally identically distributed. When considering the natural filtration, i.e. when $\mathcal{G}_n = \sigma(X_1, \ldots, X_n)$ and $\mathcal{G}_0 = \{\emptyset, \Omega\}$, (X_n) is called *naturally c.i.d.*, or, simply, c.i.d. That is for all $n \geq 1$ and all $k \geq 1$:

$$X_{n+k} \mid \mathcal{G}_n \stackrel{d}{=} X_{n+1} \mid \mathcal{G}_n$$

Then a predictive rule, to characterize and exchangeable law must be stationary and comply with (13). From this, one understands that the CID condition is more general than that of exchangeability. Clearly all of what has been said so far about exchangeable sequence holds for CID ones. Kallenberg (1988) and then Berti et al. (2004) give an equivalent condition to CID.

Proposition 2.3. *Let* $(\mathcal{G}_n)$ *be the natural filtration generated by a sequence* (X_n) *The following properties are equivalent:*

(i) The sequence (X_n) *is CID.*

(ii) The process $P_n(\cdot) = \mathbb{P}(X_{n+1} \in \cdot \mid \mathcal{G}_n)$ *is a measure-valued* $\mathcal{G}$*-martingale.*

We are now ready to give two main results that hold for CID sequences. We have hinted, when discussing DP mixtures, that we one would still want to hold on to some results about exchangeable sequences for a more general dependence form. Berti et al. (2004) provide the following results.

Theorem 2.4 (Strong Law of Large Numbers for c.i.d. sequences)**.** *Let* (X_n) *be a c.i.d. sequence with* $\mathbb{E}[|X_1|] < \infty$*. Then*

$$\frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow{a.s.} \int x\, \tilde{P}(x) \qquad \textit{and} \qquad \frac{1}{n}\sum_{k=1}^{n} X_k \xrightarrow{L^1} \int x\, d\tilde{P}(x),$$

where $\tilde{P}(\cdot) = \mathbb{P}(X_1 \in \cdot \mid \mathcal{G}_\infty)$ *is the directing random measure.*

Theorem 2.5 (Asymptotic equivalence of predictive and empirical distributions)**.** *Let* (X_n) *be a c.i.d. sequence. Then*

$$\frac{1}{n}\sum_{i=1}^{n}\delta_{X_i} \xrightarrow{w} \tilde{P} \quad \mathbb{P}\text{-}a.s. \qquad \text{and} \qquad \mathbb{P}(X_{n+1} \in \cdot \mid \mathcal{G}_n) \xrightarrow{w} \tilde{P} \quad \mathbb{P}\text{-}a.s.$$

Informally speaking, we are handed results like those of Theorem 1.3 and Theorem 1.6 for CID sequences. In the exchangeable case $\tilde{P}(\cdot)$ is the directing random measure of the sequence (or the statistical model) whose law is the de Finetti measure. With an abuse of terminology we will still refer to $\tilde{P}(\cdot)$ as the directing random measure for CID sequences even though at finite n we cannot give a representation theorem of the form defined in Theorem 1.3.

Then a natural question is what is the relation of CID sequences to exchangeability; or "how far off" is a CID sequence to an exchangeable one. It turns out that CID sequences are exchangeable in the limit as proved in Aldous (1985), Lemma 8.2.

Lemma 2.6. *Let* (X_n) *be an infinite sequence, let* $P_n(\cdot) = \mathbb{P}(X_{n+1} \in \cdot \mid \mathcal{G}_n)$ *and suppose* $P_n \to \tilde{P} \quad \mathbb{P} - a.s.$ *Then*

$$(X_{n+1}, X_{n+2}, \ldots) \xrightarrow{d} (Z_1, Z_2, \ldots) \quad as\ n \to \infty,$$

where (Z_n) *is an infinite exchangeable sequence directed by* $\tilde{P}$*.*

By Theorem 2.5, a CID sequence is asymptotically exchangeable with directing random measure $\tilde{P}$ or as Fortini and Petrone (2020) put it $X_n \mid \tilde{P} \overset{n}{\approx} \tilde{P}$ for large n. This allows us to recover, at least in the limit, a de Finetti-like hierarchical representation for a CID sequence.

2.5 Newton's model as a quasi-Bayes approximation of DP mixtures

Previously we asked in which way the sequence (X_n), whose statistical model is (11), departs from exchangeability. The c.i.d. framework established before is exactly what is needed to answer. Before doing so, we briefly motivate why this question matters by recalling the notion of *quasi-exchangeable Bayes* procedure formalized by Fortini and Petrone (2020).

The frequentist consistency of Newton's algorithm has been studied extensively. Ghosh and Tokdar (2006) proved almost sure convergence of G_n to the true mixing density g when Θ is finite, under regularity conditions on the kernel and the weights, namely $\sum_n w_n^2 < \infty$, $\sum_n w_n = \infty$, and identifiability of the kernel. Martin and Ghosh (2008) strengthened this to weak consistency almost surely via stochastic approximation arguments, and Tokdar et al. (2009) extended the result to general Θ under analogous conditions. However, as Fortini and Petrone (2020) observe, frequentist consistency alone does not capture what is specifically Bayesian about Newton's procedure. When consistent, Newton's estimator will asymptotically agree, almost surely, with any other consistent estimator of $\widetilde{G}$ under the law of i.i.d. observations from the true $f_{G_{\text{true}}}$, making them indistinguishable under this criterion. The computational advantage of Newton's recursion is preserved, but its Bayesian motivation is lost.

A genuinely Bayesian interpretation is different: when a researcher uses estimator (11), she is implicitly changing the predictive rule for θ_{n+1} and therefore using (11) that differs from the exchangeable Bayesian model (5). Following Fortini and Petrone (2020), we say that a probability law $\widetilde{\mathbb{P}}$, characterized by a sequence of predictive rules, is a *quasi exchangeable-Bayes* approximation of an exchangeable law $\mathbb{P}$ if, for a sequence $(X_n) \sim \widetilde{\mathbb{P}}$, the sequence is asymptotically exchangeable in the sense of Lemma 2.6, with limiting exchangeable law $\mathbb{P}$.

The relevance of this definition is precisely what the c.i.d. and Aldous frameworks deliver: if $(X_n) \sim \widetilde{\mathbb{P}}$ is c.i.d., then by Theorem 2.5 its predictive distributions converge almost surely to a directing random measure $\widetilde{P}$, and by Lemma 2.6 the tail of the sequence converges in distribution to an exchangeable sequence directed by $\widetilde{P}$. This recovers, at least asymptotically, the full de Finetti structure of Theorem 1.3: the researcher is acting as if she had assigned a proper prior Q on $\widetilde{G}$, and genuine Bayesian uncertainty quantification through $\widetilde{\mathbb{P}}$ becomes available , something that frequentist consistency results cannot deliver. In our case we are approximating the exchangeable model (5) with (11), and the question becomes: is Newton's law $\widetilde{\mathbb{P}}$ a quasi-Bayes approximation of the DPMM exchangeable law $\mathbb{P}$, and what is its directing random measure?

Fortini and Petrone (2020) exactly answer it by leveraging the properties of CID

sequences.

Theorem 2.7. *Let the process $((X_n, \theta_n))$ have a probability law $\mathbb{P}$ that satisfies assumptions* (11)*. Then, $\mathbb{P}$-almost surely:*

(a) The sequence (G_n) converges to a random probability measure G, i.e.

$$G_n \xrightarrow{w} \tilde{G} \quad \mathbb{P}\text{-a.s.}$$

(b) For every $n \geq 1$, measurable set A, and $k \geq 1$,

$$P(\theta_{n+k} \in A \mid X_{1:n}) = \mathbb{E}(\tilde{G}(A) \mid X_{1:n}).$$

Proof sketch. **Part (a).** By Newton's recursion (7), fix a measurable set A:

$$G_n(A) = (1 - \alpha_n)G_{n-1}(A) + \alpha_n \cdot \frac{\int_A f(x_n \mid \theta)\, dG_{n-1}(\theta)}{\int_\Theta f(x_n \mid \theta)\, dG_{n-1}(\theta)}.$$

This can be written as $G_n(A) = G_{n-1}(A) + \alpha_n(Z_n(A) - G_{n-1}(A))$, where

$$Z_n(A) = \frac{\int_A f(x_n \mid \theta)\, dG_{n-1}(\theta)}{\int_\Theta f(x_n \mid \theta)\, dG_{n-1}(\theta)} = q_{G_{n-1}}(\theta_n \in A \mid x_n)$$

is the Bayes update of G_{n-1} at the single observation x_n. Now, by model (11):

$$\mathbb{E}[G_n(A) \mid X_{1:n-1}] = (1 - \alpha_n)G_{n-1}(A) + \alpha_n \mathbb{E}[Z_n(A) \mid X_{1:n-1}].$$

One verifies that $\mathbb{E}[Z_n(A) \mid X_{1:n-1}] = G_{n-1}(A)$, so that

$$\mathbb{E}[G_n(A) \mid X_{1:n-1}] = G_{n-1}(A),$$

Thus (G_n) is a $\mathcal{G}_n$-martingale. By Lemma 7.14 in Aldous (1985) we have:

$$G_n \xrightarrow{w} \tilde{G} \quad \mathbb{P} - a.s.$$

Part (b). $(G_n(A))$ is a bounded martingale for every A thus it is closed and by (11):

$$\mathbb{P}(\theta_{n+k} \in A \mid X_{1:n}) = \mathbb{E}(\mathbb{P}(\theta_{n+k} \in A \mid X_{1:n+k-1}) \mid X_{1:n}) = \mathbb{E}(G_{n+k-1}(A) \mid X_{1:n}) = \mathbb{E}(\tilde{G}(A) \mid X_{1:n})$$

□

One can then check using the relations established by (11) that the Newton's predictive (12) is indeed a martingale with respect to $(\mathcal{G}_n)$ being the natural filtration generated by the (X_n).

Proposition 2.8. *The predictive rule* (12) *is a* $(\mathcal{G}_n)$*- martingale therefore by Theorem 2.5 the sequence* (X_n) *is CID. Moreover the (limiting) directing random measure of sequence is* $P_{\tilde{G}}(\cdot) = \int F(\cdot \mid \theta) d\tilde{G}(\theta)$.

Proof. The first term in (12) is $\mathcal{G}_{n-1}$-measurable, so it suffices to compute the conditional expectation of the second. Under Newton's model, $X_n \mid \mathcal{G}_{n-1}$ is P_{n-1}-distributed. Therefore, for any measurable set A,

$$\begin{aligned}
\mathbb{E}\Big[\int_\Theta F(A \mid \theta) dq_{G_{n-1}} \mid \mathcal{G}_{n-1}\Big] &= \int_{\mathbb{X}} \int_\Theta \frac{F(A \mid \theta)}{\int_\Theta F(x \mid \theta)\, dG_{n-1}(\theta)} \cdot \int_\Theta F(x \mid \theta')\, dG_{n-1}(\theta')\, dP_{n-1} \\
&= \int_{\mathbb{X}} \frac{F(A \mid \theta)}{\int_\Theta F(x \mid \theta)\, dG_{n-1}(\theta)} \cdot \int_\Theta F(x \mid \theta')\, dG_{n-1}(\theta') \int_\Theta F(x \mid \theta)\, dG_{n-1}(\theta) dx \\
&= P_{n-1}(A)
\end{aligned}$$

where we use Fubini's theorem and the fact that being $F(\cdot \mid \theta)$ a probability measure on $\mathbb{X}$ then $\int_{\mathbb{X}} F(x \mid \theta) dx = F(\mathbb{X} \mid \theta) = 1$. Hence,

$$\mathbb{E}[P_n(A) \mid \mathcal{G}_{n-1}] = (1 - \alpha_n)\, P_{n-1}(A) + \alpha_n\, P_{n-1}(A) = P_{n-1}(A),$$

and by Proposition 2.3 the sequence (X_n) is c.i.d.

Moreover from Theorem 2.7 we have that on a set of $\mathbb{P}$-probability 1 Ω_j one for each $t_j \in \{t_j : j \in J\}$ $P_{G_n} \to P_{\tilde{G}}$, clearly holds for all j thus over the whole class. Which is enough to have weak convergence $\mathbb{P}$-almost surely.

Then by Theorem 2.6 we have that the sequence (X_n) is asymptotically exchangeable with directing random measure $P_{\tilde{G}}$ □

This results are telling precisely what we were hoping to recover. In the language used above we can say that the sequence (X_n) whose law is defined in (11) $X_n \mid \tilde{G} \approx P_{\tilde{G}}$ for n large enough. $P_{\tilde{G}}$ plays the role of the statistical model and $\tilde{G}$ of the infinite dimensional parameter.

This is exactly the quasi-Bayes justification of Newton's algorithm that we were looking

for. Moreover under an identifiability condition on $P_{\tilde{G}}$ (i.e. $P_{\tilde{G}_1} = P_{\tilde{G}_2}$ implies $\tilde{G}_1 = \tilde{G}_2$) Fortini and Petrone (2020) prove that also the sequence (θ_n) is asymptotically exchangeable .

2.6 The prior $\tilde{G}$

We have mentioned how, by specifying predictive rules, one, by Theorem 1.3, implicitly defines a mixing distribution with its prior law. This is a similar case, G indeed is implicitly defined by $\mathbb{P}$, i.e. the law implied by (7).

Although there are no results concerning the law of G, Fortini and Petrone (2020) prove that if G_0 is absolutely continuous with respect to a σ-finite measure λ on Θ ($G_0 \ll \lambda$), then we have a corresponding recursion for densities (see (9)). They prove the following result extending onto Theorems 1 and 4 of Berti et al. (2013) which shows that, for c.i.d. processes, the directing measure is absolutely continuous with respect to λ if and only if the predictive distribution is absolutely continuous and converges in total variation. We report Fortini and Petrone (2020) result here.

Theorem 2.9. *Let G be the P-a.s. limit of the sequence of predictive rules G_n defined by (7), with $G_0 \ll \lambda$. If the following conditions hold*

$$\sum_n \alpha_n^2 < \infty, \qquad \int_K g_0(\theta)^2 \, d\lambda(\theta) < \infty, \quad \text{for every } K \text{ compact,} \tag{14}$$

and

$$\sup_{\theta_1,\theta_2 \in K} \int \frac{f(x \mid \theta_1)^2}{f(x \mid \theta_2)} \, d\mu(x) < \infty, \quad \text{for every } K \text{ compact such that } \lambda(K) < \infty, \tag{15}$$

then $G \ll \lambda$, $\mathbb{P}$-a.s. Moreover, $\mathbb{P}$-a.s., G_n converges in L^1 to $g \equiv d\tilde{G}/d\lambda$.

Assumption (15) is the most delicate and holds for $f(\cdot \mid \theta)$ being a Poisson density or a Gaussian density with fixed variance or a Gamma density with fixed shape parameter. Being $\tilde{G}$ implicitly defined through Newton's recursions, its law is not available in closed form. Yet it is still possible to obtain samples from it. The key result is Theorem 2.5, since (X_n) is c.i.d., the sequence of predictive distributions (P_n) defined in (12) converge weakly almost surely to the directing random measure $P_{\tilde{G}}$.

This structure enables the predictive-based sampling scheme described in Fortini and

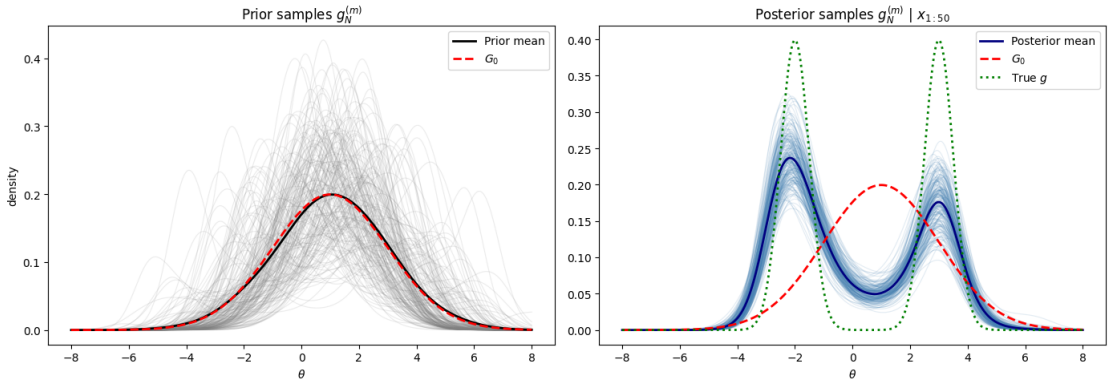


Figure 1: Prior and posterior draws from the implicit law of G under Newton's model, obtained via the predictive-based sampling scheme of Fortini and Petrone (2025, Section 2.4). *Left:* $M = 200$ prior trajectories of $G_n^{(m)}$ (gray; in fact, of its corresponding density g_n), with prior mean (black) coinciding with G_0 (red dashed) by the martingale property. *Right:* posterior draws given $n = 50$ observations from $g_{\text{true}} = 0.5\,\mathcal{N}(-2, 0.25) + 0.5\,\mathcal{N}(3, 0.25)$ (green dotted).

Petrone (2025, Section 2.4). The principle relies on Theorem 2.5: since the empirical distributions and predictive distributions of a c.i.d. sequence converge $\mathbb{P}$-a.s. to the same limit $\tilde{P}$, knowledge of the entire trajectory $\omega = (x_1, x_2, \ldots)$ pins down $\tilde{P}(\omega)$. The residual uncertainty on $\tilde{P}$ given a finite sample $(x_1, \ldots, x_n)$ is therefore exactly the uncertainty on the missing observations $(x_{n+1}, x_{n+2}, \ldots)$. Thus we can sample from prior and posterior of $\tilde{G}$ in the following way.

Sampling from the prior. For $m = 1, \ldots, M$, generate a trajectory $\omega^{(m)} = (x_1^{(m)}, x_2^{(m)}, \ldots, x_N^{(m)})$ by sampling recursively from the predictive rule (7): draw $x_1^{(m)} \sim G_0$ and $x_{n+1}^{(m)} \sim G_n^{(m)}(\cdot)$ for each subsequent step, updating $G_n^{(m)}$ via (7) along the simulated trajectory. For N sufficiently large, $G_N^{(m)}$ approximates the limit $G^{(m)}$, and

$\{G_N^{(m)}, m = 1, \ldots M\}$ constitutes a Monte Carlo sample from the prior law of $\tilde{G}$.

Sampling from the posterior. Given observed data $x_{1:n}$, one fixes the observed portion of the trajectory $\omega^{(m)}$ and generates only the missing observations $(x_{n+1}^{(m)}, \ldots, x_N^{(m)})$ from the predictive rule conditionally on $x_{1:n}$. The measure $G_N^{(m)}$, computed over the augmented trajectory $(x_1, \ldots, x_n, x_{n+1}^{(m)}, \ldots, x_N^{(m)})$, then approximates a draw from the posterior law of $\tilde{G}$ given $x_{1:n}$.

In Figure 1 the two panels illustrate the predictive-based sampling scheme applied to Newton's model , with $g_0 = \mathcal{N}(1, 2)$, $\alpha_n = 1/(\alpha + n)$, $\alpha = 5$, and a Gaussian kernel with

$\sigma^2 = 1$. These choices fall under the requirements of Theorem 2.9 so $\tilde{G}$ is $\mathbb{P}$ almost surely continuous with density g and thus we will consider the Newton recursion in terms of densities G_n (9) with respect to the Lebesgue measure on $\mathbb{X} = \mathbb{R}$.

The left panel shows $M = 200$ draws $G_n^{(m)}(t)$, for t on a fixed grid, from the prior law of $\tilde{g}$, obtained by generating trajectories of length $N = 10{,}000$ from the predictive rule and using $G_N^{(m)}$ as a proxy for the limit $\tilde{g}^{(m)}$. The prior mean (black) coincides with g_0 (red dashed), consistently with the martingale property that gives $\mathbb{E}[G_n(A)] = g_0(A)$ for all n. The high variability across trajectories reflects genuine prior uncertainty on $\tilde{g}$.

The right panel shows the corresponding posterior draws given $n = 50$ observations from a bimodal mixing distribution $g_{\text{true}} = 0.5\,\mathcal{N}(-2, 0.25) + 0.5\,\mathcal{N}(3, 0.25)$ (green dotted). The posterior samples are obtained by fixing the observed portion of the trajectory and generating only the missing observations from the conditioned predictive rule. Two features are apparent: the posterior mean (blue) correctly identifies the bimodal structure of g_{true}, departing substantially from both g_0 and the prior mean; and the posterior variability is markedly reduced relative to the prior, reflecting the information gained from the data.

Note that Newton's recursive estimate $G_n(A)$ can be regarded as a point estimate under quadratic loss of $\tilde{G}(A)$, being $G_n(A) = \mathbb{E}(\tilde{G}(A) \mid x_{1:n})$ (see point (b) of Theorem 2.7). However, for providing full Bayesian uncertainty quantification, we need the complete posterior distribution of $G(A)$ given the data; the strategy described above allows to sample from it and provide a (predictive) Monte Carlo approximation. Interestingly, it is also possible to provide an asymptotic approximation of the posterior distribution of $\tilde{G}(A)$, for any measurable set A (Fortini and Petrone, 2020). Denote by $\Phi(t \mid \mu, \sigma^2)$ the distribution function of a Gaussian with mean μ and variance σ^2 evaluated at t.

Theorem 2.10 (Asymptotic posterior distribution of $\tilde{G}(A)$)**.** *Let (X_n) follow Newton's model* (11)*. Let (α_n) satisfy $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$, and let (r_n) be a monotone sequence with $r_n \sim \left(\sum_{k>n} \alpha_k^2\right)^{-1}$, satisfying*

$$\sqrt{r_n} \sup_{k \geq n} \alpha_k \to 0, \qquad \sum_{k \geq 1} r_k^2 \alpha_{k+1}^4 < \infty.$$

Then, for every $A \in \mathcal{B}(\Theta)$,

$$\mathbb{P}\Big(\sqrt{r_n}\,(\tilde{G}(A) - G_n(A)) \le t \;\Big|\; X_{1:n}\Big) \xrightarrow{a.s.} \Phi(t \mid 0, V_A), \tag{16}$$

where

$$V_A = \int_{\{x:\, f_{\tilde{G}}(x)\neq 0\}} P_{\tilde{G}}(A \mid x)^2\, dF_{\tilde{G}}(x) - \tilde{G}(A)^2 = \mathbb{E}\Big[\Big(P_{\tilde{G}}(A \mid X) - \tilde{G}(A)\Big)^2\Big].$$

where for any distribution K *on* Θ, $P_K(A \mid x) \propto \int_A f(x \mid \theta) K(d\theta)$.
If $\alpha_n = (\alpha + n)^{-\beta}$ *with* $\beta \in (1/2, 1]$ *and* $\alpha > 0$, *then* (16) *holds with* $r_n = (2\beta - 1)n^{2\beta-1}$.

The asymptotic variance V_A is the $\mathbb{P}$-almost sure limit of $V_{A,n} = \mathbb{E}((P_{G_n}(A \mid X_{n+1}) - G_n(A))^2 \mid X_{1:n})$ which expresses the variability where P_{G_n} (as in (8)) acts as the posterior and G_n the prior. However V_A depends on $\tilde{G}$ which is unknown, thus the above result cannot be directly used to provide asymptotic credible intervals. Fortini and Petrone (2020) use a convergent estimator of V_A, as follows.

Theorem 2.11 (Asymptotic posterior distribution with estimated variance)**.** *Under the same assumptions as Theorem 2.10, define the estimator*

$$V_{A,n} = \int_{\mathbb{X}} P_{G_n}(A \mid x)^2\, dF_{G_n}(x) - G_n(A)^2,$$

which satisfies $V_{A,n} \xrightarrow{a.s.} V_A$. *Then, for almost all* $\omega = (x_1, x_2, \ldots)$ *such that* $V_A(\omega) > 0$,

$$P\left(\sqrt{r_n}\,\frac{\tilde{G}(A) - G_n(A)}{\sqrt{V_{A,n}}} \le t \;\middle|\; x_{1:n}\right) \to \Phi(t \mid 0, 1).$$

Theorem 2.11 yields computable asymptotic $(1-\gamma)$ credible interval for $G(A)$:

$$\Big[G_n(A) - z_{1-\gamma/2}\sqrt{V_{A,n}/r_n},\; G_n(A) + z_{1-\gamma/2}\sqrt{V_{A,n}/r_n}\Big].$$

A last fact conveyed by these results is the rate at which model (11) reaches exchangeabiliy. By 2.10, with weights of the type $\alpha_n = \frac{1}{(n+\alpha)^\beta}$ the rate of convergence r_n is $(2\beta-1)^{2\beta-1}$ with the fastest convergence (as $1/\sqrt{n}$) with $\beta = 1$ i.e. when α_n mimics the weights of a DPMM. This results highlights a fundamental advantage of recasting Newton's recursions as a predictive learning rule: without explicating the law over (X_n, θ_n)

implied by the algorithm, uncertainty quantification for a quasi-Bayes estimator such as G_n would not be feasible.

3 Warm start and clustering performance of Newton's algorithm

In this chapter we discuss some open questions about the implementation of Newton's recursions. In particular we discuss the role of model (11) as an approximate mixture model and, through an example, how well it can recover the partition structure of the data. This is useful to recover the clustering objective which is typical of DPMM.

The second question we discuss is how to reduce the impact of permutations of the data $x_{1:n}$ caused by the lack of the exchangeability of the sequence (X_n). We will show empirically that it may be convenient to introduce an initial MCMC step and input, as G_0 a smoothed version of the estimated (true) mixing measure in a sort of "warm start" fashion.

3.1 Clustering performance of Newton's algorithm

A natural question is whether the implicit prior law on $\tilde{G}$ induced by Newton's recursions coincides with the Dirichlet process prior $\mathrm{DP}(\alpha, G_0)$. The answer is negative. Under a $\mathrm{DP}(\alpha, G_0)$ prior, for any measurable set A the marginal prior distribution of $\tilde{G}(A)$ is

$$\tilde{G}(A) \sim \mathrm{Beta}(\alpha G_0(A),\, \alpha(1 - G_0(A)))\,.$$

Figure 2 compares the prior distribution of $G(0)$ implied by Newton's algorithm, obtained via the Monte Carlo sampling scheme of Section 2.6, with the $\mathrm{Beta}(\alpha G_0(0), \alpha(1 - G_0(0)))$ distribution of the DP. The two distributions are visibly different: Newton's implicit prior is more concentrated and shifted relative to the DP prior, confirming that the two models are distinct. This is consistent with Theorem 2.9, which establishes that the limit $\tilde{G}$ of Newton's recursion is $\mathbb{P}$-almost surely absolutely continuous, whereas DP realizations are almost surely discrete.

Despite not being a DP, Fortini and Petrone (2020) provide an interpretation of Newton's recursion as a smoothed version of a DP mixture. To make this precise, consider a sequence of latent random distributions $(\tilde{H}_n)$ such that

$$X_n \mid \tilde{H}_n \overset{\text{ind}}{\sim} P_{\tilde{H}_n}(\cdot) = \int f(\cdot \mid \theta)\, \mathrm{d}\tilde{H}_n(\theta), \tag{17}$$

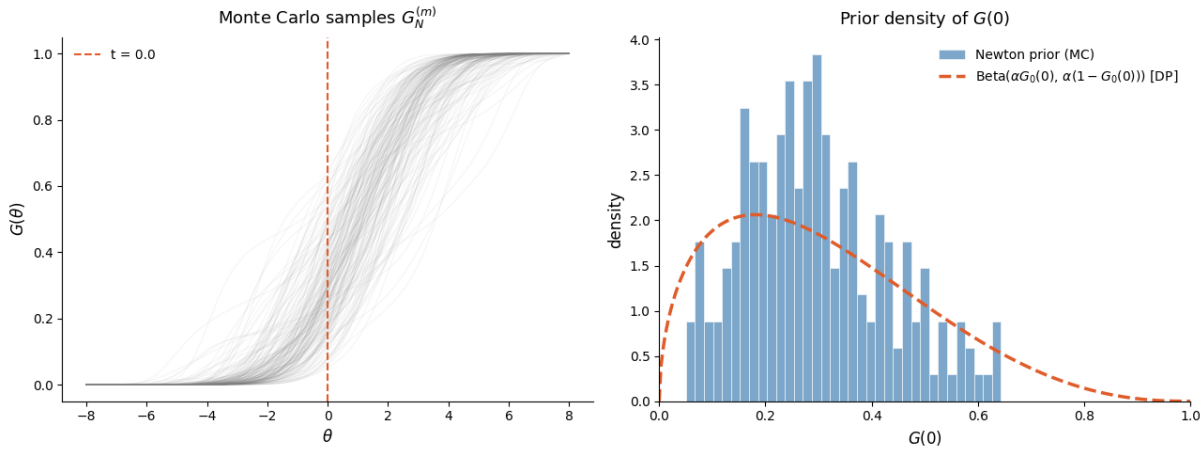


Figure 2: Comparison between the implicit prior law of G under Newton's model and the Dirichlet process prior. *Left:* $M = 200$ Monte Carlo prior trajectories $G_N^{(m)}$ (gray) obtained by simulating paths of length $N = 10{,}000$ from the predictive rule (7), with $G_0 = \mathcal{N}(1, 2)$, $\alpha_n = 1/(\alpha + n)$, $\alpha = 5$, and Gaussian kernel with $\sigma^2 = 1$. The dashed vertical line marks $\theta = 0$. *Right:* prior density of $G(0)$ under Newton's model (blue histogram with 35 bins, MC estimate) versus the Beta$(\alpha G_0(0), \alpha(1-G_0(0)))$ distribution implied by a DP(αG_0) prior (red dashed curve).

with dynamics

$$\tilde{H}_1 \sim \mathrm{DP}\left(\tfrac{1-\alpha_1}{\alpha_1} G_0\right), \qquad \tilde{H}_n \mid X_{1:n-1}, \theta_{1:n-1} \sim \mathrm{DP}\left(\tfrac{1-\alpha_n}{\alpha_n} G_{n-1}\right), \quad n > 1. \tag{18}$$

That is, at each time n, the latent mixing distribution $\tilde{H}_n$ follows a DP centered on the current Newton estimate G_{n-1}. This is a *time-dependent mixture model*, where the latent distributions evolve over time rather than remaining fixed as in a standard exchangeable mixture. The model is misspecified in a static setting, but provides a natural interpretation of the weights α_n as controlling the speed of evolution of the latent $\tilde{H}_n$: smaller α_n give a milder evolution, closer to the exchangeable case in which $\tilde{H}_n$ would remain constant.

Equations (17) and (18) are consistent with model (11) thus convergence results and quasi-Bayes interpretation carry over. Crucially, at each time n, $\tilde{H}_n$ is a DP realization, hence discrete, but its center G_{n-1} is absolutely continuous. The Newton estimate $G_n = \mathbb{E}[\tilde{G} \mid X_{1:n}]$ is in this sense a smoothed version of a DP: it inherits the recursive structure of DP updating while producing an absolutely continuous estimate. In the long run, $G_n \to \tilde{G}$ almost surely, and the model asymptotically stabilizes to a stationary mixing distribution. That is, under model (17) and (18) for n large we would result in a situation $X_n \mid \tilde{H} \sim P_{\tilde{H}}$ where $\tilde{H} \sim DP(\eta \tilde{G})$.

This smoothing interpretation motivates the approximate clustering analysis that follows, namely whether from the estimated G_n, or better its density g_n, is it possible to approximate the true partition of the data. This is important because DPMM are naturally used for clustering purposes and although Newton's recursions can only yield continuous mixing distributions, still one may want to extract some information about the underlying partition; in a sort of multiple shrinkage spirit rather than rigid clustering. Recall that, by the stick-breaking construction of Sethuraman (1994), if $\tilde{G}$ is $DP(\alpha G_0)$ distributed then

$$\tilde{G}(\cdot) \overset{a.s.}{=} \sum_{k=1}^{\infty} \pi_k \delta_{\theta_k^*}(\cdot). \tag{19}$$

In clustering exercises, DPMMs are popular for their flexibility, namely because one does not fix the number of components when specifying the statistical model. If data come from $G_{\text{true}} = \sum_{k=1}^{K} \pi_{k,\text{true}}\, \delta_{\theta^*_{k,\text{true}}}$, posterior consistency results would ensure that $\tilde{G} \mid x_{1:n}$ converges weakly to G_{true} as $n \to \infty$, concentrating mass on the true atoms $\theta^*_{k,\text{true}}$ with the correct weights $\pi_{k,\text{true}}$ and asymptotically suppressing spurious components. (It is worth noting that posterior consistency of the posterior law of the mixing distribution for DPMMs is a debated problem; see e.g. Ascolani et al. (2022)).

For the implicit prior from Newton's predictive rule, although G_n has no atoms, the smoothed-DP interpretation of equations (24)–(25) suggests, heuristically, that for large n the modal structure of g_n may reflect the discrete structure of G_{true}: modes $\hat{\theta}^*_k$ of g_n could approximate the true atoms, and the mass $G_n(C_k)$ assigned to neighborhoods of each mode could approximate the corresponding weight $\pi_{k,\text{true}}$. We explore this conjecture empirically in what follows.

Recovering the partition through g_n modes. Let $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_K$ denote the modes of g_n and define the partition of Θ induced by proximity to the estimated modes:

$$C_k = \left\{ \theta \in \Theta : \|\theta - \hat{\theta}^*_k\| \leq \|\theta - \hat{\theta}^*_{k'}\|,\ \forall\, k' \neq k \right\}, \tag{20}$$

so that $\{C_k\}_{k=1}^K$ partitions Θ into disjoint regions, each containing one estimated mode. The mixture weight π_k is then approximated by

$$\hat{\pi}_k = G_n(C_k) = \int_{C_k} g_n(\theta)\, \mathrm{d}\theta, \tag{21}$$

which converges to $\tilde{G}(C_k)$ as $n \to \infty$ by the almost sure convergence $G_n \to \tilde{G}$. For large n, when G_n is close to its limit, the mass assigned to each region C_k may approximate π_k provided the true atoms are sufficiently separated relative to σ. Each observation is then assigned to the region whose mode is nearest:

$$\hat{z}_i = \arg\min_k \|x_i - \hat{\theta}_k^*\|,$$

defining a partition of $\mathbb{X}$ induced by the structure of the modes of g_n.

Simulated example. We consider $n = 200$ observations from a Gaussian location mixture with $K = 3$ components,

$$G_{\text{true}} = 0.3\,\delta_{-3} + 0.5\,\delta_0 + 0.2\,\delta_4, \qquad X_i \mid \theta_i \sim \mathcal{N}(\theta_i, 1), \tag{22}$$

with $\alpha = 1$ and $G_0 = \mathcal{N}(1, 4)$. Newton's algorithm recovers three modes at $\hat{\theta}_1^* \approx -3$, $\hat{\theta}_2^* \approx 0$, $\hat{\theta}_3^* \approx 4$, with estimated weights $\hat{\pi}_1 = 0.341$, $\hat{\pi}_2 = 0.427$, $\hat{\pi}_3 = 0.193$, against true values 0.3, 0.5, 0.2 respectively. The partition achieves 90.5% accuracy on all 200 observations, with errors concentrated in the overlap region between C_1 and C_2, where the two Gaussian components are closest (Figure 3).

Iris dataset. We validate the approach on the classical Iris dataset, using petal length and petal width ($n = 150$, $K = 3$ species). Newton's algorithm identifies three modes of g_n corresponding to *Iris setosa*, *versicolor*, and *virginica*. The partition computed as (20) achieves 92.7% accuracy on all 150 observations, with misclassified observations concentrated in the *versicolor*–*virginica* boundary region (Figure 4).

Both experiments confirm the smoothed DP interpretation: Newton's absolutely continuous G_n approximates the discrete DP structure through its modal geometry. The accuracy of partition recovery depends on the separation between the true components relative to the kernel bandwidth σ: when the components are well separated, the modes of g_n are clearly identifiable and the estimated weights $\hat{\pi}_k$ are close to the true π_k. When components overlap, errors are localized in the boundary regions between adjacent partition cells C_k, as expected from the geometry of (20).

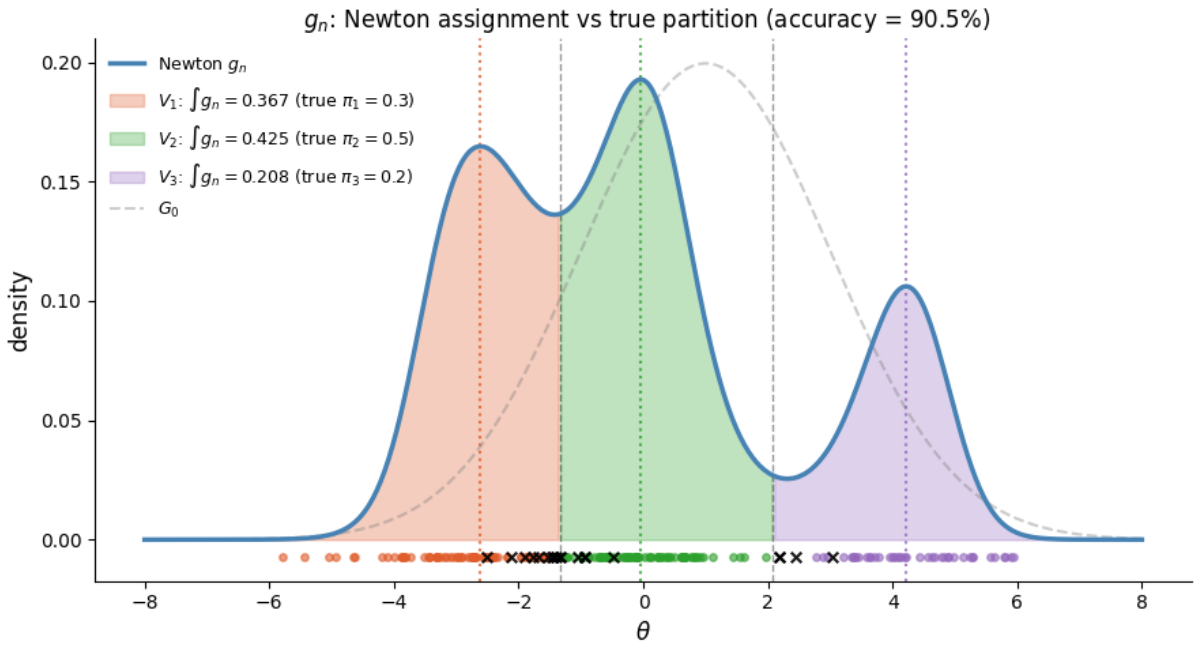


Figure 3: Mixing density estimate g_n after $n = 200$ observations from $G_{\text{true}} = 0.3\,\delta_{-3} + 0.5\,\delta_0 + 0.2\,\delta_4$. Shaded regions correspond to the partition $\{C_k\}$ induced by the modes of g_n (dotted vertical lines). Colored dots on the horizontal axis indicate correctly assigned observations; black crosses indicate misclassified observations.

This is consistent with the general principle that Newton's algorithm is a computationally fast, misspecified approximation of the DP mixture posterior, whose quality improves with data and which recovery of the latent structure depends on the kernel variance σ. However, the above analysis provides only an intuitive and not rigorous procedure to assess the properties of $\tilde{G}$ as a continuous counterpart of a DP. A rigorous discussion would require explicit results on the law of $\tilde{G}$.

Still, our study suggests that the approximate partition found by using Newton's algorithm can be employed as a warm start for MCMC algorithms. For simple datasets (as the Iris one) efficient samplers perform as well as one initialized using an estimated partition; we envisage that for more complicated datasets with many clusters a first Newton step with possibly a low kernel variance can speed up convergence of an MCMC sampler.

3.2 MCMC as a Newton's warm start

The lack of exchangeability renders the estimate G_n dependent on permutations. In particular, Fortini and Petrone (2020) show that, for learning parameters $\alpha_n = 1/(\alpha + n)$, the lack of invariance of G_n to sample permutations is heavily influenced by the choice of

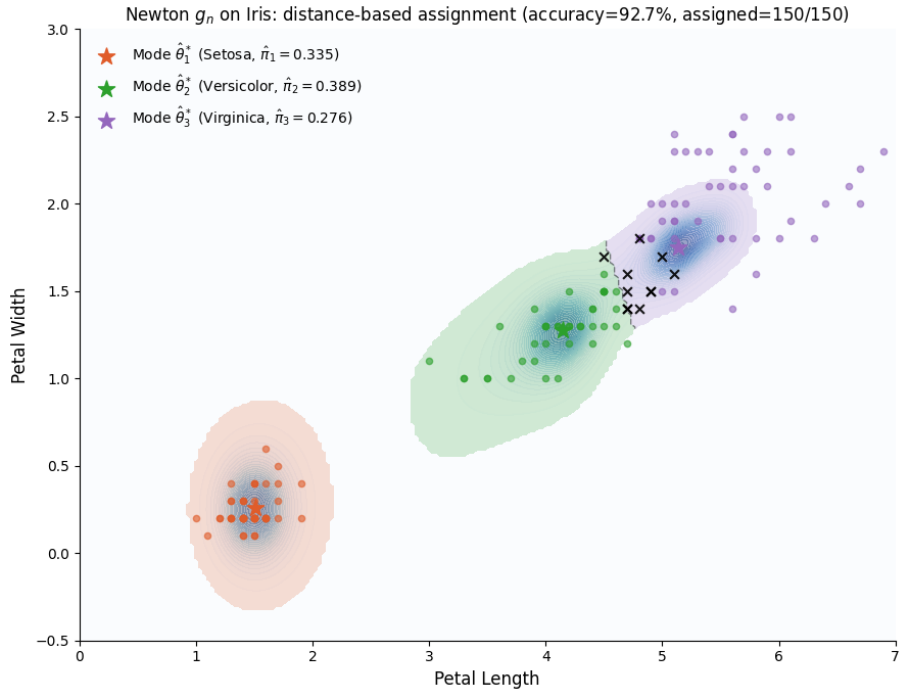


Figure 4: Newton's algorithm applied to the Iris dataset ($n = 150$, $K = 3$ species) using petal length and petal width. Stars mark the three estimated modes and wights $\hat{\theta}_1^*$, $\hat{\pi}_1$ (Setosa), $\hat{\theta}_2^*$, $\hat{\pi}_2$ (Versicolor), $\hat{\theta}_3^*$, $\hat{\pi}_3$(Virginica). Colored dots indicate correctly assigned observations; black crosses indicate misclassified observations. The partition $\{C_k\}$ is defined by proximity to the estimated modes as in (20).

α. Indeed, for small values of α the first weights α_n make the contributions of the first data in the sample disproportionally important. For example, for $\alpha = 1$ we get $\alpha_1 = \frac{1}{2}$ which means that x_1 contributes for a half (of the entire dataset) to the final estimate G_n. Thus when shuffling the observations and estimating the mixing density one would obtain a vector $(G_n^1, \ldots, G_n^s)$ over s permutations of the sample whose components would significantly differ across them (see section 5.2 of Fortini and Petrone (2020) for the simulation results).

It has been empirically shown that increasing α, i.e. increasing the uniformity of the contribution of the observations with respect to the first ones, reduces the role of permutations.

Our intuition on how, in a completely different way, we could reduce permutation sensitivity without tuning α moves from considering the role of G_0 as a prior on the mixing distribution. We propose using a fixed fraction ρ of the sample for a preliminary Gibbs sampling step (Neal's Algorithm 8 (Neal, 2000)), which yields a sequence of atom

locations ($\hat{\theta}_k$) and weights ($\hat{\pi}_k$) across post-burn-in MCMC iterations. All locations and weights are pooled and smoothed with a Gaussian kernel of bandwidth h computed via Silverman's rule of thumb,

$$h = 1.06 \cdot \hat{\sigma} \cdot n_{\text{atoms}}^{-1/5}, \tag{23}$$

where $\hat{\sigma}$ is the empirical standard deviation of the pooled atom locations and n_{atoms} is the total number of pooled atoms. The resulting continuous density g_0 is evaluated on a fixed grid and serves as the informed base measure G_0 for Newton's recursion, which is then run on the remaining $(1 - \rho)$ fraction of the sample.

The rationale draws on the predictive-based sampling scheme of Section 2.6. Figure 1 shows that the posterior draws of $\tilde{G}(A) \mid x_{1:n}$ are markedly more concentrated than the prior draws: the variance of $\tilde{G}(A) \mid x_{1:n}$ is substantially smaller than that of $\tilde{G}(A)$, for any measurable set A. This is similar to what happens when $\tilde{G}$ is DP distributed and the variance of the posterior decreases at rate $1/(\alpha + n)$. Now, by the martingale property, $G_n = \mathbb{E}[\tilde{G} \mid x_{1:n}]$: Newton's estimate at step n is the posterior mean of G. An informed G_0, already concentrated near the true atom locations, initializes the recursion closer to the posterior in the sense that the residual uncertainty on $\tilde{G}$ given G_0 (which contains the information carried by $\lfloor \rho n \rfloor$) is smaller than it would be under an uninformative G_0. Since the posterior variance is small, early observations produce only minor corrections to G_0, so the disproportionate influence of the first few data points, the source of permutation sensitivity for small α, is substantially reduced.

We simulate $n = 1000$ observations from a Gaussian location-scale mixture with $K = 5$ components,

$$g^* = 0.1\,\mathcal{N}(-3,\, 0.2) + 0.3\,\mathcal{N}(0,\, 0.1) + 0.2\,\mathcal{N}(2,\, 0.1) + 0.2\,\mathcal{N}(1,\, 0.1) + 0.2\,\mathcal{N}(4,\, 0.05), \tag{24}$$

where $\mathcal{N}(\mu, \tau^2)$ denotes a Gaussian with mean μ and variance τ^2.

Using $\alpha = 1$, for each value of $\rho \in \{0.00, 0.05, 0.15, 0.30\}$ (where ρ is the proportion of the sample used for the Gibbs step) we draw $s = 50$ independent uniformly random permutations of the entire dataset. For each permutation, the first $\lfloor \rho\, n \rfloor$ observations are used for the MCMC step and the remaining $(1-\rho)n$ for Newton's recursion with Gaussian kernel with variance $\sigma^2 = 1$, so that both the composition of the training set and the ordering of the Newton input vary across permutations. Figure 5 reports the spread of

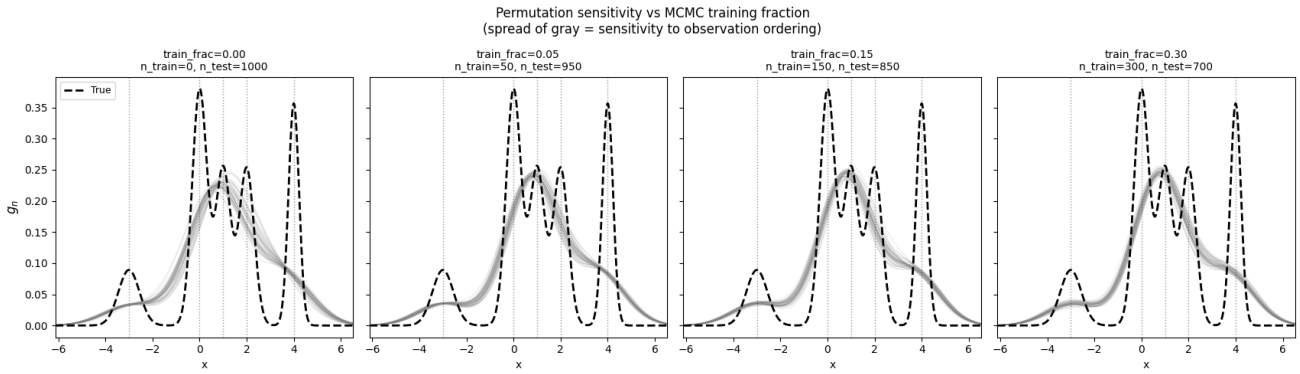


Figure 5: Permutation sensitivity analysis for an MCMC plus Newton across four different fractions ρ of an $n = 1000$ sample, used for the Gibbs step: from left to right $\rho = \{0.00; 0.05; 0.15; 0.30\}$. The DGP is as in (24) with kernel variance $\sigma = 1$. .

the resulting estimates $(G_n^1, \ldots, G_n^s)$: as ρ increases, the bundle tightens noticeably, with most of the reduction already achieved at $\rho = 0.05$. This confirms our intuition and in particular suggests that it may be enough to use just a small part of the dataset (roughly 5%) to mitigate substantially the role of permutations.

Notably, the density estimation performance of the algorithm is not affected by the MCMC step which is affected by the kernel variance which oversmooths the modes of the data generating process.

4 Partial exchangeability and Hierchical Dirichlet process

In this chapter we expand the notion of exchangeability to that of *partial* exchangeability, where data are randomly sampled form different populations which are not independent but not fully exchangeable neither. We then present the *Hierarchical Dirichlet process* (HDP), a popular nonparametric prior for a vector of random distributions, and HPD mixture models as an important example of Bayesian modeling with partially exchangeable data. For such model we will formalize the exact (up to what is computable) predictive structure and highlight how sample information is recursively used for optimal prediction. This will be at the basis of the construction of an approximate method for HDP mixture learning.

4.1 Partial exchangeability

We have seen that the natural probabilistic assessment in the context of identical repeated experiments is exchangeability. However, data may be more complex; still, for some substructures, an assumption of invariance under permutation can be made. Indeed de Finetti (1937) writes:

> *"exchangeability can still be considered, but specifying that the trials are divided into a certain number of types, and what is judged exchangeable are the events of the same type."*

Usually in Bayesian statistics partial exchangeability is assumed for a finite number of parallel random sampling experiments. In fact, Fortini and Petrone (2025) clarify that the number of experiments may be countable. Partial exchangeability should be thought of as the probabilistic judgment underlying fixed design regression where the *types* are induced by covariates, which can be continuous variables (such as temperature). Even though experiments are performed independently, one judges that information about one of them is useful for prediction or inference in the others. Again, this is a probabilistic statement regarding information, not a physical dependence between the types.
In formal terms:

Definition 4.1 (Partial Exchangeability in the sense of de Finetti)**.** A family of sequences $(X_{n,j})_{n\geq 1}$, $j = 1, \ldots, J$ of random variables, where $X_{n,j}$ describes the n-th observation of type j, $j = 1, \ldots, J$, is *partially exchangeable in the sense of de Finetti* if its probability law is invariant under separate finite permutations within each type, that is, if

$$(X_{n,j})_{n\geq 1} \stackrel{d}{=} (X_{\sigma_j(n),\, j})_{n\geq 1} \quad j = 1, \ldots, J$$

for any finite permutation σ_j, $j = 1, \ldots, J$. Here M can be finite or infinite, and the types may be taken from a continuum.

As for exchangeability, the inferential framework is based on a formulation of Theorem 1.3 for partially exchangeable sequences.

Theorem 4.2 (LLN and de Finetti representation for partially exchangeable sequences)**.** *Let $[X_{n,j}]_{n\geq 1,\, j=1,\ldots,J} \sim \mathbb{P}$ be a partially exchangeable array in the sense of de Finetti. Then:*

1. *For $n_1, \ldots, n_J \to \infty$, the vector of marginal empirical distributions $(\hat{G}_{n_1}, \ldots, \hat{G}_{n_J})$ converges weakly to a vector of random distributions $(\tilde{G}_1, \ldots, \tilde{G}_J)$, $\mathbb{P}$-a.s.*

2. *For any $n \geq 1$ and measurable sets $A_{i,j}$,*

$$\mathbb{P}\left(\bigcap_{j=1}^{M} \left(X_{1,j} \in A_{1,j}, \ldots, X_{n_j,j} \in A_{n_j,j}\right)\right) = \int \prod_{j=1,\ldots,M} \prod_{i=1,\ldots,n_j} \tilde{G}_j(A_{i,j})\, d\pi(\tilde{G}_1, \ldots, \tilde{G}_J),$$

where π is the joint probability law of $(\tilde{G}_1, \ldots, \tilde{G}_J)$.

The representation in Theorem 4.2 can be read as a conditional independence of the sequences $(X_{n,j})_{n\geq 1}$ given $(\tilde{G}_1, \ldots, \tilde{G}_J)$ and within the j-th sequence, the $X_{n,j}$ are conditionally i.i.d. with common random distribution $\tilde{G}_j$. The cross-group dependence is entirely encoded in the joint law π of $(\tilde{G}_1, \ldots, \tilde{G}_J)$: if π were a product measure, the sequences $(X_n, j)_{n\geq 1}$ would be independent across j. In the non-product case, π acts as a (nonparametric) prior that ties the groups together, formalizing the notion of *borrowing of strength* across experiments.

Because the sequence $(X_{n,j})_{n\geq 1}$ is exchangeable, clearly by Theorem 1.6 one has $P_{n,j} \to \tilde{G}_j$, yet, given the across-types dependence induced by π, a more interesting predictive

distribution is

$$Q_{n,j} := \mathbb{P}(X_{n+1,j} \mid X_{1:n,j}, \boldsymbol{X}_n^{-j}),$$

where $\boldsymbol{X}_n^{-j} := [X_{j',i}]_{(j'=1,\dots,j-1,j+1,\dots,J,\ i=1,\dots,n)}$, for which Aldous (1985) proves convergence $\mathbb{P}$-a.s. to $\tilde{G}_j$. Both $P_{n,j}$ and $Q_{n,j}$ approximate the random $\tilde{G}_j$, however the former uses less information and thus provides a worse approximation.

An analog of Theorem 1.5 specifies the conditions on the predictive rule to characterize a partially exchangeable law $\mathbb{P}$.

Theorem 4.3 (Predictive characterization of partial exchangeability (Fortini and Petrone, 2025))**.** *A family $(X_{n,j})_{n\geq 1} \quad j = 1, \dots, J$ of sequences of random variables is partially exchangeable in the sense of de Finetti if and only if for every finite $m \leq J$ and every $n \geq 0$, the following conditions hold:*

1. *For every measurable sets $A_1, \dots, A_J$ and every $i = 1, \dots, m$,*

$$P_n(A_1 \times \cdots \times A_J \mid (x_{k,j})_{k\leq n,\, j\leq m})$$

   is symmetric in $(x_{1,i}, \dots, x_{n,i})$.

2. *The set function that maps $\{A_j, B_j : j \leq m\}$ into*

$$\int_{A_1\times\cdots\times A_J} P_{n+1}(B_1 \times \cdots \times B_J \mid (x_{k,j})_{k\leq n+1,\, j\leq m}) \times dP_n((x_{n+1,1}, \dots, x_{n+1,m}) \mid (x_{k,j})_{k\leq n,\, j\leq m})$$

   is symmetric in (A_i, B_i) for every $i = 1, \dots, m$,

where P_n is the conditional distribution of $(X_{n+1,j})_{j\leq m}$ given $(X_{k,j})_{k\leq n,\, j\leq m}$, and $P_0(\cdot \mid (x_{k,j})_{k\leq 0,\, j\leq m})$ is meant as P_0.

4.2 Example: the Hierarchical Dirichlet process

As in Chapter 1, we can make a comparison between inferential and predictive schemes through an important example, namely the *Hierarchical Dirichlet process* (HDP) proposed by Teh et al. (2006) as a workable form of the law π of the vector $(\tilde{G}_1, \dots, \tilde{G}_J)$ as a nonparametric prior underlying a partially exchangeable array $[X_{n,j}]_{n\geq 1,\, j=1,\dots,J}$.

Definition 4.4 (Hierarchical Dirichlet process)**.** A vector $(\tilde{G}_1, \ldots, \tilde{G}_J)$ of random probability measures follows a Hierarchical Dirichlet process with base measure H and concentration parameters $\gamma, \alpha > 0$, written $\text{HDP}(\gamma, \alpha, H)$, if

$$\begin{aligned} \tilde{G}_0 \mid \gamma, H &\sim \text{DP}(\gamma, H), \\ \tilde{G}_j \mid \alpha, \tilde{G}_0 &\overset{\text{ind}}{\sim} \text{DP}(\alpha, \tilde{G}_0), \quad j = 1, \ldots, J. \end{aligned} \tag{25}$$

Teh et al. (2006) started by considering a clustering task with different populations; in their motivating application, clustering words across different documents . There the idea is that the distribution of the words in a dictionary changes according with the topic treated, so that clusters are induced by the topic, and topics can be shared across documents. In the HDP construction, the aim is to introduce borrowing strength across documents by having shared clusters. This is achieved by drawing an a.s. discrete measure $\tilde{G}_0$ from a DP and then introducing document-specific variability around it by drawing iid another set of discrete random measures $\tilde{G}_j \mid \tilde{G}_0$. In a compact form the prior π can be expressed as

$$\pi\big(\mathrm{d}\tilde{G}_1, \ldots, \mathrm{d}\tilde{G}_J\big) = \left[\prod_{j=1}^{J} \text{DP}_{\alpha,\tilde{G}_0}\big(\mathrm{d}\tilde{G}_j\big)\right] \text{DP}_{\gamma,H}\big(\mathrm{d}\tilde{G}_0\big).$$

A predictive characterization of the HDP is also possible, and had in fact been provided by the hierarchical Hoppe's urn construction by Beal et al. (2002) (see also Fortini and Petrone (2012)), which is also known as the *Chinese Restaurant Franchise* (CRF) (Teh et al., 2006). The CRF metaphor involves J restaurants (one per each group j), all sharing a common menu of dishes. Customers enter the restaurants sequentially: customer i in restaurant j corresponds to the latent parameter θ_{ji}. Within each restaurant, customers sit at tables; each table serves exactly one dish. The dishes themselves are drawn from a global menu that evolves as a Pólya urn across all restaurants.

Formally, let K_j denote the number of occupied tables in restaurant j after n_j customers, with $n_k^{(j)}$ customers at table k and dish $\phi_k^{(j)}$ served there. Let $\{\phi_k\}_{k=1}^{K_0}$ denote the distinct dishes (global unique values) across all restaurants, where K_0 is the total number of distinct dishes within the sample, and m_k is the number of tables serving dish k across all restaurants, and $M = \sum_{k=1}^{K_0} m_k$ is the total number of tables globally. Note

that both K_0 and M are random variables whose distributions depend on the entire array $[X_{n_j,j}]_{n_j \geq 1,\, j=1,\ldots,J}$.

The CRF operates at two levels. At the *group level*, a new customer in restaurant j either joins an existing table k with probability proportional to $n_k^{(j)}$, or opens a new table with probability proportional to α:

$$\mathbb{P}\Big(X_{n_j+1,\,j} \in A \;\Big|\; X_{1:n_j,\,j},\, \tilde{G}_0\Big) = \frac{\displaystyle\sum_{k=1}^{K_j} n_k^{(j)}\, \delta_{\phi_k^{(j)}}(A) + \alpha\, \tilde{G}_0(A)}{\alpha + n_j}. \tag{26}$$

where $\sum_{k=1}^{K_j} n_k^{(j)} = n_j$ is the total number of customers in restaurant j, so the denominator $\alpha + n_j$ is the total seating capacity already occupied plus the new-table weight α. If the new customer opens a new table, the dish served at that table is drawn from the global measure $\tilde{G}_0$. At the global level, a new dish is either an existing one ϕ_k already on the menu (with probability proportional to its global multiplicity m_k), or a fresh draw from the base measure H (with probability proportional to γ):

$$\mathbb{P}\Big(\phi_{K_0+1} \in A \;\Big|\; \{\phi_k\}_{k=1}^{K_0},\, H\Big) = \frac{\displaystyle\sum_{k=1}^{K_0} m_k\, \delta_{\phi_k}(A) + \gamma\, H(A)}{\gamma + M}. \tag{27}$$

The two-level structure encodes the hierarchical borrowing of strength: restaurants share dishes through the global menu, so observations in one group influence the dish probabilities seen by all other groups via $\tilde{G}_0$.

Unlike the single-group case, where the Pólya urn yields a closed-form predictive directly at the observation level, marginalizing (26) over $\tilde{G}_0$ requires integrating out the latent table-dish configuration. The total number of tables M appears in the denominator of (27) and is itself a random variable with an intractable distribution over the combinatorial space of all CRF configurations compatible with the observed array $[X_{n_j,j}]_{n_j \geq 1,\, j=1,\ldots,J}$.

A last consideration concerns the hyperparameters α and γ. Higher values of α increase the weight each group's predictive assigns to the global measure $\tilde{G}_0$, equivalently, more new tables are opened and more dishes are drawn from the global menu, drawing group-level estimates closer to the shared signal and amplifying cross-group borrowing. The role of γ is more subtle: higher values push $\tilde{G}_0$ towards H, which is typically diffuse, so that each new dish drawn from the global menu is increasingly distinct. As a consequence,

restaurants share fewer and fewer dishes, borrowing of strength diminishes, and the system approaches a collection of independent DPMMs with base measure H. Conversely, as $\gamma \to 0$, the global measure concentrates on a small number of shared dishes, driving all groups towards the same mixing measure.

HDP mixture models. A partially exchangeable mixture model is obtained by specifying a continuous emission kernel $f(\cdot \mid \theta)$ over the partially exchangeable latent array $[\theta_{n_j,j}]_{n_j \geq 1,\, j=1,\ldots,J}$ with an emission density that allows to model the observations $[X_{n_j,j}]_{n_j \geq 1,\, j=1,\ldots,J}$:

$$\begin{aligned} X_{n_j,j} \mid \theta_{n_j,j} &\overset{ind}{\sim} f(x_{n_j,j} \mid \theta_{n_j,j}), \\ (\theta_{n_j,j})_{n_j \geq 1} \mid (\tilde{G}_1, \ldots, \tilde{G}_J) &\overset{iid}{\sim} \tilde{G}_j \qquad j = 1, \ldots, J \\ (\tilde{G}_1, \ldots, \tilde{G}_J) &\sim HDP(\alpha, \gamma, H). \end{aligned} \tag{28}$$

4.3 Predictive structure for HDP mixtures

Consider the HDP model specified in (28). We derive the exact Bayesian predictive for the next observation in group j after observing all data $\mathbf{x}^N$ (where $N = \sum_{j=1}^{J} n_j$), following an approach analogous to that of Fortini and Petrone (2020) for the single-group DPMM.

By the CRF construction of Section 4.2, the predictive for a new latent parameter in group j, conditionally on the current table-dish configuration $\{\theta_{j,1:n_j}\}$ and the global measure $\tilde{G}_0 = G_0$ (i.e. the random $\tilde{G}_0(\omega) = G_0$) , is:

$$\mathbb{P}(\theta_{j,n_j+1} \in A \mid \{\theta_{j,1:n_j}\}, G_0) = \frac{\alpha\, G_0(A) + \sum_{i=1}^{n_j} \delta_{\theta_{ji}}(A)}{\alpha + n_j}.$$

In CRF terms: with probability $\frac{\alpha}{\alpha+n_j}$ the new customer opens a new table and draws a dish from $\tilde{G}_0$; with probability $\frac{n_j}{\alpha+n_j}$ she joins an existing table, replicating the dish already served there.

Unlike the single-group DPMM, here $\tilde{G}_0$ is shared across all groups. To obtain the predictive given only the observed data $\mathbf{x}^N$, we must marginalize over both $\tilde{G}_0$ and $\{\theta_{ji}\}$.

For $\tilde{G}_0$, each group-level unique value $\phi_k^{(j)}$ is a draw from $\tilde{G}_0$, so the M tables across all restaurants constitute a sample from $\tilde{G}_0$. By DP conjugacy, conditionally on the

global dish configuration $\{\phi_k, m_k\}_{k=1}^{K_0}$:

$$\tilde{G}_0 \mid \{\phi_k, m_k\}_{k=1}^{K_0} \sim \mathrm{DP}\left(\gamma + M,\ \frac{\gamma H + \sum_{k=1}^{K_0} m_k\, \delta_{\phi_k}}{\gamma + M}\right). \tag{29}$$

We define the posterior mean of $\tilde{G}_0$ given the observed data:

$$\tilde{G}_0^{(\mathrm{post},N)}(A) := \mathbb{E}[\tilde{G}_0(A) \mid \mathbf{x}^N] = \mathbb{E}\left[\frac{\gamma H(A) + \sum_{k=1}^{K_0} m_k\, \delta_{\phi_k}(A)}{\gamma + M} \;\middle|\; \mathbf{x}^N\right]. \tag{30}$$

This expression does not simplify to a closed form. Both K_0 and M are random variables; in CRF terms, the number of distinct dishes and the total number of tables are determined by the latent seating arrangement, not directly by the data. Computing (30) requires summing over all CRF configurations compatible with $\mathbf{x}^N$, a cost that grows combinatorially with N. This is a fundamental source of intractability in exact HDP inference.

The predictive for θ_{j,n_j+1} given the data becomes:

$$\begin{aligned}
\mathbb{P}(\theta_{j,n_j+1} \in A \mid \mathbf{x}^N) &= \mathbb{E}\Big[\mathbb{P}(\theta_{j,n_j+1} \in A \mid \{\theta_{j,1:n_j}\}, G_0) \;\Big|\; \mathbf{x}^N\Big] \\
&= \mathbb{E}\left[\frac{\alpha\, G_0(A) + \sum_{i=1}^{n_j} \delta_{\theta_{ji}}(A)}{\alpha + n_j} \;\middle|\; \mathbf{x}^N\right] \\
&= \frac{\alpha}{\alpha + n_j}\, \tilde{G}_0^{(\mathrm{post},N)}(A) + \frac{1}{\alpha + n_j} \sum_{i=1}^{n_j} \mathbb{P}(\theta_{ji} \in A \mid \mathbf{x}^N),
\end{aligned} \tag{31}$$

where the last step uses linearity of expectation and $\mathbb{E}[\delta_{\theta_{ji}}(A) \mid \mathbf{x}^N] = \mathbb{P}(\theta_{ji} \in A \mid \mathbf{x}^N)$. Integrating (31) over the kernel $f(\cdot \mid \theta)$, yields the Bayes predictive for the next observation expands as

$$\tilde{P}^{(\mathrm{Bayes},j)}(A) \equiv P(X_{j,n_j+1} \in A \mid \mathbf{x}^N) = \frac{\alpha}{\alpha + n_j}\, \tilde{P}_0^{(\mathrm{post},N)}(A) + \frac{1}{\alpha + n_j} \sum_{i=1}^{n_j} \mathbb{E}\Big[F(A \mid \theta_{ji}) \mid \mathbf{x}^N\Big]. \tag{32}$$

Throughout, for any measurable set $A \in \mathcal{B}(\mathbb{X})$, let $F(\dot{|}\theta)$ be the distribution function (d.f.) corresponding to the density $f(\cdot \mid \theta)$, and let us denote by the same symbol the d.f. and the corresponding measure; e.g. $F(A \mid \theta) = \int_A f(x \mid \theta)\, \mu(dx)$. Similarly, we use $\tilde{P}_0^{(\mathrm{post},N)}(\cdot) \equiv \int_\Theta F(A \mid \theta)\, \mathbb{E}(\tilde{G}_0(d\theta) \mid \mathbf{x}^N)$.

This decomposition reveals how the sample information shapes the predictive distribution: it is a weighted average between a global component, inherited from the posterior

distribution of $\tilde{G}_0$, and per-observation posterior contributions from the latent parameters θ_{ji}. However, neither term admits a closed form: each expectation $\mathbb{E}[F(A \mid \theta_{ji}) \mid \mathbf{x}^N]$ requires marginalizing over the posterior law of all CRF configurations compatible with $\mathbf{x}^N$, inheriting the same combinatorial intractability as (30).

4.4 Efficient use of sample information in the Bayesian approach

The exact Bayesian approach (30)–(31) uses sample information in two essential ways.

(1) Global backpropagation of information. When a new observation $x_{\ell,i}$ arrives in group ℓ, the Bayesian approach recomputes *all* posteriors $\mathbb{P}(\theta_{ji} \in \cdot \mid \mathbf{x})$ for every group j and every past observation i. In CRF terms, the new customer in restaurant ℓ may open a new table and draw a new dish, updating K_0, M, and the multiplicities $\{m_k\}$, which in turn shifts $\tilde{G}_0^{(\text{post})}$ and thus the hierarchical component of every group's predictive. This global updating ensures optimal borrowing of strength: information from data-rich restaurants immediately improves inference in data-poor ones.

(2) Exact hierarchical borrowing. The posterior mean $\tilde{G}_0^{(\text{post},N)}$ pools information from all groups through the dish multiplicities $\{m_k\}$: each dish ϕ_k contributes to the global picture weighted by the number of tables m_k serving it across all restaurants, naturally distinguishing between widely shared dishes (large m_k) and group-specific ones (small m_k).

The price of this statistical efficiency is computational. After each new observation, computing the exact predictive requires:

- Recomputing $\mathcal{O}(N)$ posteriors $\mathbb{P}(\theta_{ji} \in \cdot \mid \mathbf{x}^N)$, one per past observation across all groups;
- Computing $\tilde{G}_0^{(\text{post},N)}$ by marginalizing over all compatible CRF table-dish configurations (combinatorial cost);
- Evaluating the predictive mixture at cost $\mathcal{O}(n_j)$ for group j.

Processing a complete dataset of size N sequentially thus costs $\mathcal{O}(N^2)$ or more, depending on the complexity of the individual posterior computations.

4.5 One-step predictive for the HDP mixture

The intractability of the exact predictive (32) is a general feature of HDP inference; yet in the minimal case of a single observation per group the expression simplifies to a tractable closed form that makes the borrowing-of-strength mechanism fully explicit. Consider the synchronous setting in which exactly one observation $X_{1,j}$ has arrived in each group $j = 1, \ldots, J$, so that $n_j = 1$ for all j and $N = J$. Let $P_H(\cdot) = \int F(\cdot \mid \theta)\, H(d\theta)$ be the marginal distribution of $X_{1,j}$. Then the distribution of $X_{2,j}$ given $X_{1,j} = x$ is:

$$P_H(\cdot \mid x) \;=\; \int_\Theta F(\cdot \mid \theta)\, q_H(d\theta \mid x), \qquad \text{where} \quad q_H(d\theta \mid x) = \frac{f(x \mid \theta)\, H(d\theta)}{\int f(x \mid u)\, H(du)},$$

is the posterior of $\theta_{2,j}$ given $X_{1,j} = x$. After one observation per group (i.e. when $\mathbf{x^N} = \mathbf{x^1}$ $= (x_{1,1}, x_{1,2}, \ldots, x_{1,J})$), the exact predictive for $\theta_{j,2}$ given in (31) reduces to

$$\mathbb{P}\Big(\theta_{j,2} \in A \mid \mathbf{x}^1\Big) \;=\; \frac{\alpha}{\alpha+1}\, \tilde{G}_0^{(\text{post},1)}(A) \;+\; \frac{1}{\alpha+1}\, \mathbb{P}(\theta_{j,1} \in A \mid \mathbf{x}^1), \tag{33}$$

We now make each term explicit. Since at step $n_j = 1$ the unique customer in restaurant j necessarily opens a new table and draws a fresh dish from H, the prior on $\theta_{j,1}$ is H and its posterior given $x_{j,1}$ is:

$$\mathbb{P}(\theta_{j,1} \in A \mid \mathrm{x}^{(1)}) = \frac{\int_A f(x_{j,1}|\theta)\, H(d\theta)}{\int_\Theta f(x_{j,1}|\theta)\, H(d\theta)} := q_H(A \mid x_{1,j}). \tag{34}$$

For the term $\tilde{G}^{(\text{post},1)}$, with one observation per group the CRF configuration is deterministic: each restaurant has exactly one table ($K_j = 1$, $n_k^{(j)} = 1$) serving a fresh dish drawn from the diffuse H, so $K_0 = J$, $M = J$, and $m_k = 1$ for every k. The posterior mean of G_0 in (30) reduces to:

$$\tilde{G}_0^{(\text{post},1)}(A) = \frac{\gamma H(A) + \sum_{j'=1}^{J} q_H(A \mid x_{1,j'})}{\gamma + J}. \tag{35}$$

Substituting (34) and (35) into (33) and integrating against $f(\cdot \mid \theta)$ yields.

$$\mathbb{P}\Big(X_{2,j} \in \cdot \mid \mathbf{x}^{(1)}\Big) \;=\; \frac{1}{\alpha+1}\, P_H(\cdot \mid x_{1,j}) \;+\; \frac{\alpha}{\alpha+1} \left[\frac{\gamma\, P_H(\cdot) + \displaystyle\sum_{j'=1}^{J} P_H(\cdot \mid x_{1,j'})}{\gamma + J} \right]. \tag{36}$$

The formula decomposes into two terms. The first, with weight $1/(\alpha+1)$, is the within-group contribution: it reinforces the single local observation $X_{1,j}$ through the predictive $P_H(\cdot \mid X_{1,j})$. The second, weighted $\alpha/(\alpha+1)$, reflects the hierarchical component: it pools the prior $P_H(\cdot)$ and the predictives from all groups, including j itself, with weights γ and 1 respectively. As a consequence, $X_{1,j}$ enters the predictive through both channels once directly in the first term, and through $\tilde{G}_0^{(\text{post},1)}$ in the second with total weight $w_j = \frac{1}{\alpha+1} + \frac{\alpha}{(\alpha+1)(\gamma+J)}$. This is not redundancy: the two contributions are inferentially distinct, as $\theta_{j,1}$ informs both the group-level and the global measure G_0 through the table it opens in the CRF. Cross-group terms $P_H(\cdot \mid X_{1,j'})$, $j' \neq j$, enter with equal weight $w_{j'} = \frac{\alpha}{(\alpha+1)(\gamma+J)}$ each, mediated entirely through $\tilde{G}_0^{(\text{post},1)}$. This closed form is the benchmark against which the Newton–HDP algorithm is compared in Chapter 5.

5 Newton-HDP as an expansion of Newton recursions to the HDP information flow

In this chapter we present the Newton-HDP algorithm, a novel recursive procedure for HDP mixture estimation. The algorithm applies Newton's estimation principle at each level of the HDP hierarchy: group-level measures $G_{j,t}$ are updated as new observations arrive, while a global measure $G_{0,t}$ aggregates information across groups and re-enters each group-level update as a shared prior. This feedback loop is the mechanism through which borrowing of strength is achieved at computational cost $\mathcal{O}(NJK)$.

5.1 Temporal structure and notation

The Newton–HDP algorithm operates on a stream of observations arriving sequentially across J groups. To handle precisely the asymmetry between group specific and global updates, we introduce two distinct notions of time.

Let $t = 1, 2, \ldots$ denote the *global time index*: at each step t, exactly one observation $X_{t,j} \equiv X_{t,j(t)}$ arrives in a group $j(t) \in \{1, \ldots, J\}$, where $j(\cdot)$ is the (possibly random) group assignment sequence. The *local sample size*

$$n_{j,t} \;=\; \#\{s \le t : j(s) = j\} \tag{37}$$

records the number of observations accumulated in group j by global time t, and $N_t = \sum_{j=1}^{J} n_{j,t} = t$.

The two levels of the hierarchy are updated at different rates. The *group estimate* $G_{j,t}$ is updated only when an observation arrives in group j, that is, when $j(t) = j$; otherwise it remains frozen:

$$G_{j,t} = G_{j,t-1} \quad \text{for all } j \neq j(t). \tag{38}$$

. The *global estimate* $G_{0,t}$ by contrast, is updated at every global step t, incorporating the current state of all group measures through the aggregated target $\bar{G}_t$.

This distinction is not merely notational. Because $G_{0,t}$ aggregates information from all groups at every step, it serves as a continuously refreshed shared signal; group estimates are updated only when data in their specific group arrive, but they "benefit" on the updated $G_{0,t-1}$ which has been updated continuously in the meantime with other groups'

data. The interplay between these two clocks is the mechanism through which cross-group borrowing of strength is achieved.

5.2 From model structure to algorithm

The HDP model specifies that, conditionally on G_0, each group is an independent DPMM:

$$\tilde{G}_j \mid \tilde{G}_0 \sim \mathrm{DP}(\alpha, G_0), \qquad \theta_{ji} \mid \tilde{G}_j \sim \tilde{G}_j, \qquad X_{ji} \mid \theta_{ji} \sim f(\cdot \mid \theta_{ji})$$

Assuming $\tilde{G}_0$ were known , say G_0, each group j would be exactly the single-group setting of Newton with base measure G_0. A direct application of Newton would give:

$$G_{j,t} = \left(1 - \frac{1}{\alpha + n_{j,t}}\right) G_{j,t-1} + \frac{1}{\alpha + n_{j,t}} q_{G_{j,t-1}}(\cdot \mid X_{t,j(t)}) \tag{39}$$

using $G_{j,t-1}$ as the prior for the Bayesian step in the second term, exactly as in (7). As discussed in the previous chapter, this would be a legitimate choice, yet we would not exploit the information on $G_{j,n_{j,t}}$ brought by the other groups and would thus learn it at a slower rate. This might not be problematic for a group for which a large sample is available, given the computational speed provided by Newton's recursions; but for a group with far fewer observations we rather want to exploit all the available information to make good predictions.

We mimic the hierarchic structure of (25) by specifying global and local components, using $G_{j,t-1}$ to track only the accumulated local information and re-injecting G_0 explicitly at each step. That is, we use a different prior in the Bayesian update step in equation (39), namely:

$$\pi_{t,j} = \frac{\alpha}{\alpha + n_{j,t}} G_0 + \frac{n_{j,t}}{\alpha + n_{j,t}} G_{j,t-1} \tag{40}$$

with the Pólya-urn weights from the CRP predictive, obtaining the group-specific update:

$$G_{j,t} = \left(1 - \frac{1}{\alpha + n_{j,t}}\right) G_{j,t-1} + \frac{1}{\alpha + n_{j,t}} q_{\pi_{t,j}}(\cdot \mid X_{t,j(t)}). \tag{41}$$

That is, if G_0 were truly known and fixed, one might use it directly as the base measure in (39), replacing $G_{j,t-1}$ with $\pi_{t,j}$ as the prior for the Bayesian update step. The two approaches would in general produce different estimates, since $\pi_{t,j}$ explicitly injects global information at each step whereas (39) relies solely on the accumulated local history. The

motivation for preferring (41) is precisely this injection: groups with few observations benefit immediately from the shared signal carried by G_0, rather than waiting for their local sample to grow large enough to recover it autonomously. The role of α is essentially the same as in (25), it induces more or less borrowing strength from the other populations depending on its value.

Keeping G_0 explicit in the prior (40) suggests that we can replace it with a time-varying estimate $G_{0,t-1}$ that is continuously updated as data arrive in any all groups. Indeed, since $\tilde{G}_0$ is itself a latent random measure, the update (41) cannot be implemented directly. As discussed, computing $\tilde{G}_0^{(\text{post},N-1)}$ is computationally-wise heavy.

If we could observe $\tilde{G}_0 = G_0$ directly, each group could run an independent Newton update with G_0 as base measure (more precisely since G_0 would be discrete we shall take a smoothed counterpart as we did in Chapter 3). However, we do not observe these draws. At each global step t we have instead the group-level estimates $G_{j,t}$. In the HDP mixture each $\tilde{G}_j$ is a DP draw with base measure $\tilde{G}_0$, thus $\mathbb{E}(\tilde{G}_j \mid \tilde{G}_0 = G_0) = G_0$ for each j. An intuitive way to mimic this is through an observation-based weighted average which in accordance with the posterior structure of (29) becomes

$$\bar{G}_t = \frac{\gamma H + \sum_{j=1}^{J} n_{j,t}\, G_{j,t}}{\gamma + N_t}. \tag{42}$$

This form mirrors the DP posterior mean structure $(\gamma H + \sum \delta_\cdot)/(\gamma + n)$, but with compressed group estimates replacing individual atoms. The estimate of the global measure $\tilde{G}_0$ is then updated starting from the initial value $G_{0,0} = H$ and recursively computing

$$G_{0,t} = (1 - w_t^{(0)})\, G_{0,t-1} + w_t^{(0)}\, \bar{G}_t, \qquad w_t^{(0)} = \frac{1}{\gamma + N_t},$$

where $\bar{G}_t$ is defined as in (42) This choice is coherent with the update structure performed at the population level.

The complete algorithm is obtained by replacing the known G_0 in (40) with the current estimate $G_{0,t-1}$:

$$\pi_{t,j} = \frac{\alpha}{\alpha + n_{j,t}}\, G_{0,t-1} + \frac{n_{j,t}}{\alpha + n_{j,t}}\, G_{j,t-1}$$

This creates a feedback loop between the two levels of the hierarchy: group estimates $G_{j,t}$ feed into the global target $\bar{G}_t$, which updates $G_{0,t}$, which in turn enters the group-level

priors $\pi_{t+1,j(t+1)}$ for the next observation. Information flows across groups through this shared global estimate, providing the mechanism for borrowing of strength. Again the role of γ is similar as in (25), a higher values increases the weight of the uninformative base measure H essentially letting the populations updates take their course and producing a situation similar as in (41).

The complete algorithm is summarized in Definition 5.1.

Definition 5.1 (Newton-HDP Algorithm)**.** Let $\{X_{t,j}\}$ denote observations arriving sequentially, where at each global step t exactly one observation $X_{t,j(t)}$ arrives in group $j(t) \in \{1, \ldots, J\}$. Let $n_{j,t} = \#\{s \leq t : j(s) = j\}$ denote the local sample size of group j at global time t, and $N_t = \sum_{j=1}^{J} n_{j,t} = t$.

The **Newton-HDP algorithm** maintains recursive state estimates $(G_{0,t}, G_{1,t}, \ldots, G_{J,t})$, initialized as $G_{j,0} = G_{0,0} = H$. Upon receiving observation $X_{t,j(t)}$ in group $j = j(t)$:

1. **Form hierarchical prior:**

$$\pi_{t,j} = (1 - \eta_{t,j})\, G_{j,t-1} + \eta_{t,j}\, G_{0,t-1}$$

2. **Compute posterior kernel:**

$$q_{\pi_{t,j}}(d\theta \mid X_{t,j}) = \frac{f(X_{t,j} \mid \theta)\, \pi_{t,j}(d\theta)}{\int_\Theta f(X_{t,j} \mid u)\, \pi_{t,j}(du)}$$

3. **Update group measure** (only for $j = j(t)$, all others unchanged):

$$G_{j,t} = (1 - w_{t,j})\, G_{j,t-1} + w_{t,j}\, q_{t,j}, \qquad G_{k,t} = G_{k,t-1} \quad \forall\, k \neq j$$

4. **Update global measure:**

$$G_{0,t} = (1 - w_t^{(0)})\, G_{0,t-1} + w_t^{(0)}\, \bar{G}_t, \qquad \bar{G}_t = \frac{\gamma H + \sum_{k=1}^{J} n_{k,t}\, G_{k,t}}{\gamma + N_t}$$

The weights are defined as:

$$\eta_{t,j} = \frac{\alpha}{\alpha + n_{j,t}}, \qquad w_{t,j} = \frac{1}{\alpha + n_{j,t}}, \qquad w_t^{(0)} = \frac{1}{\gamma + N_t}$$

where $\alpha > 0$ is the local concentration parameter and $\gamma > 0$ is the global concentration parameter.

Here $G_{j,t} = G_{j,t}(x_1, \ldots, x_t)$ is a function of the observations up to time t, thus in this definition it is completely deterministic.

5.3 Information compression and computational savings

The Newton compression applied at each level of the hierarchy trades statistical efficiency for computational tractability. We now make this trade-off explicit by comparing the algorithm's components with their exact Bayesian counterparts derived in Section 4.4.

Consider the observations ordered in a global temporal sequence $t = 1, \ldots, N$, where x_t arrives in group $j(t)$ and $\mathbf{x}^{(t)}$ denotes the complete dataset across all groups up to time t. The exact predictive (31) for the latent component can be decomposed by isolating the most recent latent component in group $j = j(t)$:

$$\frac{\alpha + n_{j,t} - 1}{\alpha + n_{j,t}} \left[\frac{\alpha}{\alpha + n_{j,t} - 1} G_0^{(\mathrm{post},t)}(A) + \frac{1}{\alpha + n_{j,t} - 1} \sum_{i=1}^{n_{j,t}-1} \mathbb{P}(\theta_{ji} \in A \mid \mathbf{x}^{(t)}) \right] + \frac{1}{\alpha + n_{j,t}} \mathbb{P}\Big(\theta_{j,n_{j,t}} \in A \mid \mathbf{x}^{(t)}\Big); \quad (43)$$

whereas in the Newton-HDP it is

$$G_{j,t}(A) = \frac{\alpha + n_{j,t} - 1}{\alpha + n_{j,t}} G_{j,t-1}(A) + \frac{1}{\alpha + n_{j,t}} q_{\pi_{t,j}}(A) \quad (44)$$

where $q_{\pi_{t,j}} := q_{\pi_{t,j}}(d\theta \mid X_{t,j})$ is the one-step posterior kernel under the hierarchical prior $\pi_{t,j}$ updated with $X_{t,j(t)}$. This form follows directly from the algorithm with weights $w_{t,j} = 1/(\alpha + n_{j,t})$ and $1 - w_{t,j} = (\alpha + n_{j,t} - 1)/(\alpha + n_{j,t})$.

The structural parallel between (43) and (44) is immediate: in both cases the new observation contributes with weight $1/(\alpha + n_{j,t})$, and the remaining weight goes to the accumulated past. The difference lies entirely in what the past is summarised by:

Component	**Exact**	**Newton-HDP**
Global component	$\tilde{G}_0^{(\text{post},t)}$: posterior mean of G_0, requires marginalization over CRF table distribution	Encoded implicitly in $q_{\pi_{t,j}}$ via $\pi_{t,j} = (1 - \eta_{t,j})\, G_{j,t-1} + \eta_{t,j}\, G_{0,t-1}$; forward-only, $\mathcal{O}(JK)$
Local component	$\{\mathbb{P}(\theta_{ji} \in \cdot \mid \mathbf{x}^{(t)})\}_i$: $n_{j,t} - 1$ full-data posteriors	$G_{j,t-1}$: single compressed measure, $\mathcal{O}(K)$
New observation	$\mathbb{P}(\theta_{j,n_{j,t}} \in \cdot \mid \mathbf{x}^{(t)})$: full-data posterior	$q_{\pi_{t,j}} = P_{\pi_{t,j}}(\cdot \mid x_t)$: one-step update, $\mathcal{O}(K)$

We identify three specific sources of information loss (which mirror the problematic components highlighted for equation (31)):

Loss 1: Compression of local posteriors. At the group level, Newton's compression replaces $n_{j,t} - 1$ individual posterior distributions $\{\mathbb{P}(\theta_{ji} \in \cdot \mid \mathbf{x}^{(t)})\}_{i=1}^{n_{j,t}-1}$ with a single measure $G_{j,t-1}$. This is the same compression that Newton's method performs in the single-group case, but with an additional consequence in the hierarchical setting: it prevents the algorithm from retroactively adjusting group j's past parameter estimates when new data arrive in other groups. In the exact approach, each $\mathbb{P}(\theta_{ji} \in \cdot \mid \mathbf{x}^{(t)})$ is conditioned on data from *all* groups and is recomputed whenever any observation arrives anywhere.

Loss 2: Forward-only information propagation. When observation x_t arrives in group ℓ, the exact Bayesian approach recomputes all posteriors across all groups. Newton-HDP updates only $G_{\ell,t}$ and $G_{0,t}$. Information does flow to other groups, but only *forward in time*: the updated $G_{0,t}$ enters future priors $\pi_{s,j}$ for $s > t$, but past estimates $G_{j,s}$ for $s < t$ and $j \neq \ell$ are never revised. This is inherent to any online algorithm that processes each observation exactly once.

Loss 3: Customer-level aggregation replaces table-level structure. The exact posterior mean $G_0^{(\text{post},t)}$ in (30) requires marginalizing over the CRF table structure, which tracks how many tables exist, which dish each table serves, and how many customers sit at each table. The Newton-HDP global target $\bar{G}_t$ in (42) replaces this with a customer-level weighted average: each group estimate $G_{j,t}$ contributes proportionally to its sample

size $n_{j,t}$, without distinguishing between shared atoms and distinct atoms. This is the most significant departure from the exact model structure, and reflects the fundamental difficulty of estimating G_0 without access to the latent table assignments.

Computational savings. The information losses translate directly into computational savings:

Table 1: Computational cost comparison.

Operation	**Exact Bayesian**	**Newton-HDP**
Update local posteriors	$\mathcal{O}(N \cdot C_{\text{MCMC}})$	$\mathcal{O}(K)$
Update global measure	combinatorial (CRF tables)	$\mathcal{O}(JK)$
Evaluate predictive	$\mathcal{O}(n_j)$	$\mathcal{O}(K)$
Total per observation	$\mathcal{O}(N \cdot C_{\text{MCMC}})$	$\mathcal{O}(JK)$
Total for full dataset	$\mathcal{O}(N^2 \cdot C_{\text{MCMC}})$	$\mathcal{O}(NJK)$

K = grid size, J = number of groups, C_{MCMC} = cost of MCMC per posterior update.

The state $(G_{0,t}, G_{1,t}, \ldots, G_{J,t})$ serves as a finite-dimensional surrogate for the exact sufficient statistic (the full posterior over CRF configurations). The computational savings, from $\mathcal{O}(N^2 \cdot C_{\text{MCMC}})$ to $\mathcal{O}(NJK)$, persist at all sample sizes.

5.4 Comparison with the exact one-step predictive

The three information losses identified in Section 4.5 can be made precise by comparing the predictive produced by the Newton–HDP algorithm with the exact expression (36) derived at the end of Chapter 4. We consider the case in which each population has received an observation thus $n_{j,1} = 1$ for all j and $N_1 = J$.

From Theorem 5.1, the state of the algorithm at $t = 1$ is

$$G_{j,1} = \frac{\alpha}{\alpha+1} H + \frac{1}{\alpha+1} q_{1,j}, \qquad q_{1,j}(A) = \frac{\int_A f(x_{j,1}|\theta)\, H(d\theta)}{\int_\Theta f(x_{j,1}|\theta)\, H(d\theta)}, \tag{45}$$

$$G_{0,1} = \frac{\gamma+J-1}{\gamma+J} H + \frac{1}{\gamma+J} \bar{G}_1, \qquad \bar{G}_1 = \frac{\gamma H + \sum_{j'=1}^{J} G_{j',1}}{\gamma+J}. \tag{46}$$

Substituting the expression for $G_{j',1}$ into $\bar{G}_1$ gives

$$\bar{G}_1 = \frac{\left(\gamma + \dfrac{\alpha J}{\alpha+1}\right) H + \dfrac{1}{\alpha+1} \displaystyle\sum_{j'=1}^{J} q_{1,j'}}{\gamma+J},$$

and substituting in turn into $G_{0,1}$ isolates the coefficient of each posterior kernel $q_{1,j'}$ in

the global measure:

$$G_{0,1} \;=\; c_H^{(0)}\, H \;+\; \frac{1}{(\alpha+1)(\gamma+J)^2} \sum_{j'=1}^{J} q_{1,j'},$$

where $c_H^{(0)}$ collects all mass on H and we omit its expression. The algorithm's one-step-ahead predictive for $X_{2,j}$ is

$$\mathbb{P}\Big(X_{2,j} \in \cdot \mid \mathbf{x}^1\Big) \;=\; \frac{1}{\alpha+1}\, P_{G_{j,1}}(\cdot) \;+\; \frac{\alpha}{\alpha+1}\, P_{G_{0,1}}(\cdot), \tag{47}$$

where $P_\mu(\cdot) = \int F(\cdot \mid \theta)\, \mu(d\theta)$ denotes the predictive under μ (a probability measure on Θ). Since $P_{q_{1,j}}(\cdot) = P_H(\cdot \mid X_{1,j})$ by definition of $q_{1,j}$, collecting the coefficients of $P_H(\cdot \mid X_{1,j})$ and $P_H(\cdot \mid X_{1,j'})$ for $j' \neq j$ yields

$$P^{\text{N-HDP}}(\cdot) := \mathbb{P}\Big(X_{2,j} \in \cdot \mid \mathbf{x}^{(1)}\Big) \;=\; \tilde{w}_0\, P_H(\cdot) \;+\; \tilde{w}_j\, P_H(\cdot \mid X_{1,j}) \;+\; \tilde{w}_{j'} \sum_{j' \neq j} P_H(\cdot \mid X_{1,j'}), \tag{48}$$

where

$$\tilde{w}_j \;=\; \frac{1}{(\alpha+1)^2} + \frac{\alpha}{(\alpha+1)^2(\gamma+J)^2}, \qquad \tilde{w}_{j'} \;=\; \frac{\alpha}{(\alpha+1)^2(\gamma+J)^2}. \tag{49}$$

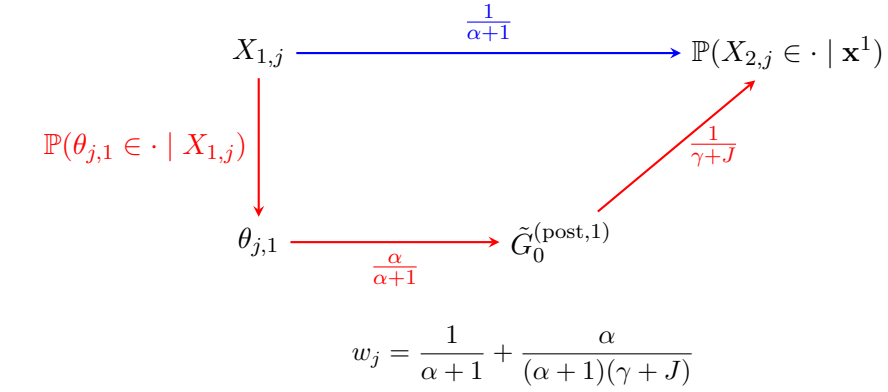
Comparing (49) with the exact weights from (36),

$$w_j \;=\; \frac{1}{\alpha+1} + \frac{\alpha}{(\alpha+1)(\gamma+J)}, \qquad w_{j'} \;=\; \frac{\alpha}{(\alpha+1)(\gamma+J)}, \tag{50}$$

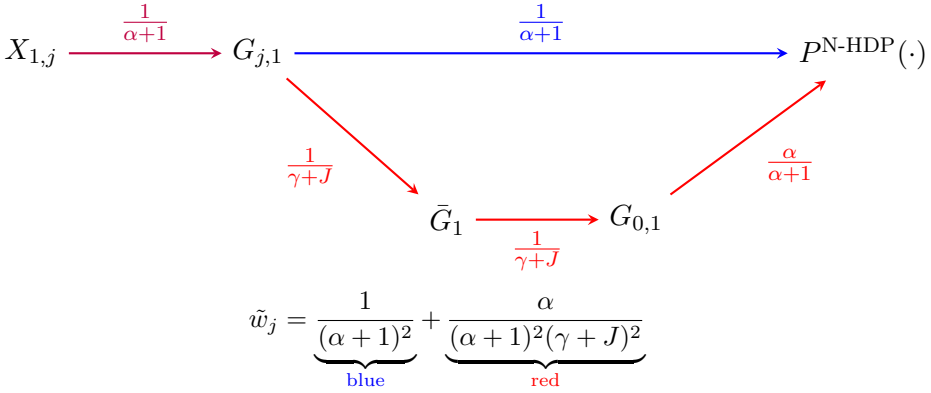
Both weights are deflated relative to the exact counterparts (50), and correspondingly $\tilde{w}_0$ is inflated. All three deviations trace back to Loss 3: $\bar{G}_1$ weights each group estimate $G_{j',1}$ proportionally to sample size $n_{j',1}$, without accounting for the fact that each $G_{j',1}$ is itself a mixture of H and $q_{1,j'}$, not a direct observation from $\tilde{G}_0$.

Figure 6 makes the deflation mechanism explicit. In the exact model, $X_{1,j}$ influences the predictive through two distinct channels from $X_{1,j}$: directly via the local term with weight $1/(\alpha+1)$, and through the latent $\theta_{j,1} \to \tilde{G}_0^{(\text{post},1)}$ with weight $\alpha/[(\alpha+1)(\gamma+J)]$. In the Newton–HDP $X_{1,j}$ enters in the definition of $G_{j,1}$: the direct path continues $G_{j,1} \to P^{\text{N-HDP}}$ with weight $1/(\alpha+1)$, while the indirect path continues $G_{j,1} \to \bar{G}_1 \to G_{0,1} \to P^{\text{N-HDP}}$ with each step multiplying by $1/(\gamma+J)$, $1/(\gamma+J)$, and $\alpha/(\alpha+1)$ in turn. Cross-group observations $X_{1,j'}$ influence $P^{\text{N-HDP}}$ only through the indirect path, so their weight $\tilde{w}_{j'} = \alpha/[(\alpha+1)^2(\gamma+J)^2]$ is deflated by the same factor relative to $w_{j'}$.

The qualitative structure of the exact predictive is nevertheless preserved at every finite time: the algorithm assigns strictly higher weight to the local observation than to any individual cross-group datum, treats all other groups symmetrically, and recovers the prior in the absence of data.



Exact HDP



Newton–HDP

Figure 6: Information flow from $X_{1,j}$ to the one-step predictive. *Exact HDP* (top): the blue arrow carries $X_{1,j}$ directly to the predictive with weight $1/(\alpha+1)$; the red path passes through the latent $\theta_{j,1}$ and $G_0^{(\text{post},1)}$, contributing weight $\alpha/[(\alpha+1)(\gamma+J)]$. The two paths are independent and their weights *add*. *Newton–HDP* (bottom): the purple arc $X_{1,j} \to G_{j,1}$ is shared through $q_{1,j}$; from $G_{j,1}$ the blue arrow goes directly to the predictive (note (47)) (weight $1/(\alpha+1)$) while the red path continues through $\bar{G}_1$ and $G_{0,1}$ (weights $1/(\gamma+J)$, $1/(\gamma+J)$, $\alpha/(\alpha+1)$). Summing the contribution this scheme yields $\tilde{w}_j < w_j$.

6 Simulations

In this chapter we employ the novel Newton-HDP algorithm and study its performance in a number of simulations. To begin with, we specify a balanced data generating process and study the behavior of the Newton-HDP estimate of the latent distributions. Then, moving closer to real datasets, we assess the algorithm's peculiarities by comparing its estimates with those obtained from plain Newton recursions applied independently for each population. Lastly we compare, in terms of speed and accuracy, its performance with respect to an exact Gibbs sampler.

6.1 Partially shared atoms and balanced populations

We begin by testing the performance of the Newton-HDP algorithm by generating data from a system of three finite mixtures defined as

$$\begin{aligned}
&\text{Group A:} \quad G_{A,\text{true}} = 0.5\,\delta_{-2} + 0.5\,\delta_{7},\\
&\text{Group B:} \quad G_{B,\text{true}} = 0.5\,\delta_{-2} + 0.5\,\delta_{-8},\\
&\text{Group C:} \quad G_{C,\text{true}} = 0.5\,\delta_{-2} + 0.5\,\delta_{10},
\end{aligned}$$

with observations $X_{j,i} \mid \theta_{j,i} \sim \mathcal{N}(\theta_{j,i}, 1)$. We have a mixture model where, in clustering language, all the three populations share a cluster with mean -2 and are characterized by a population specific component. The emission distribution $f(x \mid \theta)$ is a Gaussian with known unitary variance.

We initialized the algorithm choosing a $H = \mathcal{N}(0, 9)$. We evaluate the algorithm by comparing the true predictive distribution for each j (here j = A, B, C),

$$P_{G_{j,\text{true}}}(\cdot) = \int_{\Theta} F(\cdot \mid \theta)\, G_{j,\text{true}}(d\theta),$$

with the estimated predictive distribution

$$P_{G_{j,t}}(\cdot) \equiv \mathbb{P}(X_{n_j+1} \in \cdot \mid \mathbf{x}^N) = \int_{\Theta} F(\cdot \mid \theta)\, G_{j,t}(d\theta)$$

Since both measures are absolutely continuous with respect to the Lebesgue measure

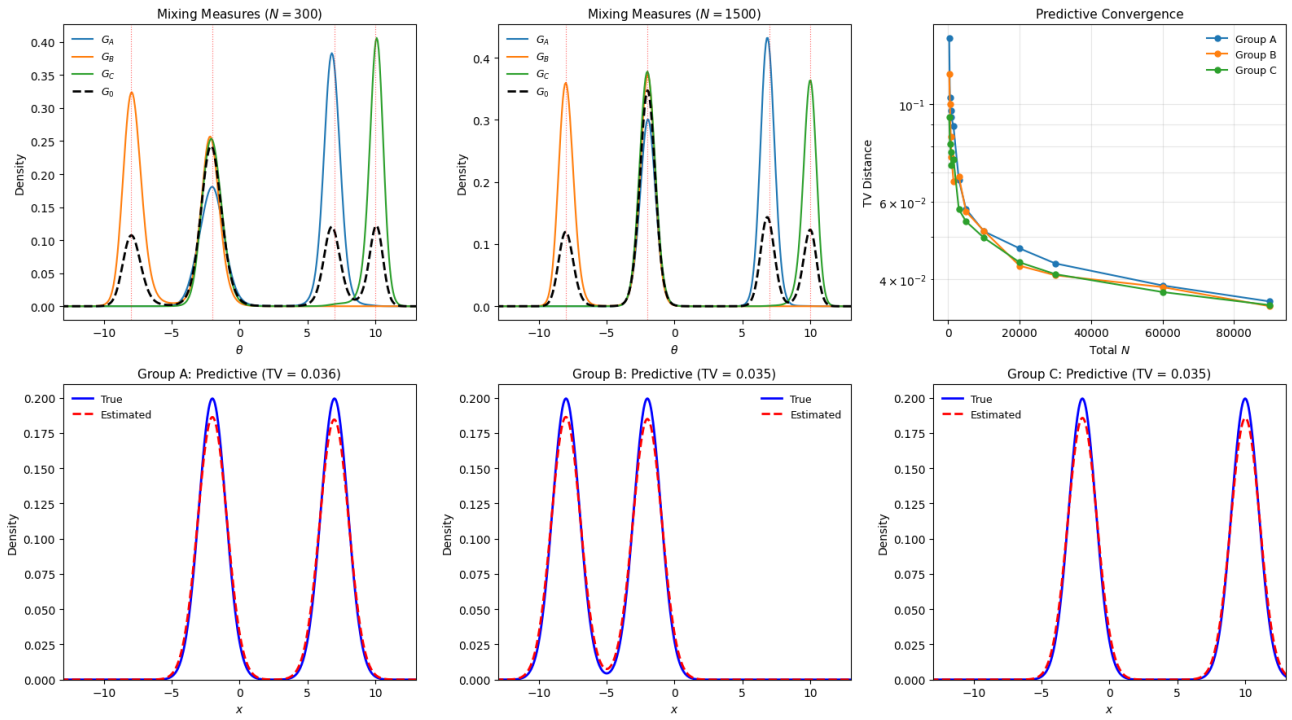


Figure 7: Newton-HDP applied to three balanced groups sharing a common component at $\theta = -2$ and group-specific components at $\theta \in \{7, -8, 10\}$, with $X_{ji} \mid \theta_{ji} \sim \mathcal{N}(\theta_{ji}, 1)$ and initial distribution $\mathcal{N}(0, 9)$. Top row: estimated mixing measures at $N = 300$, i.e. $n_j = 100$(left) and $N = 1500$, i.e. $n_j = 500$ (right) , overlaid with the global estimate of the density of G_0 (dashed black line). Since Newton-style recursions produce absolutely continuous mixing measures, the algorithm cannot recover the true discrete atoms but converges to the correct true predictive distribution. Bottom row: true (solid, blue) and estimated (dashed) predictive densities for each group at $N = 1500$, with total variation distances reported in the panel titles. Top-right panel: TV distance between true and estimated predictives as a function of total sample size N, showing that most accuracy gains occur between $n_j = 100$ and $n_j = 1000$ observations per group, after which convergence slows markedly.

λ on $\mathbb{X} = \mathbb{R}$, we measure discrepancy via the total variation distance:

$$\mathrm{TV}(P_{G_{j,true}}, P_{G_{j,t}}) = \sup_{A \in \mathcal{B}(\mathbb{X})} |P_{G_{j,true}}(A) - P_{G_{j,t}}(A)| = \frac{1}{2} \int_{\mathbb{X}} \left| p_{G_{j,true}}(x) - p_{G_{j,t}}(x) \right| d\lambda(x),$$

where $p_{G_{j,t}}$ and $p_{G_{j,true}}$ are the corresponding densities.

We note that, because, as shown in Figure 7, a Newton-style algorithm is unable to recover the true mixing structure when it is discrete, the appropriate metric for evaluation is in terms of agreement between the predictive densities rather than agreement of the mixing measures themselves. In this sense the right-top panel tracks the TV distance, for each group, between the true predictive density and the estimated one. The second row panels show the estimated and true densities at $n_j = 500$. Notably much of the gain in

terms of estimation accuracy is achieved between 100 and 1000 observations per population; then the distance tends to flatten (even though it shows still a monotone behavior) and a substantial amount of data is needed to further improve the performance. All in all this experiment shows that, if one interests is prediction even with small (balanced) datasets the algorithm can yield accurate results.

6.2 Density estimation: comparison between plain Newton and Newton-HDP

A further experiment we conduct is on the ability of the Newton-HDP to mimic the appealing properties of Bayesian models under partial exchangeability: borrowing of information. Namely the hierarchical structure assumed for the data allows to leverage information from groups with plenty of data into inference for groups where observations are few. This feature of these models is appealing for a number of reasons: data may not be as easy to obtain for one population as it is for others; one population can be much smaller than others, etc.; and in general, forms of shrinkage are desirable.
To simulate these scenarios we generate data from the following data generating process:

$$\begin{aligned} \theta_1, \dots, \theta_K \quad & \text{(shared atoms)} \\ \beta &\sim \mathrm{Dir}(\gamma/K \cdot \mathbf{1}_K) \\ \pi_j \mid \beta &\sim \mathrm{Dir}(\alpha_j \cdot \beta) \quad j = 1, \dots, J \\ z_{ji} \mid \pi_j &\sim \mathrm{Cat}(\pi_j) \qquad i = 1, \dots, n_j \\ X_{ji} \mid z_{ji} &\sim \mathcal{N}(\theta_{z_{ji}}, \sigma^2) \end{aligned} \tag{51}$$

All groups share the same K atoms θ; group-specific weights π_j are drawn from a Dirichlet centered on the global weights β, and group specific proportions are centered onto different parameters α_j to simulate heterogeneity across populations over the common components. A consequence of this, with a finite possibly unbalanced dataset, is that some components may not be present for different populations.

The parameters we use are the following:

- $K = 5$ shared atoms located at $\theta = (-8, -4, 0, 4, 8)$, with $\sigma = 1.0$
- $J = 10$ groups with sample sizes $n_j \in \{800, 600, 500, 150, 120, 100, 30, 25, 20, 15\}$

- Global concentration $\gamma_{\text{data generating process}} = 2.0$ for the data generating process
- Group-specific concentration vectors α_j designed to induce heterogeneous weight profiles across groups, ranging from balanced ($\alpha_j = 2 \cdot \mathbf{1}_K$) to strongly left- or right-skewed, bimodal, and center-peaked configurations
- For the Newton-HDP recursion: local concentration $\alpha = 4.0$, global concentration $\gamma = 0.5$.

We compare the performance in estimating the predictive density $P_{G_{j,n_j}}$ resulting from employing the plain Newton's recursions as in (7) and the proposed Newton-HDP as in Theorem 5.1. To be consistent with the temporal structure defined in Chapter 5 we randomly shuffle the whole dataset. Each measure $G_{j,t}$ is updated at the arrival of one data from population j whereas the global $G_{0,t}$ is updated at each sweep. This means that for every population different from the one receiving the *first* update the initial measure is not H but already the updated base $G_{0,t}$. Whereas by running independent Newton's recursions, for every population the algorithm starts from H. This is exactly the borrowing of information advantage encoded in the architecture of the Newton-HDP scheme, which enables populations to have already a sort of *informative* initial distribution from where to start. This is not very relevant with big populations, for which a Newton sweep is enough for estimation, but makes a difference with smaller ones where, the single algorithm is not able to learn substantially the underlying structure. The results of this exercise are summarized in Figure 8 where we show the populations with the biggest improvements in total variation distance with respect to the truth compared to Newton's. Not surprisingly those are the populations for which we have the least amount of data.

Notably, even though sometimes the improvements are of a negligible order of magnitude (see Figure 9), for each population Newton-HDP yields a better estimate. This exemplifies the advantage of using an algorithm that couples the populations in a coherent way with the assumption of partial exchangeability.

6.3 MCMC collapsed Gibbs and Newton-HDP

As it has been already much clarified before, the recursive structure specified in Theorem 5.1 is only approximate. Unlike an MCMC algorithm, for which the theory on Markov Chains, ensures that eventually we can sample from the right posterior if the algorithm

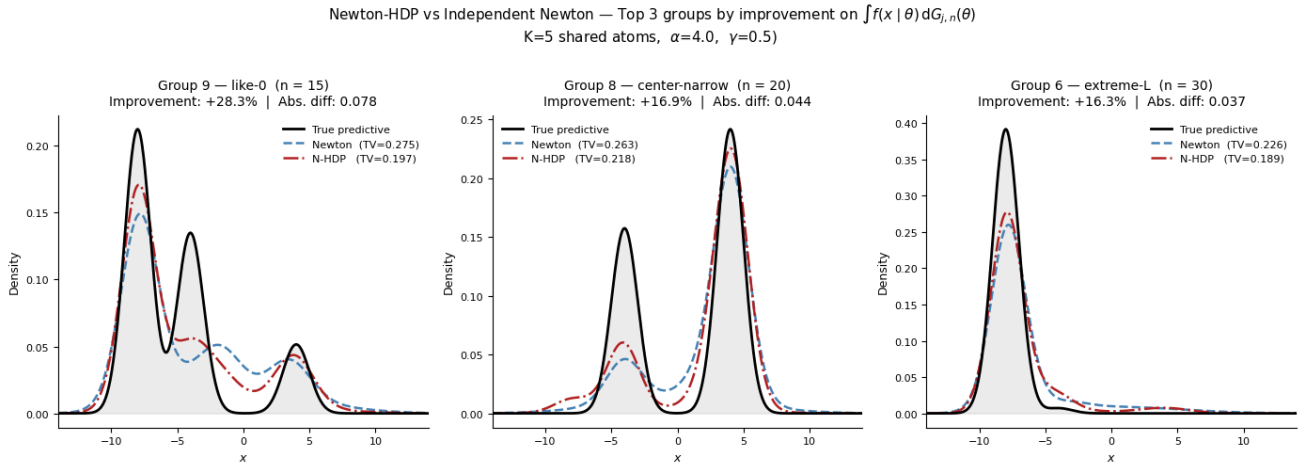


Figure 8: Density estimation comparison between Newton-HDP and independent Newton recursions for the three groups with the largest relative improvement in total variation distance. Each panel shows the true predictive density (black solid), the Newton estimate (blue dashed) and the Newton-HDP estimate (red dashed), with group index, sample size n_j, relative improvement and absolute difference in TV distance reported in the panel title. The three displayed groups ($n_j \in \{15, 20, 30\}$) are the smallest in the experiment, confirming that borrowing of strength through the hierarchical structure of Newton-HDP yields the largest gains precisely where data are scarcest. Parameters: $K = 5$ shared atoms at $\theta = (-8, -4, 0, 4, 8)$, $\alpha = 4.0$, $\gamma = 0.5$.

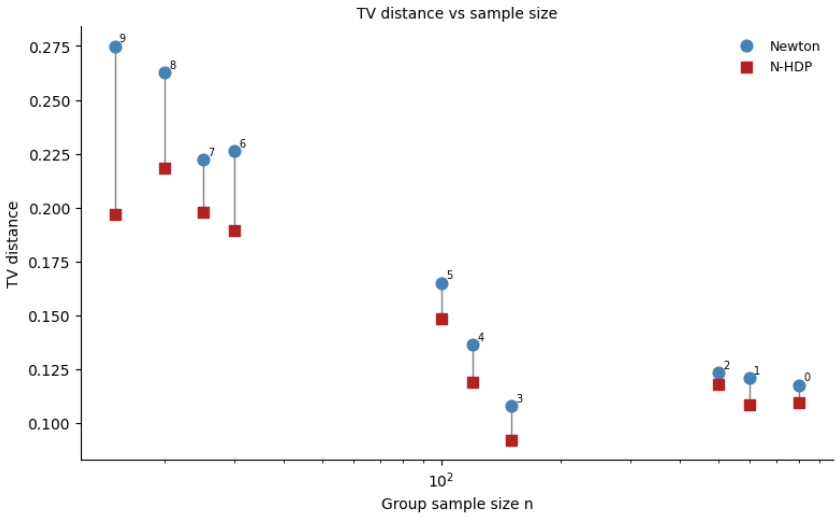


Figure 9: Total variation distance between estimated and true predictive densities as a function of group sample size n_j, comparing Newton-HDP (squares) and independent Newton recursions (circles) across all $J = 10$ groups. Each point is labelled by group index.

is built properly, for the Newton-HDP we cannot make such claim. It is thus interesting and useful to compare the performance of an exact posterior sampling scheme.
In order to do so we synthetically generate a dataset from (51) with parameters summarized in Tables 2 and 3.

Table 2: data generating process Parameters

Parameter	Value	Description
K	4	Number of shared atoms
θ	$(-6, -2, 2, 6)$	Atom locations (equally spaced)
γ	2.0	Global concentration for β
σ	1.0	Emission standard deviation
J	4	Number of groups
(n_1, n_2, n_3, n_4)	$(1000, 400, 150, 70)$	Group sample sizes

Table 3: Group-Specific Dirichlet Parameters $\alpha_j = (\alpha_{j,1}, \alpha_{j,2}, \alpha_{j,3}, \alpha_{j,4})$

Group	n_j	$\alpha_{j,1}$	$\alpha_{j,2}$	$\alpha_{j,3}$	$\alpha_{j,4}$
0	1000	2.0	2.0	2.0	2.0
1	400	0.3	3.0	3.0	0.3
2	150	4.0	0.5	0.5	0.2
3	70	0.2	0.3	0.5	4.0

The choice of α_j induces different mixture profiles: Group 0 uses all components roughly equally; Group 1 concentrates on central atoms (θ_2, θ_3); Group 2 is dominated by the leftmost atom (θ_1); Group 3 by the rightmost (θ_4). This setup tests the algorithm's ability to borrow strength when some components are rare or absent in smaller groups. The MCMC used to sample from the posterior is the efficient direct assignment sampler (section 5 of Teh et al., 2006 for a thorough explanation).

Figure 10 shows the results, in term of density estimation, of our exercise. At a first glance the performance of the algorithm is satisfying. Even though approximate it manages to recover the distributional structure of each population even for the ones with few data points. Notably the average TV distance between the estimated predictives and the true data generating process is, on average, 0.111. This is encouraging but we see that an exact sampling method is able to recover in much better accuracy the actual statistical structure of the data. Notably, however, the computational time involved in both the

estimations is enormously different. In particular the estimation time for the Newton-HDP was 1% what took the MCMC to sweep across 2500 iterations. This is probably the most interesting advantage of this kind of algorithms which grows bigger as the dataset enlarges both in terms of populations and data as well as its online nature. Indeed the only bookkeeping involved when computing the recursions are the population and global measures $G_{j,t}$, $G_{0,t}$ and the population counts $n_{j,t}$ which is much less involved than a marginal sampler (even the minimal bookkeeping version that is the direct assignment sampler).

6.3.1 Introducing a decay parameter

The estimation results of the previous paragraph are encouraging but still the "distance" from an exact sampler is relevant. Fortini and Petrone (2020) argue that introducing a decay parameter on the update weights of the Newton recursions after a fixed amount of observations processed amounts to letting the model learn more from the data as the weights decay slower to zero. We implement the same strategy, in particular we introduce the following weights:

$$w_{t,j} = (\alpha + n_{j,t})^{-\delta}, \quad w_t^{(0)} = (\gamma + N_t)^{-\delta} \tag{52}$$

where $\delta \in (0, 1]$ is the decay parameter. With $\delta = 1$ we recover the standard weights; with $\delta < 1$ the weights decay more slowly, enabling continued learning. We define the split sample weights as:

$$w_{t,j} = \begin{cases} (\alpha + n_{j,t})^{-1} & \text{if } n_{j,t} \leq \tau_j \\ (\alpha + n_{j,t})^{-\delta} & \text{if } n_{j,t} > \tau_j \end{cases} \tag{53}$$

with group-specific threshold $\tau_j = \lfloor 0.55 \cdot n_j \rfloor$ and decay exponent $\delta = 0.75$. An analogous scheme applies to the global weights $w_t^{(0)}$ where the threshold is applied to the total number of observations processed N_t with the global threshold at $\tau_0 = 0.75$.

The results in Figure 11 show a substantial improvement over the previous exercise for which our approximate estimates are very close to the ones produced by the MCMC but retaining all the computational cost advantage (again Newton-HDP is 90 times faster than the Gibbs sampler).

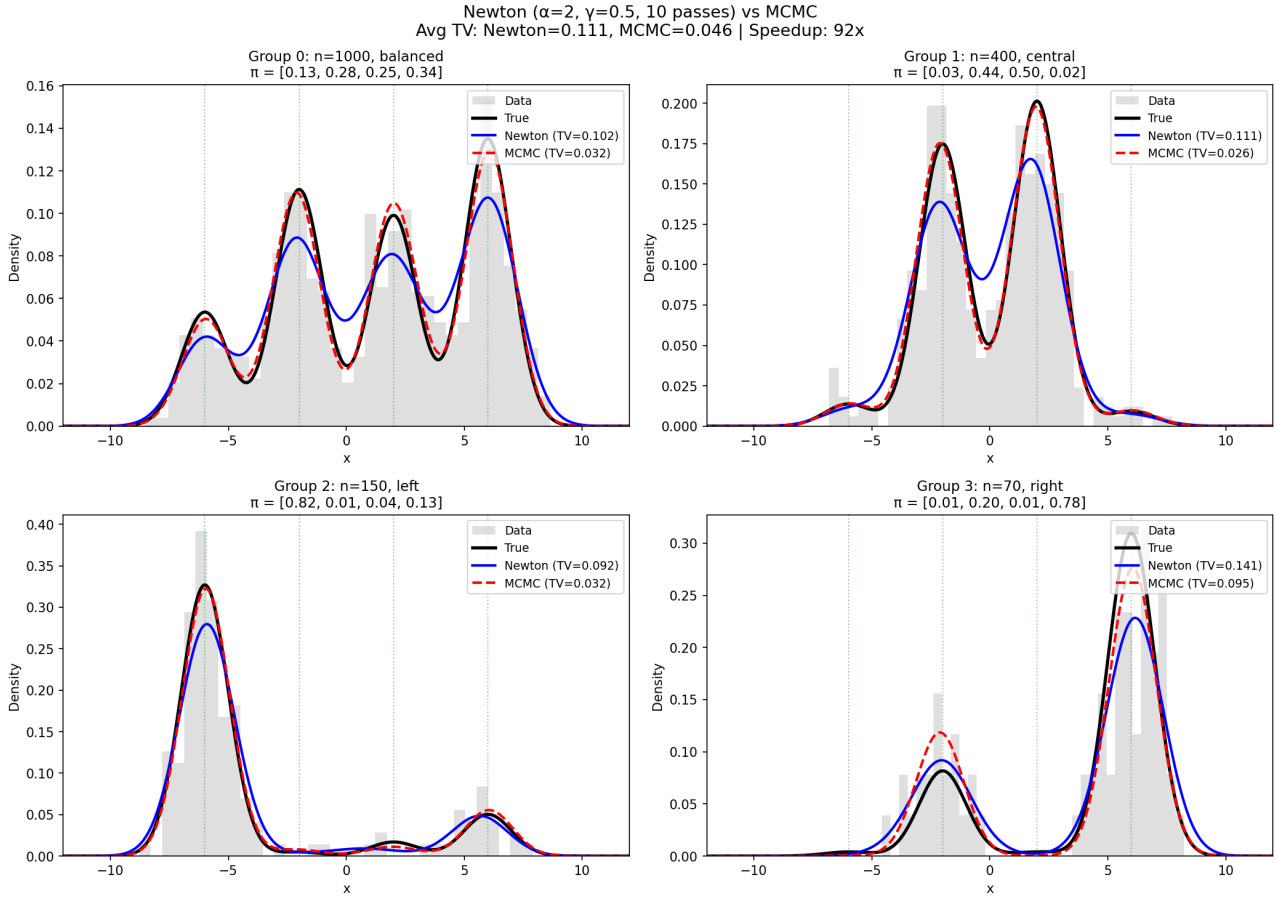


Figure 10: Predictive density estimates for the Newton-HDP recursions (blue, $\alpha = 2$, $\gamma = 0.5$, 10 passes over the dataset) and a direct-assignment collapsed Gibbs sampler (red dashed, 2500 iterations) under the data generating process of Tables 2–3: $J = 4$ groups with $K = 4$ shared atoms at $\theta = (-6, -2, 2, 6)$, heterogeneous sample sizes $(n_0, n_1, n_2, n_3) = (1000, 400, 150, 70)$, and group-specific Dirichlet weight profiles inducing balanced, central-bimodal, left-skewed, and right-skewed mixtures respectively. The true predictive density is shown in black; grey histograms display the observed data. Average TV distance: Newton-HDP = 0.111, MCMC = 0.046. Despite the approximation gap, most visible for the smallest group ($n_3 = 70$), Newton-HDP achieves this at roughly 1% of the MCMC running time.

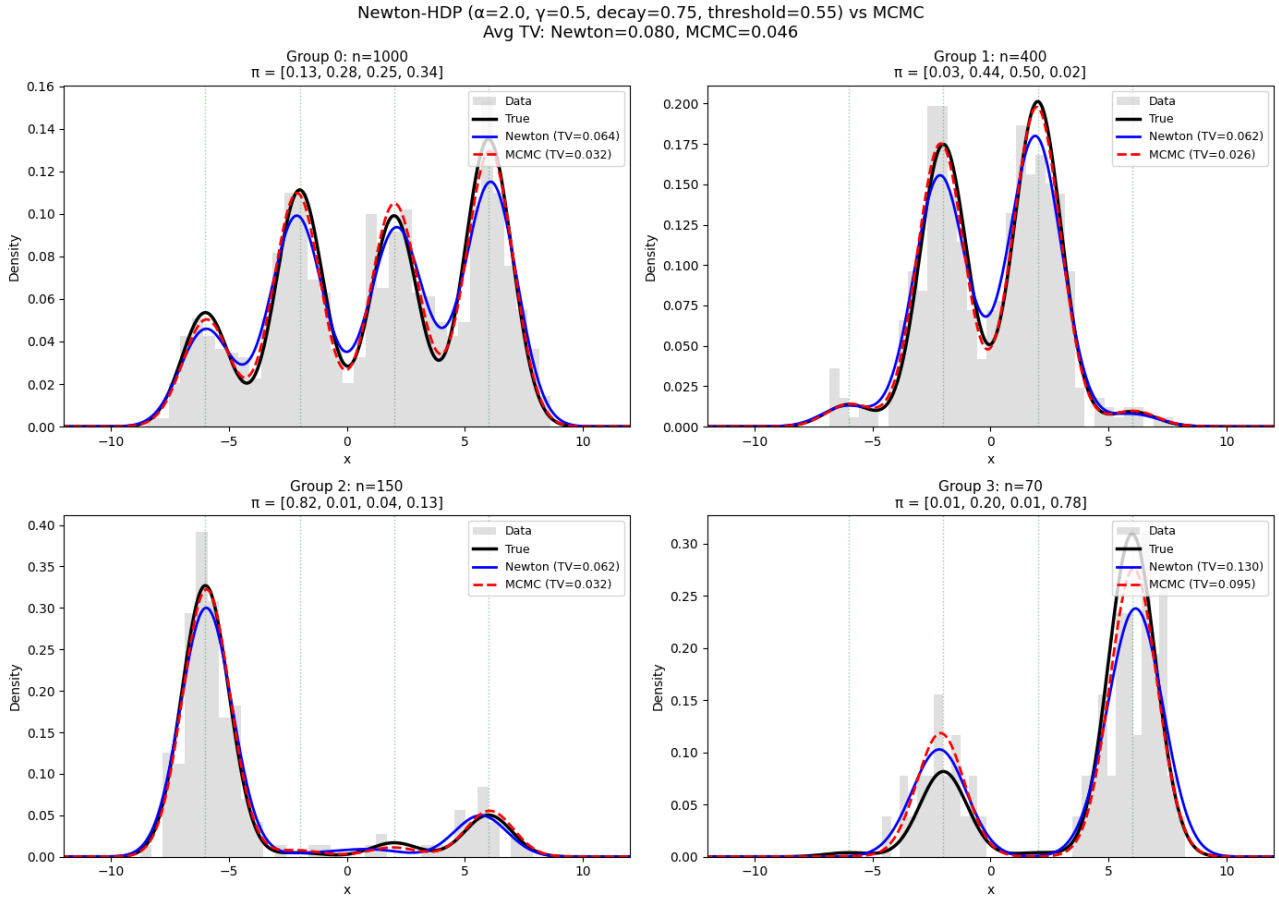


Figure 11: Predictive density estimates under the split-sample decay scheme of equation (53). Same data generating process and MCMC baseline as Figure 10 ($\alpha = 2.0$, $\gamma = 0.5$). The initial phase with $\delta = 1$ drives rapid convergence toward the asymptotic regime; the subsequent phase with $\delta < 1$ slows the weight decay, restoring sensitivity to later observations and enabling finer density recovery. Average TV distance: Newton-HDP = 0.080, MCMC = 0.046; computational speedup approximately 90×.

For the Newton case the theory presented in Chapter 2 provide insights on why this split sample definition of the weights is effective. Indeed with DP-like weights ($\delta = 1$), later observations are almost uninformative for G_n since the faster decay of $\alpha_n = (\alpha+n)^{-1}$ reduces the sensitivity of G_n to new data, but the system reaches exchangeability more rapidly (see section 2.6). The split-sample approach exploits this trade-off: using $\delta = 1$ in the first phase allows rapid convergence toward the asymptotic regime, after which switching to $\delta < 1$ slows the weight decay and re-activates the contribution of later observations to the estimation of $\tilde{G}$.

For the partially-exchangeable case the simulation in Figure 11 point towards a similar explanation. We thus follow to define a counterpart of model (11) for the Newton-HDP and we proceed in studying its properties.

7 Quasi Bayes properties of Newton-HDP model

Following the methodology developed in Fortini and Petrone (2020), in this chapter we investigate the statistical model induced by the recursions of Definition 5.1, leveraging the Bayesian predictive approach. Our aim is to give a quasi-Bayes meaning to the new algorithm; we will prove that an asymptotic partially exchangeable model can be recovered. This importantly adds a theoretical content to the simulations results of the previous chapter.

7.1 Newton-HDP recursions as predictive rules

As done in Fortini and Petrone (2020) we regard the sequential rules specified by the HDP-Newton procedure as the one step ahead predictive rule for the family of sequences $(\theta_{j,t})_{t\geq 1}$, $j = 1, \ldots, J$. For the purposes of the theoretical analysis, we adopt a synchronous convention: at each round $t = 1, 2, \ldots$, every group $j \in \{1, \ldots, J\}$ receives one observation $X_{t,j}$, so that $n_{j,t} = t$ for all j and $N_t = Jt$. This specializes the asynchronous scheme of Section 5.1, where $n_{j,t} = \#\{s \leq t : j(s) = j\}$, to the case of balanced and simultaneous arrivals. The asymptotic results of this chapter assume that $n_{j,t} \to \infty$ for each j and $N_t \to \infty$, and therefore the two assumptions on arrivals are indistinguishable in the limit. Let $\mathcal{F}_t = \sigma(X_{s,j} : s \leq t, j = 1, \ldots, J)$ denote the natural filtration generated by the observations across J populations up to time t. The estimates $G_{j,t}$ and $G_{0,t}$ are $\mathcal{F}_t$-measurable and evolve according to the Newton–HDP recursions (to ease notation we define $q_{t,j} := q_{\pi_{t,j}}$):

$$\begin{aligned} G_{j,t} &= (1 - w_{t,j})G_{j,t-1} + w_{t,j}\, q_{t,j}, \\ G_{0,t} &= (1 - w_t^{(0)})G_{0,t-1} + w_t^{(0)}\, \bar{G}_t, \end{aligned} \tag{54}$$

where

$$\begin{aligned} \pi_{t,j} &= (1 - \eta_{t,j})G_{j,t-1} + \eta_{t,j}G_{0,t-1}, \\ q_{t,j}(d\theta) &= \frac{f(X_{t,j}|\theta)\, \pi_{t,j}(d\theta)}{\int f(X_{t,j}|u)\, \pi_{t,j}(du)}, \\ \eta_{t,j} &= \frac{\alpha}{\alpha + n_{j,t}}, \qquad w_{t,j} = (\alpha + n_{j,t})^{-1}, \qquad w_t^{(0)} = (\gamma + N_t)^{-1}. \end{aligned}$$

7.2 Generative Model

Throughout this section we adopt the synchronous convention of Section 7.1, so that $\mathbf{X}_t = (X_{t,1}, \ldots, X_{t,J}) \in \mathbb{X}^J$ and $\boldsymbol{\theta}_t = (\theta_{t,1}, \ldots, \theta_{t,J}) \in \Theta^J$ are well-defined J-dimensional vectors for every $t \geq 1$.

Leveraging the scheme specified by the Bayesian predictive approach of Chapter 1 we invoke Theorem 1.4 and thus specify a law $\mathbb{P}$ on the array $(\boldsymbol{X}_t, \boldsymbol{\theta}_t)_{t\geq 1}$, thus we characterize the process for $(\boldsymbol{X}_t, \boldsymbol{\theta}_t)_{t\geq 1}$ as follows

Assumption 1 (Conditional Independence)**.** Conditionally on $\mathcal{F}_{t-1}$, the latent parameters $(\theta_{t,j})_{j=1}^J$ are independent with

$$\theta_{t,j} \mid \mathcal{F}_{t-1} \overset{ind}{\sim} (1-\eta_{t,j})G_{j,t-1} + \eta_{t,j}G_{0,t-1}.$$

Assumption 2 (Observation Model)**.** Conditionally on $(\theta_{t,j})_{j=1}^J$, the observations are independent with

$$X_{t,j} \mid \theta_{t,j} \sim f(\cdot|\theta_{t,j}),$$

where $f(\cdot \mid \theta)$ is a known emission kernel.

Thus we can define the statistical model pinned down by Newton-HDP algorithm, now regarded as a predictive learning rule, as follows:

$$\begin{cases} \theta_{t,j} \mid \mathcal{F}_{t-1} \overset{ind}{\sim} (1-\eta_{t,j})G_{j,t-1} + \eta_{t,j}G_{0,t-1} \\ X_{t,j} \mid \theta_{t,j} \overset{ind}{\sim} f(\cdot \mid \theta_{t,j}) \end{cases} \tag{55}$$

where the measures $G_{j,t}$ and $G_{0,t}$ evolve according to the recursions specified in (54) and $\theta_0, j \sim H$ with H diffuse.

In this chapter, $G_{j,t}$ and $G_{0,t}$ are treated as random variables adapted to $\mathcal{F}_{t-1}$. This is consistent with Chapter 5: there $G_{j,t}$ was viewed as a deterministic function of the observed data, but once we endow the observation sequence $(X_{t,j})$ with a probability law $\mathbb{P}$ via Theorem 1.4, the same recursions define $G_{j,t} = G_{j,t}(X_1, \ldots, X_t)$ as a measurable function of random variables, hence a random variable itself. The two interpretations are compatible: for any fixed realization $\omega = (x_1, x_2, \ldots)$, $G_{j,t}(\omega)$ coincides with the deterministic estimate computed from that sample.

The Newton–HDP recursions alone only specifies the marginal predictive distribution

of $\theta_{t,j}$ given the observable past $\mathcal{F}_{t-1}$, without fully determining the joint law of the process $(\boldsymbol{X}_t, \boldsymbol{\theta}_t)_{t\geq 1}$. Following Fortini and Petrone (2020), an obvious way to obtain a full specification (i.e. making assuptions on $\theta_{j,t} \mid \boldsymbol{X}_{t-1}, \boldsymbol{\theta}_t^{-j}$) is to additionally assume that $(\theta_{t,j})_{j=1}^J$ are conditionally independent across groups given $\mathcal{F}_{t-1}$ (they however do not impose such restriction, as in their single-group setting there is no cross-group independence to impose). This is precisely the content of Assumption 1. Indeed, the hierarchical coupling between groups is entirely mediated by $G_{0,t-1}$, which is $\mathcal{F}_{t-1}$-measurable. Conditionally on $\mathcal{F}_{t-1}$, there is therefore no residual cross-group dependence to be captured through the $\theta_{t,j}$'s, making the conditional independence assumption natural in this setting. We will discuss what is the impact of this assumption for the theoretical guarantees that will follow and which are the possible ways forward to substantially relax it without losing a quasi-Bayes interpretation. This resort in recovering, in the limit, a partially exchangeable structure mirroring that of Newton which we discussed in Section 2.5.

Uncovering the statistical model implied by the recursions, when read as predictive learning rules, allows us to study the quasi-Bayes properties of the approximation strategy. A clear implication of 1 and 2 is that we lose partial exchangeability for the $\theta_{j,i}$ and thus also for the $X_{j,i}$. Thus, someone using the recursions specified in Theorem 5.1 is implicitly using a learning scheme that has the advantage of being computationally simpler, but is different from a fully partially exchangeable one.

7.3 Convergence of $G_{j,t}$ and $G_{0,t}$

A necessary result we wish to obtain is whether the sequences $G_{j,t}$ and $G_{0,t}$ converge and if the limit is a random probability measure. There are however technical difficulties in establishing such limit behavior in full generality, particularly in ensuring the almost sure tightness of the sequences $\{G_{j,t}\}$ and $\{G_{0,t}\}$.

We will assume that Θ is a compact space (which is a reasonable assumption in many applications, and enough for our aims since, in practice, we always deal with their finite-dimensional distribution on a finite grid). Thus all the (random) measures involved are automatically tight. The following theorem assumes a.s. tightness of both sequences and proves the convergence result for general Polish spaces; we acknowledge however that a rigorous verification of tightness on non-compact spaces is an open problem that should

be investigated further.

Along the proof we will use some well known results about weak convergence: Prokhorov Theorem (see Theorem 25.10 in Billingsley (1995)) and its corollary which we cite here.

Corollary 7.1. *If $\{\mu_n\}$ is a tight sequence of probability measures, and if each subsequence that converges weakly at all converges weakly to the probability measure μ, then $\mu_n \Rightarrow \mu \quad \mathbb{P} - a.s.$.*

We now reach the main result that establish the $\mathbb{P}$-almost surely convergence of the estimates of Newton-HDP. As clarified before we assume Θ to be Polish and require the sequences of Newton-HDP measures to be almost surely tight. In practice, being Θ finite (equivalently it would work for a compact space $\Theta \subset \mathbb{R}^d$) tightness is guaranteed.

Theorem 7.2 (Almost sure convergence of group and global measures)**.** *Let Θ be a Polish space, in particular $\Theta \subset \mathbb{R}^d$. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$ be the filtered probability space generated by the Newton–HDP algorithm via the Ionesco–Tulcea theorem. Let $j \in \{1, \dots, J\}$ index groups and denote by $G_{j,t}$ and $G_{0,t}$ the group and global random probability measures produced by the algorithm in Theorem 5.1.*

Assume the step sizes satisfy, for each j,

$$\sum_{t\geq 0} w_{t+1,j}^2 < \infty, \qquad \sum_{t\geq 0} w_{t+1,j}\eta_{t+1,j} < \infty,$$

and for the global step sizes

$$\sum_{t\geq 0} (w_{t+1}^{(0)})^2 < \infty, \qquad \sum_{t\geq 0} w_{t+1}^{(0)} = \infty.$$

Lastly assume that the sequences of measures $\{G_{j,t}\}$ and $\{G_{0,t}\}$ are tight almost surely. Then:

1. *For each group j there exists an a.s.-defined random probability measure α_j with $G_{j,t} \Rightarrow \alpha_j$ almost surely as $t \to \infty$.*

2. *Suppose additionally $N_t \to \infty$ and $n_{j,t}/N_t \to \lambda_j \in [0,1]$ almost surely for each j with $\sum_{j=1}^J \lambda_j = 1$. Then there exists $\alpha_0 := \sum_{j=1}^J \lambda_j \alpha_j$ and $G_{0,t} \Rightarrow \alpha_0$ almost surely.*

Proof. The proof proceeds in two stages. First we show a.s. convergence of integral projections $Y_t^j(\varphi) := \langle \varphi, G_{j,t} \rangle = \int \varphi dG_{j,t}$ for every bounded Lipshitz $\varphi : \Theta \to \mathbb{R}$ with $\|\varphi\|_\infty \leq 1$. Second, we lift scalar convergence to weak convergence of measures using tightness and a countable separating and convergence-determining class in Polish spaces.

Stage 1: scalar convergence for each group. Fix j and a bounded Lipshitz φ with $\|\varphi\|_\infty \leq 1$ and define $Y_t := Y_t^j(\varphi) = \langle \varphi, G_{j,t} \rangle$.

Applying $\langle \varphi, \cdot \rangle$ to the population recursion gives

$$Y_{t+1} = Y_t + w_{t+1,j}\Big(\langle \varphi, q_{t+1,j} \rangle - Y_t\Big).$$

We decompose it exployting a Doob-like decomposition:

$$\Delta_{t+1} := \langle \varphi, q_{t+1,j} \rangle - Y_t, \qquad a_{t+1} := \mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t], \qquad \varepsilon_{t+1} := \Delta_{t+1} - a_{t+1}.$$

Then

$$Y_{t+1} = Y_t + w_{t+1,j} a_{t+1} + w_{t+1,j} \varepsilon_{t+1}.$$

We compute a_{t+1}. Since $q_{t+1,j}$ is the one step Bayesian posterior of $\theta_{t+1,j}$ given $(X_{t+1,j}, \mathcal{F}_t)$,

$$\mathbb{E}[q_{t+1,j}(B) \mid \mathcal{F}_t] = \mathbb{E}\Big(\mathbb{P}(\theta_{t+1,j} \in B \mid X_{t+1,j}, \mathcal{F}_t) \mid \mathcal{F}_t\Big) = \pi_{t+1,j}(B),$$

the calculations are almost identical to those for Theorem 2.8. Hence since $G_{j,t}$ is $\mathcal{F}_t$-measurable:

$$a_{t+1} = \langle \varphi, \pi_{t+1,j} - G_{j,t} \rangle.$$

Expanding $\pi_{t+1,j} = (1 - \eta_{t+1,j}) G_{j,t} + \eta_{t+1,j} G_{0,t}$ yields

$$\pi_{t+1,j} - G_{j,t} = \eta_{t+1,j}(G_{0,t} - G_{j,t}),$$

so that

$$a_{t+1} = \eta_{t+1,j} \langle \varphi, G_{0,t} - G_{j,t} \rangle.$$

Since $\|\phi\|_\infty \leq 1$ and $G_{0,t}, G_{j,t}$ are probability measures, the triangle inequality gives $|\langle \varphi, G_{0,t} - G_{j,t} \rangle| \leq |\langle \varphi, G_{0,t} \rangle| + |\langle \varphi, G_{j,t} \rangle| \leq \|\varphi\|_\infty + \|\varphi\|_\infty = 2$. Therefore $|a_{t+1}| \leq 2\eta_{t+1,j}$.

Since $w_{t+1,j}$ is $\mathcal{F}_t$-measurable (it's a deterministic sequence), the increments $w_{k+1,j}\varepsilon_{k+1}$ satisfy $\mathbb{E}[w_{k+1,j}\varepsilon_{k+1}|\mathcal{F}_k] = w_{k+1,j}\mathbb{E}[\varepsilon_{k+1}|\mathcal{F}_k] = 0$, so $M_t := \sum_{k=0}^{t-1} w_{k+1,j}\varepsilon_{k+1}$ is an $\mathcal{F}_t$-martingale.

Since $\|\varphi\|_\infty \leq 1$ and $q_{t+1,j}$, $G_{j,t}$ are probability measures, we have $|\Delta_{t+1}| \leq 2$ (same argument as above) and $|a_{t+1}| \leq 2\eta_{t+1,j}$, hence $|\varepsilon_{t+1}| \leq 4$ a.s. By orthogonality of martingale increments:

$$\mathbb{E}[M_t^2] = \sum_{k=0}^{t-1} \mathbb{E}[w_{k+1,j}^2 \varepsilon_{k+1}^2] \leq 16 \sum_{k=1}^{\infty} w_{k,j}^2 < \infty.$$

By the Martingale Convergence Theorem (see Doob (1953)), M_t converges almost surely to a finite random variable.

Now consider the drift series:

$$\sum_{k=0}^{\infty} w_{k+1,j} a_{k+1}$$

and note that by our argument above $|w_{k+1,j}a_{k+1}| \leq 2w_{k+1,j}\eta_{k+1,j}$. By hypothesis $\sum_k w_{k+1,j}\eta_{k+1,j} < \infty$, so the drift series converges absolutely.

Hence, summing the recursion telescopically,

$$Y_t = Y_0 + \sum_{k=0}^{t-1} w_{k+1,j} a_{k+1} + M_t,$$

and the right-hand side converges almost surely as $t \to \infty$. Therefore for every bounded measurable φ with $\|\varphi\|_\infty \leq 1$ we have almost sure convergence of $Y_t^j(\varphi) = \langle \varphi, G_{j,t} \rangle$ to a finite random variable ℓ.

Stage 2: from scalar convergence to weak convergence. Let $\mathcal{D} = \{\varphi_m : m \geq 1\} \subset BL_1(\Theta)$ be a countable bounded-Lipschitz convergence-determining class (such a class exists and its separating because Θ is Polish, thus separable, and BL_1 is rich enough to characterize convergence). By Stage 1, for each m the scalar sequence $\langle \varphi_m, G_{j,t} \rangle$ converges almost surely. Therefore there exists an event $\Omega_0 = \bigcap \Omega_m$ with $P(\Omega_0) = 1$ on which for every $\omega \in \Omega_0$ and every m the limit

$$\ell_m(\omega) := \lim_{t \to \infty} \langle \varphi_m, G_{j,t}(\omega) \rangle$$

exists in $\mathbb{R}$.

Fix $\omega \in \Omega_0$. By assumptions, the sequence $\{G_{j,t}(\omega)\}_{t \geq 0}$ is tight in $\mathcal{P}(\Theta)$. By

Prokhorov's theorem, it is relatively compact in the weak topology, meaning there exists a subsequence $t_r \to \infty$ and a probability measure $\mu_\omega \in \mathcal{P}(\Theta)$ such that $G_{j,t_r}(\omega) \Rightarrow \mu_\omega$.

Weak convergence implies:

$$\langle \varphi_m, G_{j,t_r}(\omega) \rangle \longrightarrow \langle \varphi_m, \mu_\omega \rangle \quad \text{for each } m.$$

Since by Stage 1 $\langle \varphi_m, G_{j,t}(\omega) \rangle \to \ell_m(\omega)$ along the full sequence, any weakly convergent subsequence with limit μ' satisfies $\langle \varphi_m, \mu' \rangle = \ell_m(\omega) = \langle \varphi_m, \mu_\omega \rangle$ for all m. Since $\mathcal{D}$ is separating, $\mu' = \mu_\omega$. By Corollary 7.1, $G_{j,t}(\omega) \Rightarrow \mu_\omega$. Defining $\alpha_j(\omega) := \mu_\omega$ for $\omega \in \Omega_0$, we have $G_{j,t} \Rightarrow \alpha_j$ $\mathbb{P}$-almost surely.

Global measure.

Fix $\varphi \in BL_1(\Theta)$ and define

$$Z_t := \langle \varphi, G_{0,t} \rangle, \qquad b_t := \langle \varphi, \bar{G}_t \rangle,$$

so the global recursion becomes

$$Z_{t+1} = (1 - w_{t+1}^{(0)}) Z_t + w_{t+1}^{(0)} b_{t+1}.$$

Limit of the targets b_t. By definition,

$$b_t = \left\langle \varphi, \frac{\gamma H + \sum_{j=1}^J n_{j,t} G_{j,t}}{\gamma + N_t} \right\rangle = \frac{\gamma}{\gamma + N_t} \langle \varphi, H \rangle + \sum_{j=1}^J \frac{n_{j,t}}{\gamma + N_t} \langle \varphi, G_{j,t} \rangle.$$

Assume $N_t \to \infty$ and $n_{j,t}/N_t \to \lambda_j$ a.s. with $\sum_j \lambda_j = 1$. Then $\gamma/(\gamma + N_t) \to 0$ a.s., and also

$$\frac{n_{j,t}}{\gamma + N_t} = \frac{n_{j,t}}{N_t} \cdot \frac{N_t}{\gamma + N_t} \;\to\; \lambda_j \quad \text{a.s.}$$

Moreover, from Part I, $\langle \varphi, G_{j,t} \rangle \to \langle \varphi, \alpha_j \rangle$ a.s. Hence

$$b_t \xrightarrow[t \to \infty]{} \sum_{j=1}^J \lambda_j \langle \varphi, \alpha_j \rangle =: \langle \varphi, \alpha_0 \rangle \qquad \text{a.s.},$$

where $\alpha_0 := \sum_{j=1}^J \lambda_j \alpha_j$.

Convergence of Z_t by weighted averaging. By assumption, the global steps satisfy $\sum_t w_{t+1}^{(0)} =$

∞ and $\sum_t (w_{t+1}^{(0)})^2 < \infty$. Applying Theorem A.1 with $a_{t+1} = b_{t+1}$ and $a = \langle \varphi, \alpha_0 \rangle$ yields

$$Z_t = \langle \varphi, G_{0,t} \rangle \;\to\; \langle \varphi, \alpha_0 \rangle \qquad \text{a.s.}$$

Weak convergence of $G_{0,t}$. By assumption $\{G_{0,t}\}_{t \geq 0}$ is a.s. tight. Let $\{\varphi_m\}_{m \geq 1} \subset BL_1(\Theta)$ be determining. Since $\langle \varphi_m, G_{0,t} \rangle \to \langle \varphi_m, \alpha_0 \rangle$ a.s. for all m, the same argument as above yields $G_{0,t} \Rightarrow \alpha_0$ $\mathbb{P}$-almost surely. □

Under the synchronous convention of Section 7.1, $n_{j,t} = t$ and $N_t = Jt$, so that $w_{t,j} = (\alpha + t)^{-1}$ and $w_t^{(0)} = (\gamma + Jt)^{-1}$. The conditions $\sum_t w_{t,j}^2 < \infty$ and $\sum_t w_{t,j} \eta_{t,j} < \infty$ follow from $\sum_t (\alpha + t)^{-2} < \infty$ and $\sum_t \alpha (\alpha + t)^{-2} < \infty$. The global conditions $\sum_t w_t^{(0)} = \infty$ and $\sum_t (w_t^{(0)})^2 < \infty$ follow from $\sum_t (\gamma + Jt)^{-1} = \infty$ and $\sum_t (\gamma + Jt)^{-2} < \infty$. Furthermore, $n_{j,t}/N_t = 1/J$ for all t and all j, so the assumption $n_{j,t}/N_t \to \lambda_j$ a.s. is satisfied deterministically with $\lambda_j = 1/J$, and the condition $\sum_j \lambda_j = 1$ holds trivially. Hence all assumptions of Theorem 7.2 are automatically satisfied in the synchronous case.

Two things are to be noticed from the result in this section. First, the limiting random measures α_j are only defined as limits of the estimates produced by Newton-HDP. We have no results for the law of such random probability measures. Secondly Theorem 7.2 establishes only a marginal result; Fortini et al. (2018) clarify that to recover a counterpart of asymptotic exchangeability through Theorem 2.6 one need to show that:

$$\mathbb{P}(\boldsymbol{X}_{t+1} \mid \mathcal{F}_t) \to P_{\boldsymbol{\alpha}}, \quad \mathbb{P}\text{-a.s.},$$

which says that the *joint* (across groups) one-step-ahead predictive converges to a limiting random model of the form $P_{\boldsymbol{\alpha}} = \otimes_{j=1}^{J} P_{\alpha_j}$, where each marginal $P_{\alpha_j}(\cdot) := \int_\Theta F(\cdot \mid \theta)\, \alpha_j(d\theta)$ is a mixture distribution with random mixing measure α_j. Thus by Theorem 2.6 $(\boldsymbol{X}_t)$ is asymptotic exchangeable and since $P_{\boldsymbol{\alpha}}$ is a product probability measure it is *asymptotically partially exchangeable* (see Fortini et al. (2018) for a more precise definition of asymptotic partial exchangeability). This is the quasi-partially exchangeable Bayes justification that we seek, and which will be the main result of the next section.

7.4 Asymptotic partial exchangeability

Before proving the main result on asymptotic partial exchangeability we need to make an additional restricting assumption on the emission kernel.

Assumption 3 (Likelihood regularity)**.** For every $g \in C_b(\mathbb{X})$, the map $\theta \mapsto \int g(x) f(x|\theta)\, dx$ is continuous on Θ.

Assumption 3 requires that the likelihood $f(x|\theta)$ depends continuously on θ in the sense that if $d_\Theta(\theta, \theta_0) < \delta \implies |\int_{\mathbb{X}} g(x) f(x|\theta) d\lambda(x) - \int_{\mathbb{X}} g(x) f(x|\theta_0) d\lambda(x)| < \varepsilon$. This is satisfied for exponential families with continuous parameterization, including Gaussian, Poisson, Binomial, and Gamma likelihoods. For the Gaussian location mixture $f(x|\theta) = \phi_\sigma(x - \theta)$ used in our simulations, the assumption is verifiable using characteristic functions.

We can now proceed by proving that the joint one step predictive distribution converges weakly to a random probability measure that is the product measure of the marginal limits

Theorem 7.3 (Asymptotic Partial Exchangeability)**.** *Let $\{(X_{t,j}, \theta_{t,j})\}_{t \geq 1, j \in \{1,\dots,J\}}$ be the array generated by the Newton–HDP model under Assumptions 1–3. Assume the conditions of Theorem 7.2 hold, so that $G_{j,t} \Rightarrow \alpha_j$ a.s. for each j. Define the mixture distributions $P_{\alpha_j}(\cdot) := \int_\Theta F(\cdot|\theta)\alpha_j(d\theta)$.*

Then for each choice of distinct $\{j_1, \dots, j_k\}$ the conditional joint predictive distributions converge weakly to the product measure:

$$\mathbb{P}\left((X_{t+1,j_1}, \dots, X_{t+1,j_k}) \mid \mathcal{F}_t\right) \Rightarrow P_{\alpha_{j_1}} \otimes \cdots \otimes P_{\alpha_{j_k}} \quad a.s.$$

Proof. The proof proceeds in three steps.

Step 1: Factorization of the joint predictive. By Assumption 1, conditionally on $\mathcal{F}_t$, the latent parameters $(\theta_{t+1,j})_{j=1}^J$ are independent with $\theta_{t+1,j} \mid \mathcal{F}_t \sim \pi_{t+1,j}$. By Assumption 2, conditionally on $(\theta_{t+1,j})_{j=1}^J$, the observations are independent with $X_{t+1,j} \mid \theta_{t+1,j} \sim f(\cdot|\theta_{t+1,j})$. Let $\{j_1, \dots, j_k\}$ be distinct in $\{1, \dots, J\}$, then

integrating out the latent parameters, for any measurable sets $A_{j_1}, \ldots, A_{j_k}$:

$$\begin{aligned}\mathbb{P}(X_{t+1,j_1} \in A_{j_1}, \ldots, X_{t+1,j_k} \in A_{j_k} \mid \mathcal{F}_t) &= \mathbb{E}\left[\prod_{i=1}^{k} \mathbb{P}(X_{t+1,j_i} \in A_{j_i} \mid \theta_{t+1,j_i}) \middle| \mathcal{F}_t\right] \\ &= \prod_{i=1}^{k} \mathbb{E}\left[F(A_{j_i}|\theta_{t+1,j_i}) \middle| \mathcal{F}_t\right] \\ &= \prod_{i=1}^{k} P_{\pi_{t+1,j_i}}(A_{j_i}),\end{aligned}$$

where again $P_\mu(\cdot) := \int F(\cdot|\theta)\mu(d\theta)$ denotes the mixture distribution induced by a mixing measure μ. The second equality uses the conditional independence of the $\theta_{t+1,j}$ given $\mathcal{F}_t$ (Assumption 1).

Thus, conditionally on $\mathcal{F}_t$, the observations $(X_{t+1,j})_{j=1}^J$ are independent with $X_{t+1,j} \mid \mathcal{F}_t \sim P_{\pi_{t+1,j}}$.

Step 2: Convergence of marginal predictives. We show that $P_{\pi_{t+1,j}} \Rightarrow P_{\alpha_j}$ a.s. for each j.

First, we establish $\pi_{t+1,j} \Rightarrow \alpha_j$ a.s. Recall that

$$\pi_{t+1,j} = (1 - \eta_{t+1,j})G_{j,t} + \eta_{t+1,j}G_{0,t},$$

where $\eta_{t+1,j} = \alpha/(\alpha + n_{j,t+1}) \to 0$ since $n_{j,t} \to \infty$. By Theorem 7.2, $G_{j,t} \Rightarrow \alpha_j$ and $G_{0,t} \Rightarrow \alpha_0$ a.s. thus by definition of weak convergence for any $\varphi \in C_b(\Theta)$:

$$\langle \varphi, \pi_{t+1,j} \rangle = (1 - \eta_{t+1,j})\langle \varphi, G_{j,t} \rangle + \eta_{t+1,j}\langle \varphi, G_{0,t} \rangle \to \langle \varphi, \alpha_j \rangle \quad a.s.$$

Hence $\pi_{t+1,j} \Rightarrow \alpha_j$ a.s.

Now, for any $g \in C_b(\mathbb{X})$, define $T_g(\theta) := \int g(x)f(x|\theta)dx$. By Assumption 3, $T_g \in C_b(\Theta)$ (boundedness comes from the fact that probability density functions integrate finitely). Therefore:

$$\int g \, dP_{\pi_{t+1,j}} = \int T_g(\theta) \, \pi_{t+1,j}(d\theta) = \langle T_g, \pi_{t+1,j} \rangle \to \langle T_g, \alpha_j \rangle = \int g \, dP_{\alpha_j} \quad a.s.,$$

where the last equality holds by Fubini and the definition of P_{α_j} and the convergence from Theorem 7.2. Since this holds for all $g \in C_b(\mathbb{X})$, we have $P_{\pi_{t+1,j}} \Rightarrow P_{\alpha_j}$ a.s.

Steps 1–2 establish that, for any finite collection of distinct group indices $j_1, \ldots, j_k$

and bounded continuous functions $g_1, \ldots, g_k \in C_b(\mathbb{X})$,

$$\mathbb{E}\left[\prod_{i=1}^{k} g_i(X_{t+1,j_i}) \,\middle|\, \mathcal{F}_t\right] \longrightarrow \prod_{i=1}^{k} \int g_i \, dP_{\alpha_{j_i}} \quad \mathbb{P}-a.s.$$

Equivalently, the joint predictive distribution converges weakly to the product measure:

$$\mathbb{P}\Big((X_{t+1,j_1}, \ldots, X_{t+1,j_k}) \mid \mathcal{F}_t\Big) \Longrightarrow P_{\alpha_{j_1}} \otimes \cdots \otimes P_{\alpha_{j_k}} \quad \mathbb{P}\text{-a.s.} \tag{56}$$

□

Thus we have shown that marginally $\mathbb{P}(X_{t+1,j} \mid \mathcal{F}_t) \to P_{\alpha_j}$ for each group but also that the joint predictive converges to a random product probability measure. Thus the sequence $(\boldsymbol{X}_t)$ is asymptotically exchangeable by Theorem 2.6 and since convergence is to a product measure, by Fortini et al. (2018) it is asymptotically partially exchangeable.

Remark 7.4 (Asymptotic partial exchangeability of latent parameters). The result for the latent array $\{\theta_{t,j}\}$ follows immediately from the proof of Theorem 7.3. Under Assumption 1, $\theta_{t+1,j} \mid \mathcal{F}_t \sim \pi_{t+1,j}$, and the same convergence $\pi_{t+1,j} \Rightarrow \alpha_j$ a.s. established in Step 2 of that proof gives

$$\mathbb{P}((\theta_{t+1,j_1}, \ldots, \theta_{t+1,j_k}) \mid \mathcal{F}_t) \;\Rightarrow\; \alpha_{j_1} \otimes \cdots \otimes \alpha_{j_k} \quad \mathbb{P}-\text{a.s.}$$

for every finite collection of distinct groups $j_1, \ldots, j_k$. No additional assumptions beyond those of Theorem 7.3 are needed.

We can now make an assessment which is the counterpart, for the Newton-HDP algorithm, of what has been said about the quasi Bayes properties of the Newton's algorithm. Notably, indeed (56) establish not only that the limit of the one-step-ahead joint predictive is a product measure, but that each term is a mixture i.e. $P_{\alpha_j}(\cdot) = \int_\Theta F(\cdot \mid \theta)\alpha_j(d\theta)$. We can indeed say that the Newton–HDP learning rule implies a probability law $\widetilde{\mathbb{P}}$ implicitly defined by (55): asymptotically, it is a partially exchangeable mixture model, in which each group-specific predictive converges to the mixture $P_{\alpha_j}(\cdot) = \int_\Theta f(\cdot \mid \theta)\, \alpha_j(d\theta)$ with random mixing measure α_j. This provides a quasi-Bayesian justification in the same sense as Newton's algorithm for the single-group case: just as $X_n \mid \tilde{G} \approx P_{\tilde{G}}$ asymptotically, here $X_{t,j} \mid \alpha_j \approx P_{\alpha_j}$ for large t and each j. Whether the implicit prior law on (α_j)

approximates the HDP prior, especially in terms of clustering behavior, remains open and, in a sense, mirrors the discussion at the end of Chapter 3.

The joint dependence structure of $(\alpha_1, \ldots, \alpha_J)$ encodes the borrowed strength: they are not independent because they share ancestry through α_0, mirroring the HDP's hierarchical tie. Their law is implicitly determined by the predictive recursions; no closed-form expression is available, but this is precisely analogous to the single-group case where the law of $\tilde{G}$ is also only implicitly defined Fortini and Petrone (2020).

7.5 Future research

The quasi-Bayes result of Section 7.4 rests crucially on Assumption 1, which imposes conditional independence of $(\theta_{t,j})_{j=1}^J$ given $\mathcal{F}_{t-1}$. This makes the joint predictive of $(X_{t+1,j})_{j=1}^J$ factorize conditionally on $\mathcal{F}_t$ already at every finite time t (Step 1 in the proof of Theorem 7.3), so that the asymptotic result follows cheaply once convergence of $G_{j,t}$ is established. Borrowing of strength is nevertheless operative at every step: each marginal prior $\pi_{t+1,j} = (1 - \eta_{t+1,j})G_{j,t} + \eta_{t+1,j}G_{0,t}$ depends on the shared global estimate $G_{0,t}$, so that information from all groups propagates into every group's predictive through the hierarchical coupling. In general a more complete results ,which would one to one match the theory established for Newton's algorithm by Fortini and Petrone (2020), would be provided by leveraging the concept of *partially conditionally identically distributed* sequences (p-CID) established in Fortini et al. (2018). This condition is, as for CID, equivalent to partial exchangeability when coupled with stationarity (see Fortini et al. (2018) for the proof). The definition of p-CID sequences is the following.

Definition 7.5 (Partially $\mathcal{G}$-c.i.d. sequence)**.** A sequence of I-th dimentsional vectors $(\mathbf{X}_n^I)$, where I can be finite or countable, is said to be *partially conditionally identically distributed* with respect to a filtration $\mathcal{G}$, if it is adapted to $\mathcal{G}$ and, for every $j \in I$,

$$\mathbb{E}[f(X_{n+1,j}) \mid \mathcal{G}_n^j] = \mathbb{E}[f(X_{n+k,j}) \mid \mathcal{G}_n^j] \tag{57}$$

for all $k \geq 1$, $n \geq 0$ and all bounded measurable functions $f : \mathbb{X} \to \mathbb{R}$, where

$$\mathcal{G}_n^j = \mathcal{G}_n \vee \sigma(X_{n+1,i} : i \neq j).$$

From the definition each sequence $(X_{j,n})_{n\geq 1}$ is $\mathcal{G}_n^j$-CID and thus it is possible to characterize p-CID sequences in a similar manner as 2.3.

Proposition 7.6. *The following are equivalent.*

1. *The sequence* $(\mathbf{X}_n^I)$ *is partially* $\mathcal{G}$*-c.i.d.;*
2. *For any* $j \in I$ *and every bounded measurable function* $f : \mathbb{X} \to \mathbb{R}$, $(\mathbb{E}[f(X_{n+1,j}) \mid \mathcal{G}_n^j])_{n\geq 0}$ *is a* $\mathcal{G}^j$*-martingale;*

In particular, condition *(2)* is the one that we would like to have to our aims. In the case of the Newton–HDP algorithm it fails due to the presence of $G_{0,t-1}$ in the group-level prior $\pi_{t,j} = (1-\eta_{t,j})G_{j,t-1} + \eta_{t,j}G_{0,t-1}$. To see why, recall that condition *(2)* requires

$$\mathbb{E}\Big[\mathbb{E}[f(X_{t+2,j}) \mid \mathcal{G}_{t+1}^j] \;\Big|\; \mathcal{G}_t^j\Big] = \mathbb{E}[f(X_{t+1,j}) \mid \mathcal{G}_t^j],$$

that is, $\big(\mathbb{E}[f(X_{t+1,j}) \mid \mathcal{G}_t^j]\big)_{t\geq 0}$ must be a $\mathcal{G}^j$-martingale. The predictive $\mathbb{E}[f(X_{t+1,j}) \mid \mathcal{G}_t^j]$ depends on $\pi_{t+1,j}$, which in turn depends on $G_{0,t}$. But $G_{0,t}$ is updated at every global step using observations from all groups at time t, including the concomitant observations $\{X_{t,i} : i \neq j\}$ that are $\mathcal{G}_t^j$-measurable but *not* $\mathcal{G}_t$-measurable. Therefore, when we compute

$$\mathbb{E}\Big[\mathbb{E}[f(X_{t+2,j}) \mid \mathcal{G}_{t+1}^j] \;\Big|\; \mathcal{G}_t^j\Big],$$

the inner expectation involves $G_{0,t+1}$, which depends on $X_{t+1,i}$ for $i \neq j$ which are observations that are not in $\mathcal{G}_t^j$ but will be in $\mathcal{G}_{t+1}^j$. This asymmetry means the tower property does not reduce the expression to $\mathbb{E}[f(X_{t+1,j}) \mid \mathcal{G}_t^j]$, and the $\mathcal{G}^j$-martingale property therefore fails. Intuitively, the group-j predictive at time $t+1$ is not determined by $\mathcal{G}_t^j$ alone, because $G_{0,t}$, which enters $\pi_{t+1,j}$, will shift once the concomitant observations at time $t+1$ arrive and update the global measure.

P-CID sequences have very nice properties that are particularly suited to establish quasi-Bayes results for approximate algorithms for partially-exchangeable models. Indeed Fortini et al. (2018) prove that a p-CID sequence is asymptotically partially exchangeable and provide central limit theorems for p-CID sequences; something that we have yet to provide. In particular entering the p-CID framework would also be theoretically consistent with the structure found for Newton's recursion for which exchangeability was relaxed

to CID and was recovered in the limit. In the same way a p-CID algorithm would relax partial exchangeability and recover it in the limit.

Conclusions

Within this work we have taken a Bayesian predictive approach to study the quasi-Bayes properties of approximate recursive algorithms for two classes of semi-parametric Bayesian models.

The first is the popular Newton algorithm for Dirichlet process mixtures, for which we have presented the theoretical results of Fortini and Petrone (2020): by recasting Newton's recursions as predictive rules, the sequence (X_n) is shown to be conditionally identically distributed and hence asymptotically exchangeable, providing a genuine quasi-Bayesian justification of the algorithm and enabling uncertainty quantification through the asymptotic posterior of $\tilde{G}$. We complemented this with two empirical contributions, a clustering procedure exploiting the asymptotic structure of the modes of the mixing density $\tilde{g}_n$, and a warm-start scheme that reduces permutation sensitivity of the estimate $\tilde{g}_n$, only adding a small computational cost.

The main contribution of this work is the Newton-HDP algorithm, a novel recursive procedure extending Newton's estimation principle to Hierarchical Dirichlet process mixtures. The algorithm couples group-specific and global estimates through a feedback loop that enables borrowing of strength across populations at a computational cost of $\mathcal{O}(NJK)$, compared to $\mathcal{O}(N^2 \cdot C_{\text{MCMC}})$ for exact inference. We have characterized the information losses relative to the exact HDP predictive, validated the algorithm empirically by showing that improvements over independent Newton recursions are largest precisely where data are scarcest and by comparing its estimation performance with that of an exact Gibbs sampler. Lastly we established its quasi-Bayesian foundations: under regularity conditions, the array $(X_{t,j})$ is asymptotically partially exchangeable with directing measures P_{α_j}.

The most natural direction for future work is establishing whether a p-CID characterization in the sense of Fortini et al. (2018) can be recovered for a suitably modified version of the Newton-HDP. This would provide a one-to-one theoretical parallel with the single-group case, unlock central limit theorems to approximate the group-specific posterior distributions, and enable uncertainty quantification in the hierarchical setting, completing the quasi-Bayesian picture that this work has begun to develop. This work, alongside the much cited Fortini and Petrone (2020) and Fortini and Petrone (2025), establish a methodology and a theoretical corpus (through also the cited literature) to

study, in a statistically principled way, the properties of other predictive algorithms, in particular employed to approximate the learning scheme of other Bayesian models.

A Appendix

Lemma A.1 (Convergence of weighted averages)**.** *Let $(a_t)_{t\geq 1}$ be a sequence of real-valued random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$a_t \longrightarrow a \qquad \text{almost surely,}$$

Let $(w_t)_{t\geq 1}$ be a deterministic sequence with $w_t \in (0,1)$ satisfying

$$\sum_{t=1}^{\infty} w_t = \infty$$

Define the recursion

$$z_{t+1} = (1 - w_{t+1})z_t + w_{t+1}a_{t+1}, \qquad t \geq 0,$$

where z_0 is an arbitrary (possibly random) initial condition.
Then

$$z_t \longrightarrow a \qquad \text{almost surely.}$$

Proof. Fix ω in the event of probability one on which $a_t(\omega) \to a(\omega)$. For this fixed ω, consider the deterministic sequences $a_t^\omega := a_t(\omega)$, $a^\omega := a(\omega)$, and $z_t^\omega := z_t(\omega)$.

Rewrite the recursion as

$$z_{t+1}^\omega - a^\omega = (1 - w_{t+1})(z_t^\omega - a^\omega) + w_{t+1}(a_{t+1}^\omega - a^\omega).$$

Iterating,

$$z_t^\omega - a^\omega = \left(\prod_{\ell=1}^{t}(1 - w_\ell)\right)(z_0^\omega - a^\omega) + \sum_{k=1}^{t}\left(\prod_{\ell=k+1}^{t}(1 - w_\ell)\right)w_k(a_k^\omega - a^\omega).$$

Since $\sum_\ell w_\ell = \infty$, we have $\prod_{\ell=1}^{t}(1 - w_\ell) \to 0$. Fix $\varepsilon > 0$. Because $a_k^\omega \to a^\omega$, there exists $K(\omega)$ such that $|a_k^\omega - a^\omega| \leq \varepsilon$ for all $k \geq K(\omega)$. Split the sum accordingly:

$$\sum_{k=1}^{t} = \sum_{k=1}^{K(\omega)-1} + \sum_{k=K(\omega)}^{t}.$$

For the tail sum,

$$\sum_{k=K(\omega)}^{t}\left(\prod_{\ell=k+1}^{t}(1-w_\ell)\right)w_k|a_k^\omega - a^\omega| \le \varepsilon \sum_{k=K(\omega)}^{t}\left(\prod_{\ell=k+1}^{t}(1-w_\ell)\right)w_k \le \varepsilon,$$

since

$$\sum_{k=1}^{t}\left(\prod_{\ell=k+1}^{t}(1-w_\ell)\right)w_k = 1 - \prod_{\ell=1}^{t}(1-w_\ell) \le 1.$$

For the finite sum $k < K(\omega)$, each term vanishes as $t \to \infty$ because $\prod_{\ell=k+1}^{t}(1-w_\ell) \to 0$.

Hence

$$\limsup_{t\to\infty} |z_t^\omega - a^\omega| \le \varepsilon.$$

Since ε is arbitrary, $z_t^\omega \to a^\omega$. As ω was arbitrary in a probability-one set, the result holds almost surely. □

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, I used *Claude* to improve the language and readability of the text, to transfer calculations from handwritten notes into LaTeX, and to double-check certain mathematical computations. Additionally I used the aforementioned tool to support the debugging and refinement of the Python code used in the simulations. After using this tool, I carefully reviewed and edited the content as needed and take full responsibility for the final version of the thesis.

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